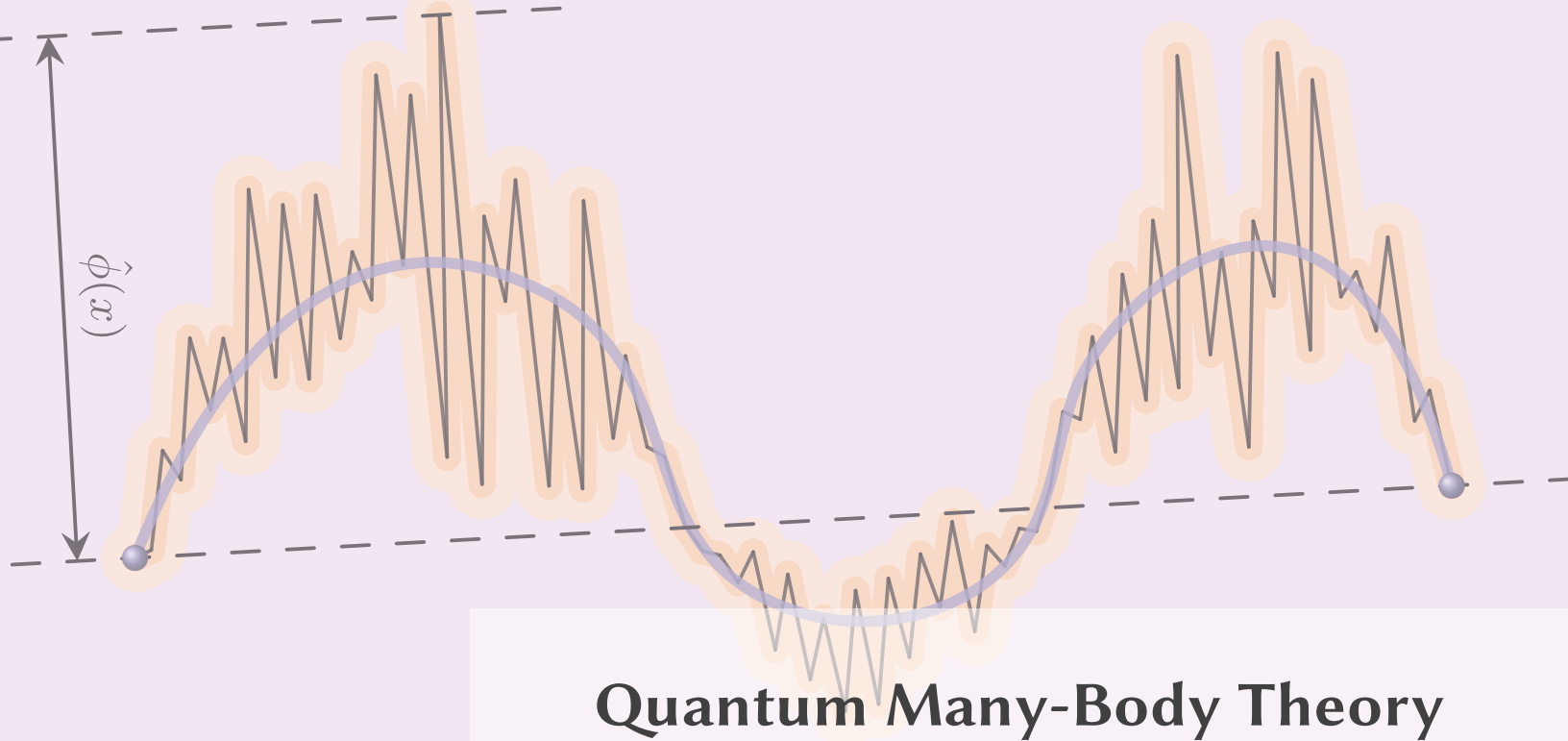




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CHAPTER 1 Introduction

Prerequisite

Quantum Mechanics QM II and / or Advanced QM Double wave (hard): instanton gas

Statistical Mechanics

Solid State Physics

Mechanics (formal theory)

Electrodynamics mostly for transport theory

Reference Texts

See the **References** chapter: [1–10]. Textbooks by Ding [1], Coleman [2], Stefanucci and Leeuwen [5], and Bogolubov Jr [7] are recommended.

In the Lecture Notes (Ding [1])

- Weak coupling theory will be standard
- Strong coupling ARE NOT standard, in developing

1.1 What QMB study

From Poisson Brackets $\{ \ }_{PO} \Leftrightarrow$ Quantum commutators $[\]$.

CHAPTER 2 Gaussian Integral

2.1 Free Gaussian Integral

$$I = \int_{-\infty}^{+\infty} dx e^{-x^2} = \sqrt{\pi}, \quad I^2 = \pi$$

2.1.1 Perturbation expansion of $\int_{-\infty}^{+\infty} dx e^{-x^2 - gx^4}$

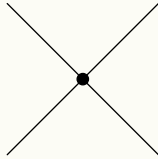
$$Z(g) = \sum_{n=0}^{\infty} \frac{(-g)^n}{n!} \int dx x^{4n} e^{-x^2} \quad (2.1)$$

$$\int x^{2m} e^{-x^2} dx \quad (2.2)$$

Perturbation expansion is actually divergent: $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \Rightarrow \infty$
 The series $Z(g)$ has zero convergence radius.

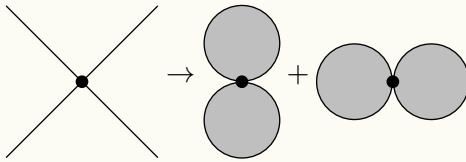
2.2 Feynman Diagram

1. Vertex: gx^4

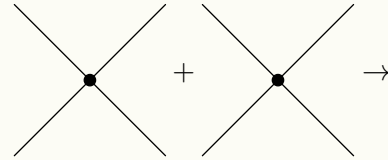


2. Propagator: $\langle x, x \rangle = \frac{1}{2}$

(a) $n = 1$:



(b) $n = 2$:

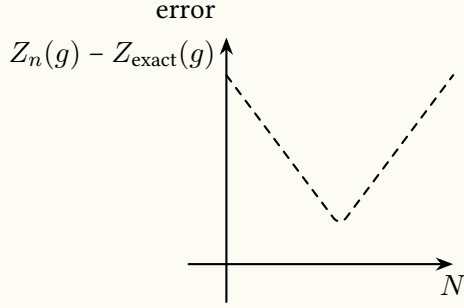


Symmetry factor.

2.3 Finite n still good for g -expansion

$$Z_{\text{exact}}(g) = \frac{1}{2} \sqrt{\frac{\pi}{g}} e^{1/(8g)} K_{1/4} \left(\frac{1}{8g} \right) \quad (2.3)$$

$K_{1/4}$ is modified Bessel function.



Around $N = 13$: it starts going out.

Kondo Problem: to 1 or 2 order.

From (2.1), we have

$$Z(g) = \frac{1}{g^{1/4}} \int dy e^{-y^4 - y^2/g^{1/2}} \quad (2.4)$$

$$= \frac{1}{g^{1/4}} \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} \frac{1}{g^{m/2}} \underbrace{\int dy y^{2m} e^{-y^4}}_{\frac{1}{2} \frac{\Gamma(\frac{m}{2} + \frac{1}{4})}{g^{m/2 + 1/4}}} \quad (\text{Taylor Expansion}) \quad (2.5)$$

$$\left| \frac{a_{m+1}}{a_m} \right| \rightarrow \frac{1}{\sqrt{2g}} \cdot \frac{1}{\sqrt{m}} \rightarrow 0 \quad (2.6)$$

Then, $Z(1/g)$ converges $(0, +\infty)$

Large y , no matter how small g is, y^4 term dominates.

2.4 Multi-variable Gaussian Integral

Just means we have found

$$\int dx_1 dx_2 \dots dx_N e^{-[x][M][x]} = \frac{1}{2} \int dy_1 \dots dy_N e^{-[y]u^\dagger M u [y]} \quad (2.7)$$

(a) $[x] = (x_1, \dots, x_N)$

(b) $[M] = (M_{ij})$.

Diagonalise M : $[x] \rightarrow u[y]$, $M \rightarrow \begin{pmatrix} \ddots & & \\ & \ddots & \\ & & \ddots \end{pmatrix}$, $u^\dagger M u = \begin{pmatrix} \ddots & & \\ & \ddots & \\ & & \ddots \end{pmatrix}$.

Add interaction:

$$\int dx_1 \dots dx_N e^{-\sum_{i=1}^N x_i^4 g} \quad (2.8)$$

Consider strong coupling expansion:

$$\int dx_1 \dots dx_n e^{-\sum x_{i1}, x_{i2}, \dots, x_{in} g_{12}^{34}} \quad (2.9)$$

$$\int dy_1 \dots dy_N = e^{-y^4} \quad (2.10)$$

CHAPTER 3 Second Quantization

3.1 QM of single particle

Dirac notation will be used in this book: ket $|\Phi\rangle$, bra $\langle\Phi|$, and operator $\hat{O}$. Wave function should be $\langle x|\Phi\rangle$.

For single particle QM, we have six axioms (postulates)

With hamiltonian $\hat{H}$

1. Description of a state $|\Phi\rangle \in \mathcal{H}$:
 - Using Vector Space
 - List
 - Orthogonality
 - Completeness
2. Observables: $\hat{A} \in \mathcal{H}$, $\hat{A}^\dagger = \hat{A}$.
- 3.
4. Measurements: Spectral Decomposition
 - a_n , eigenvalue of $\hat{A}$.
 - $P_n = |\langle a_n|\Phi\rangle|^2$
5. Post-measurement: collapse: $|\Phi\rangle_{\text{pre}} \xrightarrow{\text{measurement}} |a_n\rangle$
6. $i\hbar \partial_t |\Psi\rangle = \mathcal{H}|\Psi\rangle$

Canonical(?) Quantization rules $x \rightarrow \hat{x}$, $p \rightarrow \hat{p}$, and $[\hat{x}, \hat{p}] = i\hbar$,

Then you can define Poisson bracket $\{\bar{x}, p\} = \mathbb{I} \Rightarrow [\hat{x}, \hat{p}] = \mathbb{I}$.

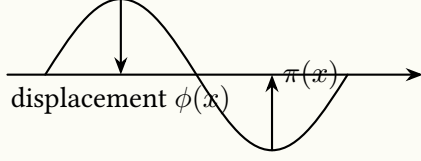
3.2 Canonical Quantisation

3.2.1 Overview

$|x\rangle$, $[\hat{x}, \hat{p}] = i\hbar$, $\hat{p} = -i\hbar \partial_x$, $\langle x|\Phi\rangle$.

$\langle \hat{p}|\Phi\rangle$, $\hat{p} \rightarrow p$, $\hat{x} \rightarrow i\hbar \partial p$.

For classical string like



$$\mathcal{H} = \int dx \left(\frac{T}{2} (\nabla_x \phi(x))^2 + \frac{1}{2\rho} \pi^2(x) \right)^1 \quad (3.1)$$

Exactly, $i\{\tilde{\phi}, \tilde{\pi}\} = 1$.

Quantisation $[\hat{\phi}(x), \hat{\pi}(y)] = i\hbar\delta(x-y)$.

3.2.2 Harmonic oscillator: zero-dim field theory

The hamiltonian, eigen energy and state are

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2, \quad E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \quad |n\rangle = \frac{1}{\sqrt{n!}}(a^\dagger)^n|0\rangle \quad (3.2)$$

Let

$$\xi = \frac{x}{\Delta x}, \quad p_\xi = \frac{p}{\Delta p}, \quad \Delta x = \sqrt{\frac{\hbar}{m\omega}}, \quad \Delta p = \sqrt{m\omega\hbar}$$

then we have

$$\begin{cases} a^\dagger = \frac{1}{\sqrt{2}}(\xi - ip_\xi) \\ a = \frac{1}{\sqrt{2}}(\xi + ip_\xi) \end{cases} \quad \text{and} \quad \begin{cases} [a, a^\dagger] = 1 \\ [a, a] = [a^\dagger, a^\dagger] = 0 \end{cases} \quad (3.3)$$

Do it differently, the time evolution

$$x(t) = \langle \phi_n(t) | \hat{x} | \phi_n(t) \rangle = \langle \phi_n | e^{iHt/\hbar} \hat{x} e^{-iHt/\hbar} | \phi_n \rangle = \langle \phi_n | \hat{x}(t) | \phi_n \rangle \quad (3.4)$$

and the EOM (Equation of Motion) is

$$-i\hbar \frac{da(t)}{dt} = [H, a(t)] = \omega a(t) \quad (3.5)$$

So,

$$\hat{x}(t) = \frac{\Delta x}{\sqrt{2}}[a(t) + a^\dagger(t)], \quad \hat{p}(t) = \frac{\Delta p}{i\sqrt{2}}[a(t) - a^\dagger(t)] \quad (3.6)$$

These type of operators are so called *field operators*.

$$\int_{-\Delta}^{+\Delta} dx e^{-x^2 - gx^4} \text{Herry Poncô} = \lim_{N \rightarrow \infty} \lim_{\Delta \rightarrow \infty} \quad (3.7)$$

1. $j, (\pi_j)^2 + (D_j - D_{j+1})^2$: chiral phonon, theormal hall.

¹Physically, it should have more terms

3.3 QM of many particles

Seventh Axiom: If you don't consider anything: we want to remove exchange degeneracy $\Rightarrow$ symmetrization / antisymmetrization rule.

Two types wavefunctions. For Boson: permanent; for Fermion Slater det.

Each state is one particle in particle label $\varphi_1^{\epsilon_1}, \varphi_2^{\epsilon_2}, \dots, \varphi_m^{\epsilon_m}$, which $\epsilon_i \approx \epsilon_j$, and state label $1, 2, \dots, N$.

$$\begin{matrix} & \varphi_1^{\epsilon_1} & \varphi_2^{\epsilon_2} & \dots & \varphi_m^{\epsilon_m} \\ \begin{matrix} 1 \\ 2 \\ \vdots \\ N \end{matrix} & \left(\begin{array}{cccc} |1, \varphi_1^{\epsilon_1}\rangle & |1, \varphi_2^{\epsilon_2}\rangle & & \\ |2, \varphi_1^{\epsilon_1}\rangle & & & \\ & & & \\ & & & \end{array} \right) \end{matrix}$$

symmetrical polynomials.

We can get into a newer occupation number representation

label any state $|\varphi\rangle = |n_{\epsilon_1}(\text{occupation / energy No.}), n_{\epsilon_2}, \dots, n_{\epsilon_n}, \dots\rangle_{\text{boson / fermion}}$

occupation number representation: dependent on underlying first quantization basis.

$$N = n_{\epsilon_1} + \dots + n_{\epsilon_n} + \dots \quad (3.8)$$

3.4 Fock space

$$F = \{\mathcal{H}_0, \mathcal{H}_1, \mathcal{H}_2, \dots, \mathcal{H}_n, \dots\} \quad (3.9)$$

Rock matrix:

$$\begin{pmatrix} |\varphi, N=0\rangle & |\varphi, N=1\rangle & & \\ H_0 & & & \\ & H_1 & & \\ & & \ddots & \\ & & & H_N \\ & & & & \ddots \end{pmatrix} \quad (3.10)$$

3.5 Field operators

Suppose we have

$$|\Psi\rangle = | \quad \rangle; \quad (3.11)$$

$\Psi(x): \langle x|\Psi\rangle$.

Each state has a particular wave function,

$|x_1\rangle$ is given by $\Psi^\dagger(x_1)|0\rangle \Rightarrow \hat{x}|0\rangle = \hat{a} + \hat{a}^\dagger|0\rangle = |1\rangle$.

We are creating $a^\dagger|0\rangle$, $|x\rangle\langle x|1\rangle = |x_1\rangle\psi_1(x_1)$.

Project $a^\dagger$ onto $|x\rangle\langle x|$, that is

$$\psi^\dagger(x) = a^\dagger|x\rangle\langle x|$$

Similarly, if u want two particles at two different positions, u get

$$\psi^\dagger(x_2)\psi^\dagger(x_1)|0\rangle = |x_1, x_2\rangle = \psi^\dagger(x_2)|x_1\rangle \quad (3.12)$$

They are commutation

$$\psi^\dagger(x)\psi^\dagger(y) = \pm\psi^\dagger(y)\psi^\dagger(x) \Leftrightarrow [\psi^\dagger(x), \psi^\dagger(y)]_{\mp}^2 = 0 \quad (3.13)$$

And finally, introduce

$$[\hat{\psi}(x), \hat{\psi}(y)] = 0, \hat{\psi}(x)|0\rangle = 0$$

For any state like $|\Psi\rangle \in F$, u also see like $\langle\Psi|\psi(x)|\psi\rangle = \langle\Psi|\psi^\dagger(x)|\psi\rangle = 0$.

$\langle\varphi, N|\varphi, N-1\rangle = 0$. And finally, we want to be able to $\langle y|x\rangle = \delta(x-y) = \langle y|\psi^\dagger(x)|0\rangle = \langle 0|\hat{\psi}(y)|x\rangle$. u can prove that: $\hat{\psi}(x)|y\rangle = \delta(x-y)$, and finally $[\hat{\psi}(x), \hat{\psi}(y)]_{\mp} = \delta(x-y)$.

3.6 Creation / annihilation operators

For any state $\langle x|n\rangle = \varphi_n(x)$, and u can also $|n\rangle = \int dx$.

IDENTITY IS VERY IMPORTANT! $\mathbb{1}$

$$\mathbb{1} = \int dx|x\rangle\langle x|$$

Then we have

$$|n\rangle = \int dx|x\rangle\langle x|n\rangle = \int dx\varphi_n(x)|x\rangle = \int dx\varphi_n(x)\psi^\dagger(x)|0\rangle$$

u can define $\Rightarrow d_n^\dagger = \int dx\varphi_n(x)\psi^\dagger(x) \Rightarrow d_n^\dagger|0\rangle = |n\rangle$.

Similarly, $\Rightarrow \hat{d}_n \equiv \int dx\varphi_n^*(x)\tilde{\psi}(x) \Rightarrow \hat{d}_n|n\rangle = |0\rangle$. So $\hat{d}_n|n\rangle \rightarrow \hat{d}_n\hat{d}_n^\dagger|0\rangle$ (Homework)

Also prove that $[d_n, d_m^\dagger]_{\pm} = \langle n|m\rangle = \delta_{n,m}$

In other words, ...

Occupation number rep.

for $|x_1, \dots, x_n\rangle$, if u change any particle like (reverse x_1 and x_n), u will get $(t)^p|x_1, x_2, \dots, x_n\rangle$. $(t)^p$: boson & fermion.

interchange: $1\ 2\ 3 \rightarrow 2\ 1\ 3$ (Permutation) $\#P$ ($P_{1\rightarrow 2}, P_{2\rightarrow 1}$): $\#$ of perm operation needed.

When take $\text{mod } 2(p)$,

$h_o : \epsilon_1, \epsilon_2, \dots, \epsilon_i \rightarrow |n_1, n_2, \dots, n_i, \dots\rangle_{b(P)} \Rightarrow$ Fock space: $H_0 \oplus H_1 \oplus \dots H_N \oplus \dots$

Creation and annihilation operators: $d_{n_i}^\dagger, d_{n_i} \Leftrightarrow \psi^\dagger(x)|0\rangle = |x\rangle$

$$^2[A, B]_{\mp} = AB \mp BA$$

$$d_n^\dagger|0\rangle = |n\rangle, d_n^\dagger = \frac{1}{\sqrt{n!}}(a^\dagger)^n$$

Identity

$$\mathbb{1} = \frac{1}{N!} \sum_{n_1, \dots, n_i} |n_1, n_2, n_3, \dots\rangle \langle \dots| = \int dx |x\rangle \langle x| = \int dx \psi^\dagger(x) |0\rangle \langle 0| \psi(x) = \int dx_1 \dots dx_n |x_1, \dots, x_i\rangle \langle \dots|$$

3.7 Hamiltonians in second quantization

Operators of total number of particles

$$\hat{N} = \int dx \hat{n}(x), \hat{N}(x) = \psi^\dagger(x) \psi(x), \hat{N} = \sum_n d_n^\dagger d_n, \hat{N}_n = d_n^\dagger d_n \quad (3.14)$$

$$\hat{n}(x)|n_1 \dots \dots\rangle \rightarrow \sum |x_1, \dots, x_i, \dots\rangle$$

Eignevalue is equal to $(\sum_i \delta(x - x_i))|x_1, \dots, x_i, \dots, x_N\rangle$

$$\hat{h}_0: \hat{h}_0|\epsilon_1\rangle = \epsilon_i|\epsilon_i\rangle$$

If we have N particles, for each of the particle, we have to specify a Hamiltonian

$$H_0 = \sum_{i=1}^N \hat{h}_0(x) i$$

In some occupation states, we have specify the occupation numbers

$$|n_1, \dots, n_j, \dots\rangle, \quad \text{while} \quad \sum n_j = N$$

and

$$H_0|n_1, \dots, n_j, \dots\rangle = \sum_j \epsilon_j n_j |\dots\rangle$$

$\oplus$ in book $\sum \epsilon_j |\dots\rangle$.

Then we can define $H_0|n \dots\rangle = (\epsilon_{n_1} + \epsilon_{n_N})|\dots\rangle$

$$H_0 = \int dx dx' \psi^\dagger(x) \langle x | \hat{h}_0 | x \rangle \hat{\psi}(x')$$

If $\hat{h}_0 = \hat{p}^2/2m$, then $\langle x | \hat{h}_0 | x' \rangle = \delta(x - x_0)(-\frac{\nabla^2 k^2}{2m})$ Then,

$$H_0 = \sum_n \int dx dx' \psi^\dagger(x) \langle x | \hat{h}_0 | n \rangle \langle n | x \rangle \hat{\psi}(x') \quad (3.15)$$

$$= \sum_n \int dx dx' \epsilon_n \varphi_n(x) \psi^\dagger(x); \varphi_n^*(x') \hat{\psi}(x') \quad (3.16)$$

$$= \sum_n \epsilon_n d_n^\dagger d_n = \sum_n \epsilon_n \hat{N}_n \quad (3.17)$$

$$[\hat{N}_n, d_m^\dagger]_- = \delta_{nm} d_m^\dagger, [\hat{N}_n, d_m]_- = -\delta_{nm} d_n$$

Consider $[H_0, \hat{d}_m^\dagger] = \epsilon_m d_m^\dagger$, $[H_0, d_m] = -\epsilon_n d_m$, the we get spectral geneeratiy algtroon $[H_0, \hat{O}] \propto \hat{O}$ while $H_o = \sum \epsilon_0 |n\rangle \langle n|$ and $|m\rangle \langle m|$

Finally, we can write

$$H_0 = \int dr \psi^\dagger(r) h_{r_0}(r, \nabla_r, \sim) \psi(r)$$

Here, $h = \langle r | \hat{h}_0 | r \rangle$.

3.7.1 Interactivity Hamiltonian

Consider two body $\nu(\hat{x}_1, \hat{x}_2)$

$$H_{\text{int}}|x_1, \dots, x_{N..}\rangle = \frac{1}{2} \left(\sum_{i \neq j} \nu(\hat{x}_i, \hat{x}_j) \right) |x_1, \dots, x_{N..}\rangle = \left(\sum_{i > j} \nu(\hat{x}_i, \hat{x}_j) \right) |x_1, \dots, x_{N..}\rangle$$

$$\hat{n}(x_1, \dots, x_n) = \hat{n}(x)$$

We write $v(\hat{x}_i, \hat{x}_j) = \int dx' dx \delta(x - x_i) \delta(x' - x_j) v(x, x')$ and substitute it into H_{int} , then using $(\sum_i \delta(x - x_i))|x_1, \dots, x_i, \dots, x_N\rangle$ inversely, then we have

$$H_{\text{int}} = \frac{1}{2} \int dx dx' (\nu(x, x') n(x) n(x') - \nu(x, x) n(x)) \quad (3.18)$$

$$= \frac{1}{2} \int dx dx' \nu(x, x') (\psi^\dagger(x) \psi(x) \psi^\dagger(x') \psi(x') - \delta(x - x') \psi^\dagger(x) \psi(x)) \quad (3.19)$$

$$= \frac{1}{2} \int dx dx' \nu(x, x') \psi^\dagger(x) \psi^\dagger(x') \psi(x) \psi(x') \quad (3.20)$$

3.7.2 Total Hamiltonian

Such as

$$\begin{aligned} \mathcal{H} &= \int dx dx' \psi^\dagger(x) \langle x | h_0 | x' \rangle \psi'(x') + \frac{1}{2} \int dx dx' \nu(x, x') \psi^\dagger(x) \psi^\dagger(x') \psi(x) \psi(x') \\ &= \sum_{ij} h_{ij} d_i^\dagger d_j + \frac{1}{2} \sum_{ijmn} \nu_{ijmn} d_i^\dagger d_j^\dagger d_m d_n \end{aligned}$$

Here

$$h_{ij} = \langle i | \hat{h} | j \rangle = \int dr \varphi_i^*(r, \nabla_r, \sim) \varphi_j(r) = h_{ji}^*, \quad \langle x | i \rangle = \varphi_i(x)$$

and

$$\nu_{ijmn} = \int dx dx' \varphi_i^*(x) \varphi_j^*(x') \nu(x, x') \varphi_m(x') \varphi_n(x)$$

Homework #1 [2025-09-02]

Problem 1.1. Prove the following equations

$$[\hat{d}_n, \hat{d}_m^\dagger]_{\mp} = \langle n|m \rangle = \delta_{nm},$$

$$\hat{d}_n|n\rangle = |0\rangle.$$

Solution. *Proof.* Reviewing the definitions of $\hat{d}_n^\dagger$ and $\hat{d}_n$

$$\hat{d}_n = \int dx \varphi_n^*(x) \hat{\psi}(x), \quad \text{and} \quad \hat{d}_n^\dagger = \int dx \varphi_n(x) \hat{\psi}^\dagger(x),$$

then substitute them into the (anti)commutator

$$\begin{aligned} [\hat{d}_n, \hat{d}_m^\dagger]_{\mp} &= \left(\int dx \varphi_n^*(x) \hat{\psi}(x) \right)^\dagger \left(\int dx' \varphi_m(x') \hat{\psi}(x') \right)_{\mp} \left(\int dx' \varphi_m(x') \hat{\psi}(x') \right) \left(\int dx \varphi_n^*(x) \hat{\psi}(x) \right)^\dagger \\ &= \int dx \int dx' \varphi_n^*(x) \varphi_m(x') [\hat{\psi}^\dagger(x) \hat{\psi}(x') \mp \hat{\psi}(x') \hat{\psi}^\dagger(x)] \\ &= \int dx \int dx' \varphi_n^*(x) \varphi_m(x') [\hat{\psi}^\dagger(x), \hat{\psi}(x')]_{\mp}. \end{aligned}$$

Since the (anti)commutator of the field operator is

$$[\hat{\psi}^\dagger(x), \hat{\psi}(x')]_{\mp} = \delta(x - x'),$$

integrate over x' , and then recognize the inner product expression. Since the functions φ_n, φ_m are orthonormal, we arrive at

$$[\hat{d}_n, \hat{d}_m^\dagger]_{\mp} = \int dx \int dx' \varphi_n^*(x) \varphi_m(x') \delta(x - x') = \int dx \varphi_n^*(x) \varphi_m(x) = \langle \varphi_n | \varphi_m \rangle = \delta_{nm}.$$

The integration above used the *Dirac measure* of the δ -function.

Proof. The one-particle ket $|n\rangle$ can be expanded in the position-spin kets using the identity

$$|n\rangle = \int dx |x\rangle \langle x|n\rangle = \int dx \varphi_n(x) |x\rangle = \int dx \varphi_n(x) \hat{\psi}^\dagger(x) |0\rangle.$$

Since $\hat{\psi}(x)|0\rangle = 0$, that is *the annihilation operator kills the vacuum*, then we have

$$\hat{\psi}(x) \hat{\psi}^\dagger(x') |0\rangle = [\hat{\psi}(x), \hat{\psi}^\dagger(x')]_{\mp} |0\rangle = \delta(x - x') |0\rangle,$$

then substitute the relation above, the expression of $\hat{d}_n$ and the expansion of $|n\rangle$ into $\hat{d}_n|n\rangle$

$$\begin{aligned} \hat{d}_n|n\rangle &= \int dx \varphi_n^*(x) \hat{\psi}(x) \int dx' \varphi_n(x') \hat{\psi}^\dagger(x') |0\rangle = \int dx \int dx' \varphi_n^*(x) \varphi_n(x') \delta(x - x') |0\rangle \\ &= \int dx \varphi_n^*(x) \varphi_n(x) |0\rangle = \int dx |\varphi_n(x)|^2 |0\rangle = |0\rangle. \end{aligned}$$

The integration above used *the sifting property of the δ -function*, and the normalization of $\varphi_n(x)$.

Problem 1.2. Prove the inverse relations (3.6.4) i.e., the inverse relations of

$$\hat{\psi}(\mathbf{x}) = \sum_n \varphi_n(\mathbf{x}) \hat{d}_n, \quad \text{and} \quad \hat{\psi}^\dagger(\mathbf{x}) = \sum_n \varphi_n^*(\mathbf{x}) \hat{d}_n^\dagger,$$

meaning expressing $\hat{d}_n$ and $\hat{d}_n^\dagger$ by $\hat{\psi}(\mathbf{x})$.

Solution. To express $\hat{d}_n$ (and $\hat{d}_n^\dagger$), we can use the orthogonality $\int d\mathbf{x} \varphi_m^*(\mathbf{x}) \varphi_n(\mathbf{x}) = \delta_{mn}$ to eliminate itself on the right side. Left multiply both sides of $\varphi_m^*(\mathbf{x})$ by $\hat{\psi}(\mathbf{x})$, and then integrate over $\mathbf{x}$

$$\int d\mathbf{x} \varphi_m^*(\mathbf{x}) \hat{\psi}(\mathbf{x}) = \int d\mathbf{x} \varphi_m^*(\mathbf{x}) \left(\sum_n \varphi_n(\mathbf{x}) \hat{d}_n \right).$$

Since the integral and sum actions are exchangeable in this case, then we arrive at

$$\int d\mathbf{x} \varphi_m^*(\mathbf{x}) \hat{\psi}(\mathbf{x}) = \sum_n \hat{d}_n \int d\mathbf{x} \varphi_m^*(\mathbf{x}) \varphi_n(\mathbf{x}) = \sum_n \delta_{mn} \hat{d}_n = \hat{d}_m.$$

Similarly, for the creation operator $\hat{d}_n^\dagger$, repeat the actions above to $\hat{\psi}(\mathbf{x})^\dagger$ as following

$$\int d\mathbf{x} \varphi_m(\mathbf{x}) \hat{\psi}(\mathbf{x})^\dagger = \sum_n \hat{d}_n^\dagger \int d\mathbf{x} \varphi_m(\mathbf{x}) \varphi_n(\mathbf{x})^* = \sum_n \delta_{mn} \hat{d}_n^\dagger = \hat{d}_m^\dagger.$$

To summarize, the inverse relations of $\hat{\psi}(\mathbf{x})$ and $\hat{\psi}(\mathbf{x})^\dagger$ for mode n are given by

$$\hat{d}_n = \int d\mathbf{x} \varphi_n^*(\mathbf{x}) \hat{\psi}(\mathbf{x}), \quad \text{and} \quad \hat{d}_n^\dagger = \int d\mathbf{x} \varphi_n(\mathbf{x}) \hat{\psi}(\mathbf{x})^\dagger.$$

Problem 1.3. Let $|n\rangle = |\mathbf{p}\tau\rangle$ be a momentum-spin ket so that $\langle \mathbf{x} | \mathbf{p}\tau \rangle = e^{i\mathbf{p}\cdot\mathbf{r}} \delta_{\sigma\tau}$.³

$$\langle \mathbf{x} | \mathbf{p}\sigma' \rangle = \delta_{\sigma\sigma'} \langle \mathbf{r} | \mathbf{p} \rangle \quad \text{with} \quad \langle \mathbf{r} | \mathbf{p} \rangle = e^{i\mathbf{p}\cdot\mathbf{r}}.$$

Show that the (anti)commutation relation in (3.6.2) then reads

$$[\hat{d}_{\mathbf{p}\tau}, \hat{d}_{\mathbf{p}'\tau'}^\dagger]_{\mp} = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}') \delta_{\tau\tau'},$$

and that the expansion (3.6.4) of the field operators in terms of the $\hat{d}$ -operators is

$$\hat{\psi}(\mathbf{x}) = \int \frac{d^3\mathbf{p}}{(2\pi)^3} e^{i\mathbf{p}\cdot\mathbf{r}} \hat{d}_{\mathbf{p}\sigma}, \quad \text{and} \quad \hat{\psi}^\dagger(\mathbf{x}) = \int \frac{d^3\mathbf{p}}{(2\pi)^3} e^{-i\mathbf{p}\cdot\mathbf{r}} \hat{d}_{\mathbf{p}\sigma}^\dagger.$$

Solution. *Proof.* When $|n\rangle = |\mathbf{p}\tau\rangle$, $\varphi_n(\mathbf{x})$ becomes $\varphi_{\mathbf{p}\tau}(\mathbf{x}) = \langle \mathbf{x} | \mathbf{p}\tau \rangle = \delta_{\sigma\tau} e^{i\mathbf{p}\cdot\mathbf{r}}$. Using the combined coordinate $\mathbf{x} \equiv (\mathbf{r}, \sigma)$ that provides a complete description of a particle's state: continuous spatial position $\mathbf{r}$ and discrete spin projection quantum number σ . The expression of the operator $\hat{d}_{\mathbf{p}\tau}$ in terms of $\hat{\psi}(\mathbf{r}, \sigma)$ should be *not only integrate over $\mathbf{r}$, but also sum over spin σ* . That is

$$\begin{aligned} \hat{d}_{\mathbf{p}\tau} &= \int d^3\mathbf{x} \varphi_{\mathbf{p}\tau}^*(\mathbf{x}) \hat{\psi}(\mathbf{x}) = \int d^3\mathbf{r} \sum_{\sigma} \delta_{\sigma\tau} e^{-i\mathbf{p}\cdot\mathbf{r}} \hat{\psi}(\mathbf{r}, \sigma) = \int d^3\mathbf{r} e^{-i\mathbf{p}\cdot\mathbf{r}} \hat{\psi}(\mathbf{r}, \tau), \\ \hat{d}_{\mathbf{p}\tau}^\dagger &= \int d^3\mathbf{x} \varphi_{\mathbf{p}\tau}(\mathbf{x}) \hat{\psi}(\mathbf{x})^\dagger = \int d^3\mathbf{r} \sum_{\sigma} \delta_{\sigma\tau} e^{i\mathbf{p}\cdot\mathbf{r}} \hat{\psi}(\mathbf{r}, \sigma)^\dagger = \int d^3\mathbf{r} e^{i\mathbf{p}\cdot\mathbf{r}} \hat{\psi}(\mathbf{r}, \tau)^\dagger. \end{aligned}$$

³[see 5, eq. (1.10)]

Promoting $[\hat{\psi}^\dagger(x), \hat{\psi}(x')]_{\mp} = \delta(x - x')$ to “4D” (means 3D coordinate r + 1D spin σ (or the variable τ))

$$[\hat{\psi}^\dagger(\mathbf{x}), \hat{\psi}(\mathbf{x}')]_{\mp} = \delta^3(\mathbf{r} - \mathbf{r}')\delta_{\tau\tau'},$$

we get a “4D” δ -result. Then substitute $\hat{d}_{\mathbf{p}\tau}$ and $\hat{d}_{\mathbf{p}\tau}^\dagger$ into the commutator

$$\begin{aligned} [\hat{d}_{\mathbf{p}\tau}, \hat{d}_{\mathbf{p}',\tau'}^\dagger]_{\mp} &= \int d^3\mathbf{r} \int d^3\mathbf{r}' e^{-i(\mathbf{p}\cdot\mathbf{r} - \mathbf{p}'\cdot\mathbf{r}')} [\hat{\psi}(\mathbf{r}, \tau), \hat{\psi}(\mathbf{r}', \tau')^\dagger]_{\mp} \\ &= \int d^3\mathbf{r} \int d^3\mathbf{r}' e^{-i(\mathbf{p}\cdot\mathbf{r} - \mathbf{p}'\cdot\mathbf{r}')} \delta^3(\mathbf{r} - \mathbf{r}')\delta_{\tau\tau'} \\ &= \int d^3\mathbf{r} e^{-i(\mathbf{p}\cdot\mathbf{r} - \mathbf{p}'\cdot\mathbf{r})} \delta_{\tau\tau'} = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}')\delta_{\tau\tau'}. \end{aligned}$$

The last line of the above derivation used the normalization of the plane waves, with a factor of $(2\pi)^3$.

Proof. Similar to Problem 1.1, we need to left multiply both sides by $\varphi_{\mathbf{p}\tau}(\mathbf{x})$, sum over τ , and integrate over $\mathbf{p}$ to $\hat{d}_{\mathbf{p}\tau}(\mathbf{x}')$ (In order to arrive at $\hat{\psi}(\mathbf{x})$, a normalization factor $1/(2\pi)^3$ is applied)

$$\sum_{\tau} \int \frac{d^3\mathbf{p}}{(2\pi)^3} \varphi_{\mathbf{p}\tau}(\mathbf{x}) \hat{d}_{\mathbf{p}\sigma} = \sum_{\tau} \int \frac{d^3\mathbf{p}}{(2\pi)^3} \varphi_{\mathbf{p}\tau}(\mathbf{x}) \int d^3\mathbf{x}' \varphi_{\mathbf{p}\tau}^*(\mathbf{x}') \hat{\psi}(\mathbf{x}').$$

We arrive at the identity of $\varphi_{\mathbf{p}\tau}(\mathbf{x})$. Now, calculate it.

$$\begin{aligned} \text{Identity} &= \sum_{\tau} \int \frac{d^3\mathbf{p}}{(2\pi)^3} \varphi_{\mathbf{p}\tau}(\mathbf{x}) \varphi_{\mathbf{p}\tau}^*(\mathbf{x}') = \sum_{\tau} \int \frac{d^3\mathbf{p}}{(2\pi)^3} (e^{i\mathbf{p}\cdot\mathbf{r}} \delta_{\sigma\tau}) (e^{-i\mathbf{p}\cdot\mathbf{r}'} \delta_{\sigma'\tau}) \\ &= \sum_{\tau} \int \frac{d^3\mathbf{p}}{(2\pi)^3} e^{i\mathbf{p}\cdot(\mathbf{r}-\mathbf{r}')} \delta_{\sigma\tau} \delta_{\sigma'\tau} = \int \frac{d^3\mathbf{p}}{(2\pi)^3} e^{i\mathbf{p}\cdot(\mathbf{r}-\mathbf{r}')} \delta_{\sigma\sigma'} = \delta^3(\mathbf{r} - \mathbf{r}')\delta_{\sigma\sigma'}, \end{aligned}$$

When handling $\delta_{\sigma\tau} \delta_{\sigma'\tau}$, we can consider choosing all of the τ s (points) that equal to σ and σ' at the same time, it is equal to find filtering all the points on the axis that satisfy $\sigma = \sigma'$, that is $\sum_{\tau} \delta_{\sigma\tau} \delta_{\sigma'\tau} = \delta_{\sigma\sigma'}$.

Now, return to the main topic. Substituting the identity and expand $\int d^3\mathbf{x}$

$$\sum_{\tau} \int \frac{d^3\mathbf{p}}{(2\pi)^3} \varphi_{\mathbf{p}\tau}(\mathbf{x}) \hat{d}_{\mathbf{p}\sigma} = \sum_{\sigma'} \int d^3\mathbf{r}' \hat{\psi}(\mathbf{r}', \sigma') \delta^3(\mathbf{r}' - \mathbf{r}) \delta_{\sigma\sigma'} = \hat{\psi}(\mathbf{r}, \sigma) = \hat{\psi}(\mathbf{x}),$$

that is

$$\hat{\psi}(\mathbf{x}) = \sum_{\tau} \int \frac{d^3\mathbf{p}}{(2\pi)^3} (e^{i\mathbf{p}\cdot\mathbf{r}} \delta_{\sigma\tau}) \hat{d}_{\mathbf{p}\sigma} = \int \frac{d^3\mathbf{p}}{(2\pi)^3} e^{i\mathbf{p}\cdot\mathbf{r}} \hat{d}_{\mathbf{p}\sigma}.$$

Similarly to $\hat{\psi}^\dagger(\mathbf{x})$. To summarize

$$\hat{\psi}(\mathbf{x}) = \int \frac{d^3\mathbf{p}}{(2\pi)^3} e^{i\mathbf{p}\cdot\mathbf{r}} \hat{d}_{\mathbf{p}\sigma}, \quad \text{and} \quad \hat{\psi}^\dagger(\mathbf{x}) = \int \frac{d^3\mathbf{p}}{(2\pi)^3} e^{-i\mathbf{p}\cdot\mathbf{r}} \hat{d}_{\mathbf{p}\sigma}^\dagger.$$

CHAPTER 4 Model Hamiltonians

4.1 Hierarchy of Hamiltonians

For solids like

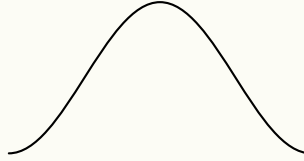
... ○ ○ ○ ○ ○ ○ ○ ...

$\sum_i N(R_i) \rightarrow$ periodic potential ($\leftarrow$ single electron $\leftrightarrow$ *quantum geometry* $\rightarrow$ phonons).
 $h_0 \xrightarrow{a \rightarrow 0} \partial_x \rightarrow k(-\infty, +\infty) \in (-\frac{\pi}{a}, \frac{\pi}{a}) \rightarrow \text{Moiré } Nlm$

4.2 Pariser-Parr-Pople Model (tight-binding + interaction)

4.2.1 Hydrogen molecule

For two hydrogen atoms



$$h_0 = \left(-\frac{\nabla^2}{2m}\right), \nu(x_1, x_2)$$

$$h_0 = \sum_{1,2,\dots,n} \hat{n}_i(\epsilon_n^0) \epsilon_N^0 \rightarrow \epsilon(\hat{n}_1 + \hat{n}_2) + \sum_{\sigma,\sigma',nm} T_{1n}(R)(d_{1n\sigma}^\dagger d_{2n\sigma} + d_{2M'_\sigma} d_{M_\sigma})$$

$$|\varphi_1^n\rangle, |\varphi_2^n\rangle, \langle\varphi_1^n|n_0|\varphi_2^m\rangle.$$

$$h_{\text{int}} = v(\hat{n}_{1\uparrow}n_{1\downarrow} + \hat{n}_{2\uparrow}n_{2\downarrow}) + \nu_{12}(\hat{n}_1 - 1)(\hat{n}_2 - 1).$$

4.3 Ammonia NH₃

4.4 Bloch electrons in 2nd Quantization

Single particle basis: Bloch state $|\mathbf{k}\sigma\rangle$. Wave function $\langle\mathbf{r}|\mathbf{k}\sigma\rangle = \psi_{\mathbf{k}\sigma}(\mathbf{r})$, $\psi_{\mathbf{k}} = u_{\mathbf{k}}(r) e^{i\mathbf{k}\cdot\mathbf{r}}$

For single particle Hamiltonian

$$\mathcal{H}_0 = \sum_{\substack{k,\sigma \\ k',\sigma'}} \langle k\sigma | h_0 | k'\sigma' \rangle d_k^\dagger d_{k'\sigma'} \Rightarrow \sum_{k\sigma} \epsilon(k) d_{k\sigma}^\dagger d_{k\sigma} \quad (4.1)$$

For tight-binding $\sum_{ij} t_{ij} |i\rangle \langle j|$

$$H_0 = \sum_{\substack{\sigma\sigma' \\ ij}} T_{ij\sigma\sigma'} \hat{d}_{i\sigma}^\dagger \hat{d}_{j\sigma'} \quad (4.2)$$

$$\langle r|\mathbf{k}\sigma\rangle \Leftrightarrow \langle r|i\sigma\rangle \quad (4.3)$$

which is called (*Maximally-localized*) *Wannier function*: exponentially atomic.

4.4.1 Generic lattice

The hamiltonian

$$\mathcal{H}_0 = \sum_{ij} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + \sum_i \epsilon_0 c_i^\dagger c_i, \quad H = \sum_k c_{k\alpha}^\dagger H_{\alpha\beta} \hat{C}_{k\beta}, \quad \hat{C}_k = \frac{1}{\sqrt{N}} \sum_{i,\alpha} e^{-i\mathbf{k}(\mathbf{R}+\mathbf{b}_\alpha)} \hat{c}_{i,\mathbf{b}_\alpha} \quad (4.4)$$

Here,

$$\mathbf{R}_{i,A} = \sum_{n_1, n_2} n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + \mathbf{b}_A, \quad \mathbf{R}_{i,B} = \sum_{n_1, n_2} n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + \mathbf{b}_B, \quad \mathbf{b}_A = (a, 0), \quad \mathbf{b}_B = (0, 1)$$

α, β : sublattice, spins; orbitals index.

4.4.2 Graphene

$$\mathbf{a}_1 = \frac{a}{2}(3, \sqrt{3}), \quad \mathbf{a}_2 = \frac{a}{2}(3, -\sqrt{3}), \quad \mathbf{b}_B = a(1, 0), \quad \delta_1 = \mathbf{a}_1 + \mathbf{b}_B, \quad \delta_2 = \frac{a}{2}(1 - \sqrt{3}), \quad \delta_3 = a(1, 0). \quad (4.5)$$

$$\mathcal{H} = -t \sum (a_i^\dagger b_j + \text{h.c.}) \quad (4.6)$$

while $a_i = C_{R_{i,A}}, b = C_{R_{i,B}}$, and $h_k \approx \sum_\delta e^{ik\delta} + e^{-ik\delta}$ Then

$$H = \begin{pmatrix} a_k^\dagger & b_k^\dagger \end{pmatrix} \begin{pmatrix} 0 & h_k \\ h_k & 0 \end{pmatrix} \begin{pmatrix} a_k \\ b_k \end{pmatrix}$$

pumping: 

$$h_k = t \left[e^{i(1,0) \cdot \mathbf{k}} + e^{-i(1,0) \cdot \mathbf{k}} + e^{i(0,1) \cdot \mathbf{k}} + \dots \right] = -2t(\cos k_x + \cos k_y) \quad (4.7)$$

and

$$H_k = - \sum 2t(\cos k_x + \cos k_y) c_k^\dagger c_k \quad (4.8)$$

h_k near some point (Dirac point): $h_k \sim (\mathbf{k} - \mathbf{K}) \frac{\partial h_k}{\partial \mathbf{k}}$, so it can be written as $h_k \sim \mathbf{v}_F \cdot \delta \mathbf{k} \leftrightarrow \partial_x$

4.4.3 Flat-bannd

$h_k = \text{const}$ now.

Landau levels

Landau gauge:

Symmertic Gauge $|\phi_{LG}\rangle \Rightarrow \sum |\Phi_{SG}\rangle$

magnetic translation T_x, T_y : $a_x, a_y \propto \hat{l}_b$

4.4.4 Electron phonon coupling: BCS

Hamiltonian with solid:

$$\mathcal{H}_{\text{solid}} \begin{cases} \mathcal{H}_{\text{phono}} = \sum_{\mathbf{q}} \mathcal{E}(\mathbf{q}) b_{\mathbf{q}}^{\dagger} b_{\mathbf{q}} \\ \mathcal{H}_{\text{Bloch}} = \sum_{\mathbf{k}} \mathcal{E}(\mathbf{k}) c_{\mathbf{k}}^{\dagger} c_{\mathbf{k}} \end{cases} \quad (4.9)$$

$$V(R_i, x_i)$$

$$H_{e-p} = \sum_{k_1} \sum_{q \mathbf{G}} T(b_q^{\dagger} + b_{-q}^{\dagger}) \hat{a}_{k+q+\mathbf{G}}^{\dagger} \hat{a}_k \quad (4.10)$$

$$\bar{x}_j^* = \frac{1}{n} \sum_{i=1}^n x_{ij} - \bar{x}_j = \frac{1}{n} \left(\sum_{i=1}^n x_{ij} - \sum_{i=1}^n \bar{x}_j \right)$$

4.5 QM identical partical

1. $|1 : \phi_1\rangle, |2 : \chi_2\rangle$ -i occupation num. of condition

- $|n_{\phi_1}, n_{\chi_2}\rangle$ in real space: $\phi \rightarrow \hat{x}$;

After interchange particle label, there should be a sign: $(\pm 1)^p$: minus - fermion; boson: trivial. so we have

$$(-1)^p \leftrightarrow [d_n^{\dagger}, d_m]_{\mp} : d_n^{\dagger} d_m \mp d_m d_n^{\dagger}$$

$$[d_n, d_m]_{\mp} = 0 = [d_n^{\dagger}, d_m^{\dagger}] \cdot d_n^{\dagger} d_m^{\dagger} |0\rangle = -d_m^{\dagger} d_n^{\dagger} |0\rangle$$

4.6 Electron-phonon coupling

$$V_{\text{ion-ion}} = \sum V(R_i - R_j) \Rightarrow \sum_{\mathbf{q}} \omega_{\mathbf{q}} a_{\mathbf{q}}^{\dagger} a_{\mathbf{q}},$$

$$V_{\text{el-el}} = \sum V(r_i - r_j) \Rightarrow \sum_i u n_{i\sigma} n_{iq} = + \dots \sum_{ij} V n_i n_j,$$

$$V_{\text{el-ion}} = \sum V(R_i - r_j) \Rightarrow \sum_{\sigma q} (b_q^{-1} + b_{q-}) c_{k+q\sigma}^{\dagger} c_{k\sigma}$$

corpspication: feed back effect (self-consistent eq.) == DFT

$$H_{ep} = \sum_{kq\sigma} T_{kq} (b_q + \hat{b}_{-q}) G_{k+q,\sigma}^{\alpha} a_{k\sigma}$$

MW_{ene} -field BCS

4.7 Effective $H_{a,g(\text{femons})}$

4.8 canonical transformaission

Also called: Foldy-Wyen

4.9 effecton H canonical transf

$$H = H_0 + \delta H$$

Broad prodive $\hat{T} = e^{-\hat{S}} \mathcal{H} e^{\hat{S}}$. Expand:

$$\tilde{H} = H + [H, \hat{S}]_- + \frac{1}{2!} [[\hat{H}, \hat{S}]_- S]_- = H_0 + \delta H + [H_0, \hat{S}] + [\delta H, \hat{S}]$$

find $\hat{S} : [H_0, \hat{S}] + \Delta H = 0$. S is usually called generator.

$$\Rightarrow H_0 + [\Delta H, S] + \frac{1}{2!} ([H_0, S] + [\delta H, S]), S = H_0 + [\delta H, S] + \frac{1}{2} [-\delta H, S] + O(S^2) = H_0 + \frac{1}{2} [\delta H, S]$$

So, from δH u get. $H = H_0 + \frac{1}{2} [\delta H, S]$.

4.10 For el-ph coupling $\hat{S} = \sum_{kq\sigma} \tau_q (x b_q + y b_{-q}^\dagger) a_{k+q\sigma}^* a_k$

$$H_0 = \sum \epsilon_k a_{k\sigma}^\dagger a_{k\sigma}$$

The commutator to S is

$$\begin{aligned} [H_0, \hat{S}] &= \sum \tau_q \{ x(\epsilon(k+1) - \epsilon(k) - \hbar\omega_q) b_q + y(\epsilon(k+q) - \epsilon(k) + \hbar\omega_q) b_{-q}^\dagger \} a_{k+q\sigma}^\dagger a_{k\sigma} \\ &= -H_{\text{el-ph}} \Rightarrow \begin{cases} x = (\epsilon(k) - \epsilon(k+1) + \hbar(\omega)q^{-1}) \\ y = (\epsilon(k) - \epsilon(k+1) - \hbar(\omega)q^{-1}) \end{cases} \end{aligned}$$

then, the effective Hamiltonian

$$H_{\text{eff}} = \frac{1}{2} [\delta H, \hat{S}] = \frac{1}{2} \sum_{\substack{kk' \\ q\sigma\sigma'}} \tau_{-q} \tau_q (q - x) a_{k+q}^\dagger a_{k'-q} a_{k'\sigma'} a_{k\sigma}$$

$$x = \frac{1}{\epsilon(k) - \epsilon(k+q) - \hbar\omega_q} \approx \delta(\epsilon(k) - \epsilon(k+1) - \hbar\omega_q).$$

Consider this δ -function

$$\begin{aligned} \int dE &= \frac{1}{E \pm \hbar\omega_q} \delta(E - \hbar\omega_q) \\ H_0 &= \sum \epsilon_n a_n^\dagger a_n, \hat{G} = (E - H_0)^{-1} = G[E] \end{aligned}$$

Green's function:

$$\begin{aligned} (E - H_{ph})^{-1} &= G_{Ph} \\ \tilde{H} &= H_0 + H_{eff} = \sum_k \epsilon_k \tilde{a}_{k\sigma}^\dagger \tilde{a}_{k\sigma} + \frac{1}{2} \sum_{kk', q\sigma\sigma'} T_q T_{-q} (y - x) \tilde{a}_{k+q}^\dagger \tilde{a}_{k'-q}^\dagger \tilde{a}_{k'\sigma'} \tilde{a}_{k\sigma} \\ e^{-\hat{S}} H_0 e^{\hat{S}} &= e^{-S} a_k^\dagger (e^S e^{-S}) a_k e^S \end{aligned}$$

$$\tilde{a}_k^\dagger = e^{-S} a_k^\dagger e^S = a_k^\dagger + [a_k^\dagger, S] + \dots = a_k^\dagger + \sum_q (x b_q + y b_{-q})^\dagger, \tilde{a}_k = e^S a_k e^{-S}$$

4.11 From el-ph to BCS

at low-T, $k \in \text{near } ES$. $|T_q|^2(y - x) \sim < 0$, means for particular part of fermion, the attraction become attractive.

$$H = \sum_k \epsilon_k c_{k\sigma}^\dagger c_{k\sigma} - v \sum_{kk'} c_{k\uparrow}^\dagger c_{k\downarrow}^\dagger c_{k'\downarrow} c_{k'\uparrow}.$$

This is called Richardson = RCS model In none meanfield 1. not a sc. qs = $\hat{c}$ CFT 2. prove bcs, superconductivity spontaneous sym. breaking: add some like Bogoliubov.

Meanfield BCS model.

$$\text{SBB: } H_{GB} = \sum \epsilon_k c_{k\sigma}^\dagger c_{k\sigma}$$

4.12 Holstein model

1. $H_c = \sum_s \omega_0 (a_s^\dagger a_s + \frac{1}{2})$
2. $H_{\text{int}} = -q \sum_{s\sigma} d_{s\sigma}^\dagger d_{s\sigma} (a_s + a_s^*)$
3. $H_{el} = \sum_i (t_{ij} d_{i\sigma}^\dagger d_{j\sigma} + h.c.)$
4. Strong coupling limit. Leads to heavy polaron.
5. Lang-Firsov transformation, (c.t.)

$$\hat{S} = -i \frac{g}{\omega_0} \sum \hat{n}_{s\sigma} (a_s^\dagger - a_s)$$

$$H_{cp} = T \sum (B_s^\dagger B_s)$$

$$\hat{B}_s = e^{q/\omega_0}$$

4.13 elec-el interaction

$$V_{el-el} = \sum V(r_i - r_j)$$

.

$$\Rightarrow H_A = \sum \epsilon_k c_{k\sigma}^{-1} c_{k\sigma} + u \hat{n}_{d\uparrow} n_{d\downarrow} + \sum \epsilon_d d^\dagger d + V \sum_k (c_{k\sigma}^\dagger d_\sigma + d_\sigma^\dagger c_{k\sigma})$$

$$S = \sum_{k\sigma} (A_k c_{k\sigma}^\dagger d_\sigma - A_k d_\sigma^\dagger c_{k\sigma}) + \sum (B_A d_\sigma c_k^\dagger - B_k^\dagger c_{k\sigma} d_\sigma^\dagger)$$

$$A_K = \frac{V_k}{\epsilon_k - \epsilon_d}, \quad B_k = \frac{V_k}{\epsilon_k - (\epsilon_d + b)}$$

then

$$H_{\text{eff}} = \frac{1}{2} [\delta H, S] = \frac{1}{2} [\delta H, S(H_0)]$$

while $H_0 = H_c + H_{n,d}$.

$$H_{\text{eff}} = J_k \mathbf{S}_d \cdot \mathbf{S}_{c,o} = \sum_{kk'} J_{kk'} (\mathbf{S}_d \cdot \sum_{\alpha\beta} c_{k'\alpha}^\dagger \frac{\boldsymbol{\sigma}_{\alpha\beta}}{2} c_{k\beta} + \frac{1}{4} \sum_{\sigma} c_{k'\sigma}^\dagger c_{k\sigma})$$

$$\mathbf{S}_d = \sum_{\alpha\beta} \hat{d}_\alpha \frac{\boldsymbol{\sigma}_{\alpha\beta}}{2} \hat{d}_\beta$$

So,

$$S_z = \frac{1}{2}(d_\uparrow^\dagger d_\uparrow - d_\downarrow^\dagger d_\downarrow), S_z^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = d_\uparrow^\dagger d_\downarrow, S^- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = d_\downarrow^\dagger d_\uparrow$$

$$\text{Hkondo-impurity} = J_k \mathbf{S}_d \mathbf{S}_{c,o}$$

for $J_K > 0$, $J_K = \frac{4V^2}{U}$.

After the treatment, from anderson module to kondo modo:

$$H_{\text{And}} \rightarrow H_{\text{Kondom}} = \sum \epsilon_k c_k^\dagger c_k + J_k \mathbf{S}_\alpha \cdot \mathbf{S}_{c,N}$$

S_d is free moment, formation of a local moment.

J_k can become negative, so change the kondo limit, Anti FM – compling, to becoma a FM.

4.14 Hubbard model

$$H = \sum_{\sigma k} \epsilon_k c_{k\sigma}^\dagger c_{k\sigma} + N \sum_i (n_\uparrow - \frac{1}{2})(n_\downarrow - \frac{1}{2}) + (\sum_{\sigma k} -\mu c_{k\sigma}^\dagger c_{k\sigma}) = - \sum_{ij\sigma} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} \quad (4.11)$$

4.14.1 Schrieff-Wolff transformation (C.T.)

For $n = 1$, deal with Heisenberg model

$$H_{\text{Heis}} = \sum_{ij} J_{ij} \mathbf{S}_i \mathbf{S}_j \quad (4.12)$$

while $J_{ij} = \frac{4t_{ij}t_{ji}}{N} = \frac{4t^2}{N}$.

When doping, $t - J$ model

$$H = \sum_{ij} t_{ij} \tilde{c}_{ik} \tilde{c}_{j\sigma} + \sum J \mathbf{S}_i \cdot \mathbf{S}_j.$$

so-called Ggutzwillen projection, project out double-occupancy.

Gutzwiller approximation $\delta \sum t \hat{c}_i^\dagger c_{j\sigma} + \sum J(1 - 2\delta) \mathbf{S}_i \cdot \mathbf{S}_j$.

4.15 Many layers of QMB problems

1. reduce degrees of freedom (wrong way: CT, quasi particles; polaron)
2. methods to reduce DoF: link/crosscheck different methods path tn: Effective theory EFT.

Homework #2 [2025-09-09]

Problem 2.1. Prove

$$\hat{H}_{\text{int}} = \frac{1}{2} \int dx \int dx' v(x, x') \hat{n}(x) \hat{n}(x') - \frac{1}{2} \int dx v(x, x) \hat{n}(x),$$

from $\hat{H}_{\text{int}}|x_1 \dots x_N\rangle = \left(\frac{1}{2} \sum_{i \neq j} v(x_i, x_j)\right)|x_1 \dots x_N\rangle$ as hinted in class, beginning with inserting $\int dx \sum_i \delta(x - x_i)$ to the original, first-quantization expression and make use of Eq. (3.8.5).

$$\begin{aligned} \hat{n}(x) \underbrace{\hat{\psi}^\dagger(x_N) \hat{\psi}^\dagger(x_{N-1}) \dots \hat{\psi}^\dagger(x_1)}_{|x_1 \dots x_N\rangle} |0\rangle &= [\hat{n}(x), \hat{\psi}^\dagger(x_N)]_- \hat{\psi}^\dagger(x_{N-1}) \dots \hat{\psi}^\dagger(x_1) |0\rangle \\ &\quad + \hat{\psi}^\dagger(x_N) [\hat{n}(x), \hat{\psi}^\dagger(x_{N-1})]_- \dots \hat{\psi}^\dagger(x_1) |0\rangle \\ &\quad \vdots \\ &\quad + \hat{\psi}^\dagger(x_N) \hat{\psi}^\dagger(x_{N-1}) \dots [\hat{n}(x), \hat{\psi}^\dagger(x_1)]_- |0\rangle \\ &= \underbrace{\left(\sum_{i=1}^N \delta(x - x_i) \right)}_{\substack{\text{density of } N \text{ particles} \\ \text{in } x_1, \dots, x_N}} |x_1 \dots x_N\rangle. \end{aligned} \tag{3.8.5}$$

Solution. *Proof.* Rewrite the first-quantized interaction $\frac{1}{2} \sum_{i \neq j} v(x_i, x_j)$ first: separate it into two terms

$$\sum_{i \neq j} v(x_i, x_j) = \sum_{i, j} v(x_i, x_j) - \sum_i v(x_i, x_i),$$

then inserting $\int dx \sum_i \delta(x - x_i)$ to the two terms above

$$\begin{aligned} \sum_{i, j} v(x_i, x_j) &= \int dx \int dx' v(x, x') \sum_i \delta(x - x_i) \sum_j \delta(x' - x_j), \\ \sum_i v(x_i, x_i) &= \int dx v(x, x) \sum_i \delta(x - x_i). \end{aligned}$$

To simplify, define $|\mathcal{X}\rangle = |x_1 \dots x_N\rangle = \hat{\psi}^\dagger(x_N) \dots \hat{\psi}^\dagger(x_1) |0\rangle$.

1. To generate the first term of $\hat{H}_{\text{int}}$, acting $\hat{n}(x)$ and $\hat{n}(x')$ on $|\mathcal{X}\rangle$, and use the eigenvalue of (3.8.5)

$$\hat{n}(x') \hat{n}(x) |\mathcal{X}\rangle = \sum_i \delta(x - x_i) \sum_j \delta(x' - x_j) |\mathcal{X}\rangle,$$

then multiply $v(x, x')$ and integrate over x and x' and using the sifting property of δ -function

$$\begin{aligned} \int dx \int dx' v(x, x') \hat{n}(x) \hat{n}(x') |\mathcal{X}\rangle &= \int dx \int dx' v(x, x') \sum_i \delta(x - x_i) \sum_j \delta(x' - x_j) |\mathcal{X}\rangle \\ &= \sum_{i, j} v(x_i, x_j) |\mathcal{X}\rangle, \end{aligned}$$

match the first term of $\hat{H}_{\text{int}}$.

2. Concerning the second term of $\hat{H}_{\text{int}}$, acting $\hat{n}(x)$ on $|\mathcal{X}\rangle$, then multiply $v(x, x)$ and integrate over x

$$\int dx v(x, x) \hat{n}(x) |\mathcal{X}\rangle = \int dx v(x, x) \sum_i \delta(x - x_i) |\mathcal{X}\rangle = \sum_i v(x_i, x_i) |\mathcal{X}\rangle.$$

In summary,

$$\begin{aligned} \left(\frac{1}{2} \int dx dx' v(x, x') \hat{n}(x) \hat{n}(x') - \frac{1}{2} \int dx v(x, x) \hat{n}(x) \right) |\mathcal{X}\rangle &= \frac{1}{2} \left(\sum_{i,j} v(x_i, x_j) - \sum_i v(x_i, x_i) \right) |\mathcal{X}\rangle \\ &= \frac{1}{2} \sum_{i \neq j} v(x_i, x_j) |\mathcal{X}\rangle. \end{aligned}$$

Then we arrive at

$$\hat{H}_{\text{int}} = \frac{1}{2} \int dx dx' v(x, x') \hat{n}(x) \hat{n}(x') - \frac{1}{2} \int dx v(x, x) \hat{n}(x).$$

Problem 2.2 (Exercise 1.4 of [5]). Let $\hat{n}_n \equiv \hat{d}_n^\dagger \hat{d}_n$ be the occupation operator for particles with quantum number n , see (1.74). Prove that in the fermionic case

$$\hat{n}_n^2 = \hat{n}_n.$$

and hence that the eigenvalues of $\hat{n}_n$ are either 0 or 1 — that is, it is not possible to create two fermions in the same state $|n\rangle$. This is a direct consequence of the Pauli exclusion principle.

Solution. *Proof.* As is known to all, for fermions, the creation and annihilation operators satisfy the following relations

$$\begin{cases} \{\hat{d}_n, \hat{d}_n^\dagger\} = \hat{d}_n \hat{d}_n^\dagger + \hat{d}_n^\dagger \hat{d}_n = 1, \\ \{\hat{d}_n, \hat{d}_n\} = 0 \Rightarrow \hat{d}_n \hat{d}_n = 0, \\ \{\hat{d}_n^\dagger, \hat{d}_n^\dagger\} = 0 \Rightarrow \hat{d}_n^\dagger \hat{d}_n^\dagger = 0. \end{cases}$$

Now calculating $\hat{n}_n^2$

$$\hat{n}_n^2 = \hat{d}_n^\dagger \hat{d}_n \hat{d}_n^\dagger \hat{d}_n = \hat{d}_n^\dagger (1 - \hat{d}_n^\dagger \hat{d}_n) \hat{d}_n = \hat{d}_n^\dagger \hat{d}_n - \hat{d}_n^\dagger \hat{d}_n^\dagger \hat{d}_n \hat{d}_n = \hat{d}_n^\dagger \hat{d}_n = \hat{n}_n.$$

Proof. The eigenfunctions for $\hat{n}_n$ and $\hat{n}_n^2$ are

$$\hat{n}_n |\psi\rangle = \lambda |\psi\rangle, \quad \text{and} \quad \hat{n}_n^2 |\psi\rangle = \lambda^2 |\psi\rangle.$$

Since $\hat{n}_n^2 = \hat{n}_n$, we have

$$\hat{n}_n |\psi\rangle = \lambda^2 |\psi\rangle.$$

Thus, $\lambda = \lambda^2$ and $\lambda = 0$ or 1 , means it is impossible to have more than one fermion in the same quantum state $|n\rangle$, which is the Pauli exclusion principle.

Problem 2.3 (Exercise 1.5 of [5]). Prove that the total number of particle operators $\hat{N} = \int dx \hat{\psi}^\dagger(x) \hat{\psi}(x)$ can also be written as $\hat{N} = \sum_n \hat{d}_n^\dagger \hat{d}_n$ for any orthonormal basis $|n\rangle$. Calculate the action of $\hat{N}$ on a generic ket $|\Psi_N\rangle$ with N particles ($|\Psi_N\rangle \in H_N$) and prove that

$$\hat{N} |\Psi_N\rangle = N |\Psi_N\rangle.$$

Solution. *Proof.* Recalling the definitions of the field operators

$$\begin{aligned}\hat{\psi}(x) &= \sum_n \langle x|n\rangle \hat{d}_n = \sum_n \phi_n(x) \hat{d}_n, \\ \hat{\psi}(x)^\dagger &= \sum_n \langle n|x\rangle \hat{d}_n^\dagger = \sum_n \phi_n^*(x) \hat{d}_n^\dagger.\end{aligned}$$

then substitute them into $\hat{N}$ and use the orthonormality of $\{|n\rangle\}$

$$\begin{aligned}\hat{N} &= \int dx \hat{\psi}^\dagger(x) \hat{\psi}(x) = \int dx \sum_m \phi_m^*(x) \hat{d}_m^\dagger \sum_n \phi_n(x) \hat{d}_n \\ &= \sum_{m,n} \hat{d}_m^\dagger \hat{d}_n \int dx \phi_m^*(x) \phi_n(x) = \sum_{m,n} \hat{d}_m^\dagger \hat{d}_n \delta_{m,n} = \sum_n \hat{d}_n^\dagger \hat{d}_n,\end{aligned}$$

means that $\hat{N}$ can be written as $\hat{N} = \sum_n \hat{d}_n^\dagger \hat{d}_n$ for any orthonormal basis $|n\rangle$.

Proof. When $\hat{N}$ act on $|\Psi_N\rangle$

$$\hat{N}|\Psi_N\rangle = \int dx \hat{\psi}^\dagger(x) \hat{\psi}(x) |x_1, x_2, \dots, x_N\rangle = \int dx \sum_{i=1}^N \delta(x - x_i) |x_1, x_2, \dots, x_N\rangle = N |x_1, x_2, \dots, x_N\rangle,$$

So, we proved that $\hat{N}|\Psi_N\rangle = N|\Psi_N\rangle$.

The total Hamiltonian of a system of interacting identical particles is the sum of $\hat{H}_0$ and $\hat{H}_{\text{int}}$. Expressing $\hat{H}$ in terms of the $\hat{d}$ -operators as $\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}} = \sum_{ij} h_{ij} \hat{d}_i^\dagger \hat{d}_j + \frac{1}{2} \sum_{ij,n} v_{ijmn} \hat{d}_i^\dagger \hat{d}_j^\dagger \hat{d}_m \hat{d}_n$.

Problem 2.4 (Exercise 1.6 of [5]). Prove that $\hat{N}$ commutes with $\hat{H}_0$ and $\hat{H}_{\text{int}}$ – that is,

$$[\hat{N}, \hat{H}_0]_- = [\hat{N}, \hat{H}_{\text{int}}]_- = 0.$$

This means that the eigenkets of $\hat{H}$ can be chosen as kets with a fixed number of particles.

Solution. *Proof.* Since the annihilation operator $\hat{d}_m$ destroys a particle, so it decreases the number by 1: $[\hat{N}, \hat{d}_m] = -\hat{d}_m$; in contrast, the creation operator $\hat{d}_m^\dagger$ creates a particle, so $[\hat{N}, \hat{d}_m^\dagger] = \hat{d}_m^\dagger$. Then

$$\begin{aligned}[\hat{N}, \hat{H}_0] &= \hat{N} \hat{H}_0 - \hat{H}_0 \hat{N} = \sum_{ij} h_{ij} [\hat{N}, \hat{d}_i^\dagger \hat{d}_j] = \sum_{ij} h_{ij} \hat{d}_i^\dagger [\hat{N}, \hat{d}_j] + \sum_{ij} h_{ij} [\hat{N}, \hat{d}_i^\dagger] \hat{d}_j \\ &= \hat{d}_i^\dagger (-\hat{d}_j) + (\hat{d}_i^\dagger) \hat{d}_j = 0,\end{aligned}$$

similarly, since

$$[\hat{N}, \hat{d}_i^\dagger \hat{d}_j^\dagger \hat{d}_m \hat{d}_n] = (\hat{d}_i^\dagger) \hat{d}_j^\dagger \hat{d}_m \hat{d}_n + \hat{d}_i^\dagger (\hat{d}_j^\dagger) \hat{d}_m \hat{d}_n + \hat{d}_i^\dagger \hat{d}_j^\dagger (-\hat{d}_m) \hat{d}_n + \hat{d}_i^\dagger \hat{d}_j^\dagger \hat{d}_m (-\hat{d}_n) = 0,$$

we arrive at $[\hat{N}, \hat{H}_{\text{int}}] = 0$.

Problem 2.5 (Exercise 1.7 of [5]). Let $n = s\sigma$ and $\sigma = \uparrow, \downarrow$ be the spin projection for fermions of spin $\frac{1}{2}$. We consider the operators

$$\hat{S}_s^z \equiv \frac{1}{2}(\hat{n}_{s\uparrow} - \hat{n}_{s\downarrow}), \quad \hat{S}_s^+ \equiv \hat{d}_{s\uparrow}^\dagger \hat{d}_{s\downarrow}, \quad \hat{S}_s^- \equiv \hat{d}_{s\downarrow}^\dagger \hat{d}_{s\uparrow} = (\hat{S}_s^+)^\dagger.$$

Using the anticommutation relations,

$$\{\hat{d}_{s\sigma}, \hat{d}_{s'\sigma'}^\dagger\} = \delta_{ss'}\delta_{\sigma\sigma'}, \quad \{\hat{d}_{s\sigma}, \hat{d}_{s'\sigma'}\} = 0, \quad \{\hat{d}_{s\sigma}^\dagger, \hat{d}_{s'\sigma'}^\dagger\} = 0.$$

prove that the action of the above operators on the kets $|s\sigma\rangle \equiv \hat{d}_{s\sigma}^\dagger|0\rangle$ is

$$\hat{S}_s^z|s\uparrow\rangle = \frac{1}{2}|s\uparrow\rangle, \quad \hat{S}_s^+|s\uparrow\rangle = |\emptyset\rangle, \quad \hat{S}_s^-|s\uparrow\rangle = |s\downarrow\rangle,$$

and

$$\hat{S}_s^z|s\downarrow\rangle = -\frac{1}{2}|s\downarrow\rangle, \quad \hat{S}_s^+|s\downarrow\rangle = |s\uparrow\rangle, \quad \hat{S}_s^-|s\downarrow\rangle = |\emptyset\rangle,$$

To what operator do $\hat{S}_s^z, \hat{S}_s^+, \hat{S}_s^-$ correspond?

Solution. Using the occupation operator $\hat{n}_{s\sigma} = \hat{d}_{s\sigma}^\dagger \hat{d}_{s\sigma}$ and the relation $\hat{d}_{s\sigma}|0\rangle = 0, \hat{d}_{s\sigma}^\dagger|0\rangle = |s\sigma\rangle$.

1. $\hat{S}_s^z|s\uparrow\rangle = \frac{1}{2}(\hat{d}_{s\uparrow}^\dagger \hat{d}_{s\uparrow} \hat{d}_{s\uparrow}^\dagger - \hat{d}_{s\downarrow}^\dagger \hat{d}_{s\downarrow} \hat{d}_{s\uparrow}^\dagger)|0\rangle = \frac{1}{2}[\hat{d}_{s\uparrow}^\dagger(1 - \hat{d}_{s\uparrow}^\dagger \hat{d}_{s\uparrow}) - (-\hat{d}_{s\downarrow}^\dagger \hat{d}_{s\uparrow}^\dagger \hat{d}_{s\downarrow})]|0\rangle = \frac{1}{2}|s\uparrow\rangle. \quad \square$
2. $\hat{S}_s^+|0\rangle = \hat{d}_{s\uparrow}^\dagger \hat{d}_{s\downarrow} \hat{d}_{s\uparrow}^\dagger|0\rangle = -\hat{d}_{s\uparrow}^\dagger \hat{d}_{s\uparrow}^\dagger \hat{d}_{s\downarrow}|0\rangle = 0$, so $\hat{S}_s^+|s\uparrow\rangle = |\emptyset\rangle. \quad \square$
3. $\hat{S}_s^-|s\uparrow\rangle = \hat{d}_{s\downarrow}^\dagger \hat{d}_{s\uparrow} \hat{d}_{s\uparrow}^\dagger|0\rangle = \hat{d}_{s\downarrow}^\dagger(1 - \hat{d}_{s\uparrow}^\dagger \hat{d}_{s\uparrow})|0\rangle = |s\downarrow\rangle. \quad \square$
4. $\hat{S}_s^z|s\downarrow\rangle = \frac{1}{2}(\hat{d}_{s\uparrow}^\dagger \hat{d}_{s\uparrow} \hat{d}_{s\downarrow}^\dagger - \hat{d}_{s\downarrow}^\dagger \hat{d}_{s\downarrow} \hat{d}_{s\downarrow}^\dagger)|0\rangle = \frac{1}{2}[-\hat{d}_{s\uparrow}^\dagger \hat{d}_{s\downarrow}^\dagger \hat{d}_{s\uparrow} - \hat{d}_{s\downarrow}^\dagger(1 - \hat{d}_{s\downarrow}^\dagger \hat{d}_{s\downarrow})]|0\rangle = -\frac{1}{2}|s\downarrow\rangle. \quad \square$
5. $\hat{S}_s^+|s\downarrow\rangle = \hat{d}_{s\uparrow}^\dagger \hat{d}_{s\downarrow} \hat{d}_{s\downarrow}^\dagger|0\rangle = \hat{d}_{s\uparrow}^\dagger(1 - \hat{d}_{s\downarrow}^\dagger \hat{d}_{s\downarrow})|0\rangle = |s\uparrow\rangle. \quad \square$
6. $\hat{S}_s^-|0\rangle = \hat{d}_{s\downarrow}^\dagger \hat{d}_{s\uparrow} \hat{d}_{s\downarrow}^\dagger|0\rangle = -\hat{d}_{s\downarrow}^\dagger \hat{d}_{s\downarrow}^\dagger \hat{d}_{s\uparrow}|0\rangle = 0$, so $\hat{S}_s^-|s\downarrow\rangle = |\emptyset\rangle. \quad \square$

The operators $\hat{S}_s^z, \hat{S}_s^+$, and $\hat{S}_s^-$ correspond to the spin operators

- Since $\hat{S}_s^z$ will not change the state, just “returns” different eigenvalues, so it measures the z -component of spin.
- Since $\hat{S}_s^+$ annihilates $|\uparrow\rangle$ and raises $|\downarrow\rangle$ to $|\uparrow\rangle$, so it is the raising operator.
- Since $\hat{S}_s^-$ lowers $|\uparrow\rangle$ and annihilates $|\downarrow\rangle$, so it is the lowering operator.

Problem 2.6 (Exercise 1.8 of [5]). Let us define the spin operators along the x and y directions as

$$\hat{S}_s^x \equiv \frac{1}{2}(\hat{S}_s^+ + \hat{S}_s^-), \quad \hat{S}_s^y \equiv \frac{1}{2i}(\hat{S}_s^+ - \hat{S}_s^-),$$

and the spin operator $\hat{S}_s^z$ along the z direction as in (1.91). Prove that these operators can also be written as

$$\hat{S}_s^j = \frac{1}{2} \sum_{\sigma\sigma'} \hat{d}_{s\sigma}^\dagger \sigma_{\sigma\sigma'}^j \hat{d}_{s\sigma'}, \quad j = x, y, z$$

with

$$\sigma^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

the Pauli matrices. Using the anticommutation relations, verify that

$$[\hat{S}_s^i, \hat{S}_{s'}^j]_- = i\delta_{ss'} \sum_{k=x,y,z} \epsilon_{ijk} \hat{S}_s^k,$$

where ϵ_{ijk} is the Levi-Civita tensor.

Solution. *Proof.* Verify every directions: x, y, z respectively.

For $j = x$

$$\begin{aligned} \frac{1}{2} \sum_{\sigma,\sigma'} \hat{d}_{s\sigma}^\dagger \sigma_{\sigma\sigma'}^x \hat{d}_{s\sigma'} &= \frac{1}{2} [\hat{d}_{s\uparrow}^\dagger(0) \hat{d}_{s\uparrow} + \hat{d}_{s\uparrow}^\dagger(1) \hat{d}_{s\downarrow} + \hat{d}_{s\downarrow}^\dagger(1) \hat{d}_{s\uparrow} + \hat{d}_{s\downarrow}^\dagger(0) \hat{d}_{s\downarrow}] \\ &= \frac{1}{2} (\hat{d}_{s\uparrow}^\dagger \hat{d}_{s\downarrow} + \hat{d}_{s\downarrow}^\dagger \hat{d}_{s\uparrow}) = \hat{S}_s^x, \end{aligned}$$

similarly, for $j = y$ and $j = z$

$$\begin{aligned} \frac{1}{2} \sum_{\sigma,\sigma'} \hat{d}_{s\sigma}^\dagger \sigma_{\sigma\sigma'}^y \hat{d}_{s\sigma'} &= \frac{1}{2} [\hat{d}_{s\uparrow}^\dagger(0) \hat{d}_{s\uparrow} + \hat{d}_{s\uparrow}^\dagger(-i) \hat{d}_{s\downarrow} + \hat{d}_{s\downarrow}^\dagger(i) \hat{d}_{s\uparrow} + \hat{d}_{s\downarrow}^\dagger(0) \hat{d}_{s\downarrow}] = \hat{S}_s^y, \\ \frac{1}{2} \sum_{\sigma,\sigma'} \hat{d}_{s\sigma}^\dagger \sigma_{\sigma\sigma'}^z \hat{d}_{s\sigma'} &= \frac{1}{2} [\hat{d}_{s\uparrow}^\dagger(1) \hat{d}_{s\uparrow} + \hat{d}_{s\uparrow}^\dagger(0) \hat{d}_{s\downarrow} + \hat{d}_{s\downarrow}^\dagger(0) \hat{d}_{s\uparrow} + \hat{d}_{s\downarrow}^\dagger(-1) \hat{d}_{s\downarrow}] = \hat{S}_s^z. \end{aligned}$$

Then, $\hat{S}_s^j = \frac{1}{2} \sum_{\sigma\sigma'} \hat{d}_{s\sigma}^\dagger \sigma_{\sigma\sigma'}^j \hat{d}_{s\sigma'}$, $j = x, y, z$ is valid.

Consider the simple situation $s' = s$, that is

$$[\hat{S}_s^i, \hat{S}_s^j] = i \sum_k \epsilon_{ijk} \hat{S}_s^k.$$

Write $\hat{S}_s^i, \hat{S}_s^j$ in terms of the representation just proved

$$\hat{S}_s^i = \frac{1}{2} \sum_{\alpha\beta} \hat{d}_{s\alpha}^\dagger \sigma_{\alpha\beta}^i \hat{d}_{s\beta}, \quad \hat{S}_s^j = \frac{1}{2} \sum_{\gamma\delta} \hat{d}_{s\gamma}^\dagger \sigma_{\gamma\delta}^j \hat{d}_{s\delta}.$$

and substitute them into the commutator

$$[\hat{S}_s^i, \hat{S}_s^j] = \hat{S}_s^i \hat{S}_s^j - \hat{S}_s^j \hat{S}_s^i = \frac{1}{4} \sum_{\alpha\beta\gamma\delta} [\hat{d}_{s\alpha}^\dagger \sigma_{\alpha\beta}^i \hat{d}_{s\beta} \hat{d}_{s\gamma}^\dagger \sigma_{\gamma\delta}^j \hat{d}_{s\delta} - \hat{d}_{s\gamma}^\dagger \sigma_{\gamma\delta}^j \hat{d}_{s\delta} \hat{d}_{s\alpha}^\dagger \sigma_{\alpha\beta}^i \hat{d}_{s\beta}].$$

Consider the product $\hat{d}_{s\beta} \hat{d}_{s\gamma}^\dagger = \delta_{\beta\gamma} - \hat{d}_{s\gamma}^\dagger \hat{d}_{s\beta}$ and the relation $\delta_{\beta\gamma} \sigma_{\gamma\delta}^j = \sigma_{\beta\delta}^j$, the first term gets

$$\begin{aligned} \hat{d}_{s\alpha}^\dagger \sigma_{\alpha\beta}^i \hat{d}_{s\beta} \hat{d}_{s\gamma}^\dagger \sigma_{\gamma\delta}^j \hat{d}_{s\delta} &= \hat{d}_{s\alpha}^\dagger \sigma_{\alpha\beta}^i [\delta_{\beta\gamma} - \hat{d}_{s\gamma}^\dagger \hat{d}_{s\beta}] \sigma_{\gamma\delta}^j \hat{d}_{s\delta} = \hat{d}_{s\alpha}^\dagger \sigma_{\alpha\beta}^i \delta_{\beta\gamma} \sigma_{\gamma\delta}^j \hat{d}_{s\delta} - \hat{d}_{s\alpha}^\dagger \sigma_{\alpha\beta}^i \hat{d}_{s\gamma}^\dagger \hat{d}_{s\beta} \sigma_{\gamma\delta}^j \hat{d}_{s\delta} \\ &= \hat{d}_{s\alpha}^\dagger \sigma_{\alpha\beta}^i \sigma_{\beta\delta}^j \hat{d}_{s\delta} - \hat{d}_{s\alpha}^\dagger \sigma_{\alpha\beta}^i \hat{d}_{s\gamma}^\dagger \hat{d}_{s\beta} \sigma_{\gamma\delta}^j \hat{d}_{s\delta}, \end{aligned}$$

similarly, for the other term in the commutator

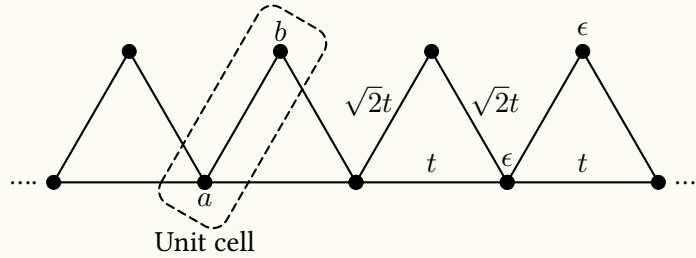
$$\hat{d}_{s\gamma}^\dagger \sigma_{\gamma\delta}^j \hat{d}_{s\delta} \hat{d}_{s\alpha}^\dagger \sigma_{\alpha\beta}^i \hat{d}_{s\beta} = \hat{d}_{s\gamma}^\dagger \sigma_{\gamma\delta}^j [\delta_{\delta\alpha} - \hat{d}_{s\alpha}^\dagger \hat{d}_{s\delta}] \sigma_{\alpha\beta}^i \hat{d}_{s\beta} = \hat{d}_{s\gamma}^\dagger \sigma_{\gamma\delta}^j \sigma_{\delta\beta}^i \hat{d}_{s\beta} - \hat{d}_{s\gamma}^\dagger \sigma_{\gamma\delta}^j \hat{d}_{s\alpha}^\dagger \hat{d}_{s\delta} \sigma_{\alpha\beta}^i \hat{d}_{s\beta}.$$

The terms with four operators cancel, since the spin operators are bilinear in fermions and the commutation should be close to bilinear.

For the situation $s \neq s'$, the operators commute because they act on different sites. So a δ -function will be inserted. Therefore

$$\begin{aligned} [\hat{S}_s^i, \hat{S}_s^j] &= \frac{1}{4} \sum_{\alpha\beta} \hat{d}_{s\alpha}^\dagger [\sigma^i, \sigma^j]_{\alpha\beta} \hat{d}_{s\beta} = \frac{1}{4} \sum_{\alpha\beta} \hat{d}_{s\alpha}^\dagger \left(2i \sum_k \epsilon_{ijk} \sigma^k \right)_{\alpha\beta} \hat{d}_{s\beta} \\ &= \frac{i}{2} \sum_k \epsilon_{ijk} \sum_{\alpha\beta} \hat{d}_{s\alpha}^\dagger \sigma_{\alpha\beta}^k \hat{d}_{s\beta} = i \sum_k \epsilon_{ijk} \hat{S}_s^k. \end{aligned}$$

Problem 2.7 (Exercise 2.2 of [5]). Consider the one-dimensional crystal below,



with matrices $h = \begin{pmatrix} \epsilon & \sqrt{2}t \\ \sqrt{2}t & \epsilon \end{pmatrix}$ and $T = \begin{pmatrix} t & \sqrt{2}t \\ 0 & 0 \end{pmatrix}$. Show that for $t > 0$ the two bands are

$$\begin{aligned} \epsilon_{k1} &= \epsilon - 2t, \\ \epsilon_{k2} &= \epsilon + 2t + 2t \cos k, \end{aligned}$$

with $k \in (-\pi, \pi)$. The first band is therefore perfectly flat. If we have half an electron per unit cell, then the ground state is highly degenerate. For instance, the states obtained by occupying each k -level of the flat band with an electron of either spin up or down all have the same energy. This degeneracy is lifted by the electron-electron interaction and the ground state turns out to be the one in which all electrons have parallel spins. The crystal is then a ferromagnet. The ferromagnetism in flat-band crystals was proposed by Mielke and Tasaki [11,12,13] and is usually called flat-band ferromagnetism.

Solution. For a periodic system, the Bloch Hamiltonian $H(k)$ is

$$\mathcal{H}(k) = h + T e^{ik} + T^\dagger e^{-ik} = \begin{pmatrix} \epsilon + 2t \cos k & \sqrt{2}t(1 + e^{ik}) \\ \sqrt{2}t(1 + e^{-ik}) & \epsilon \end{pmatrix}.$$

To find the eigenvalues ϵ_k , let $\det(\mathcal{H}(k) - \epsilon_k \mathbb{1}) = 0$

$$\begin{vmatrix} \epsilon + 2t \cos k - \epsilon_k & \sqrt{2}t(1 + e^{ik}) \\ \sqrt{2}t(1 + e^{-ik}) & \epsilon - \epsilon_k \end{vmatrix} = 0,$$

and let $\Delta = \epsilon_k - \epsilon$ for simplicity. Then we get $\Delta_1 = 2t \cos k + 2t$, $\Delta_2 = t \cos k - t(\cos k + 2) = -2t$. Thus, the two bands are

$$\epsilon_{k1} = \epsilon - 2t, \quad \epsilon_{k2} = \epsilon + 2t + 2t \cos k.$$

Homework #3 [2025-09-16]

Problem 3.1. Compute the Schrieffer-Wolff transformation for the *single impurity Anderson model (SIAM)*. The single impurity Anderson Hamiltonian in second quantized notation

$$\mathcal{H} = \sum_{k\sigma} \epsilon_k c_{k\sigma}^\dagger c_{k\sigma} + \sum_{\sigma} \epsilon_d d_{\sigma}^\dagger d_{\sigma} + \sum_{k\sigma} V_k (c_{k\sigma}^\dagger d_{\sigma} + d_{\sigma}^\dagger c_{k\sigma}) + U n_{d\uparrow} n_{d\downarrow}.$$

The Hamiltonian has one off-diagonal term, which we call $\mathcal{H}_v$ and diagonal terms, which together we call $\mathcal{H}_0$

$$\begin{aligned} \mathcal{H}_0 &= \sum_{k\sigma} \epsilon_k c_{k\sigma}^\dagger c_{k\sigma} + \sum_{\sigma} \epsilon_d d_{\sigma}^\dagger d_{\sigma} + U n_{d\uparrow} n_{d\downarrow}, \\ \mathcal{H}_v &= \sum_{k\sigma} V_k (c_{k\sigma}^\dagger d_{\sigma} + d_{\sigma}^\dagger c_{k\sigma}). \end{aligned}$$

The **generator** of SWT for SIAM is given by

$$\mathcal{S} = \sum_{k\sigma} (A_k + B_k n_{d\bar{\sigma}}) V_k (c_{k\sigma}^\dagger d_{\sigma} - d_{\sigma}^\dagger c_{k\sigma}),$$

where A_k and B_k are coefficients to be determined.

Remark. Denote

$$\begin{aligned} \Psi_k^\dagger &= \begin{pmatrix} c_{k\uparrow}^\dagger \\ c_{k\downarrow}^\dagger \end{pmatrix} \quad \Psi_k = \begin{pmatrix} c_{k\uparrow} \\ c_{k\downarrow} \end{pmatrix} \quad \Psi_d^\dagger = \begin{pmatrix} d_{\uparrow}^\dagger \\ d_{\downarrow}^\dagger \end{pmatrix} \quad \Psi_d = \begin{pmatrix} d_{\uparrow} \\ d_{\downarrow} \end{pmatrix} \\ 4(\Psi_k^\dagger \mathcal{S} \Psi_{k'}) (\Psi_d^\dagger \mathcal{S} \Psi_d) &= \Psi_k^\dagger \Psi_{k'} (\sigma \cdot \sigma) \Psi_d^\dagger \Psi_d = \Psi_k^\dagger \Psi_{k'} (\sigma \cdot \sigma) \Psi_d^\dagger \Psi_d \\ &= (\Psi_k^\dagger \sigma_z \Psi_k) (\Psi_d^\dagger \sigma_z \Psi_d) + \frac{1}{2} [(\Psi_k^\dagger \sigma^+ \Psi_k) (\Psi_d^\dagger \sigma^- \Psi_d) + (\Psi_k^\dagger \sigma^- \Psi_k) (\Psi_d^\dagger \sigma^+ \Psi_d)], \end{aligned}$$

then compute and identify each of these three terms in the last line. You will be getting other extra terms besides the Kondo exchange term.

Solution. Apply Schrieffer-Wolff transformation to $\mathcal{H}$ to second order. Using BCH, we have

$$\tilde{\mathcal{H}} = e^{\hat{\mathcal{S}}} \mathcal{H} e^{-\hat{\mathcal{S}}} = \mathcal{H}_0 + \mathcal{H}_v + [\mathcal{S}, \mathcal{H}_0] + [\mathcal{S}, \mathcal{H}_v] + \frac{1}{2} ([\mathcal{S}, [\mathcal{S}, \mathcal{H}_0]] + [\mathcal{S}, [\mathcal{S}, \mathcal{H}_v]]) + \dots$$

To ensure the Hamiltonian is diagonal to first order in $\mathcal{H}_v$, the generator $\mathcal{S}$ should satisfy

$$\mathcal{H}_v + [\mathcal{S}, \mathcal{H}_0] = 0, \quad [\mathcal{S}, \mathcal{H}_0] = -\mathcal{H}_v.$$

Since $\mathcal{S} \sim \mathcal{H}_v \sim \mathcal{O}(V^1)$, the transformed Hamiltonian is

$$\tilde{\mathcal{H}} = \mathcal{H}_0 + [\mathcal{S}, \mathcal{H}_v] + \frac{1}{2} [\mathcal{S}, -\mathcal{H}_v] + \mathcal{O}(V^3) = \mathcal{H}_0 + \frac{1}{2} [\mathcal{S}, \mathcal{H}_v] + \mathcal{O}(V^3).$$

Since the **generator** of SWT for SIAM is given by

$$\mathcal{S} = \sum_{k\sigma} (A_k + B_k n_{d\bar{\sigma}}) V_k (c_{k\sigma}^\dagger d_{\sigma} - d_{\sigma}^\dagger c_{k\sigma}).$$

Next, we need to use the identity $[\mathcal{S}, \mathcal{H}_0] = -\mathcal{H}_v$ to determine the coefficients A_k and B_k for the generator $\mathcal{S}$, and then calculate the commutator $[\mathcal{S}, \mathcal{H}_v]$.

1. Determine A_k and B_k .

We need to calculate $[S, \mathcal{H}_0]$ first. To distinguish the variables to be summed in the commutator, we need to express $\mathcal{H}_0$ as

$$\mathcal{H}_0 = \underbrace{\sum_{k'\sigma'} \epsilon_{k'} c_{k'\sigma'}^\dagger c_{k'\sigma'}}_{\text{Bath term}} + \underbrace{\sum_{\sigma'} \epsilon_d d_{\sigma'}^\dagger d_{\sigma'}}_{\text{Impurity level term}} + \underbrace{U n_{d\uparrow} n_{d\downarrow}}_{\text{Interaction term}},$$

then calculate every term accordingly.

- i. Bath term.

The main part of the commutator is

$$[c_{k\sigma}^\dagger d_\sigma - d_\sigma^\dagger c_{k\sigma}, c_{k'\sigma'}^\dagger c_{k'\sigma'}] = [c_{k\sigma}^\dagger d_\sigma, c_{k'\sigma'}^\dagger c_{k'\sigma'}] - [d_\sigma^\dagger c_{k\sigma}, c_{k'\sigma'}^\dagger c_{k'\sigma'}],$$

and it can be expanded with the identity of the commutator

$$\cancel{c_{k\sigma}^\dagger [d_\sigma, c_{k'\sigma'}^\dagger c_{k'\sigma'}]} + [c_{k\sigma}^\dagger, c_{k'\sigma'}^\dagger c_{k'\sigma'}] d_\sigma - d_\sigma [c_{k\sigma}, c_{k'\sigma'}^\dagger c_{k'\sigma'}] - \cancel{[d_\sigma^\dagger, c_{k'\sigma'}^\dagger c_{k'\sigma'}] c_{k\sigma}},$$

where c and d are commutate. Since they are fermions, we need to generate anticommutators from the four terms above to obtain the δ -terms

$$\star [c_{k\sigma}^\dagger, c_{k'\sigma'}^\dagger c_{k'\sigma'}] d_\sigma = \left(\cancel{\{c_{k\sigma}^\dagger, c_{k'\sigma'}^\dagger\} c_{k'\sigma'}} - c_{k'\sigma'}^\dagger \{c_{k\sigma}^\dagger, c_{k'\sigma'}\} \right) d_\sigma = -\delta_{kk'} \delta_{\sigma\sigma'} c_{k'\sigma'}^\dagger d_\sigma$$

$$\star d_\sigma [c_{k\sigma}, c_{k'\sigma'}^\dagger c_{k'\sigma'}] = d_\sigma \left(\{c_{k\sigma}, c_{k'\sigma'}^\dagger\} c_{k'\sigma'} - c_{k'\sigma'}^\dagger \cancel{\{c_{k\sigma}, c_{k'\sigma'}\}} \right) = \delta_{kk'} \delta_{\sigma\sigma'} d_\sigma c_{k'\sigma'}$$

After summing, the Bath term will give a factor of

$$\sum_{k'\sigma'} \epsilon_{k'} \left(-\delta_{kk'} \delta_{\sigma\sigma'} c_{k'\sigma'}^\dagger d_\sigma - \delta_{kk'} \delta_{\sigma\sigma'} d_\sigma c_{k'\sigma'} \right) = -\epsilon_k \left(c_{k\sigma}^\dagger d_\sigma + d_\sigma^\dagger c_{k\sigma} \right).$$

- ii. Impurity level term

Since $[n_{d\bar{\sigma}}, d_\sigma] = 0$, we do not consider the $B_k n_{d\bar{\sigma}}$ factor. Similarly, the commutator with $d_{\sigma'}^\dagger d_{\sigma'}$

$$[c_{k\sigma}^\dagger d_\sigma - d_\sigma^\dagger c_{k\sigma}, d_{\sigma'}^\dagger d_{\sigma'}] = \delta_{\sigma\sigma'} \left(c_{k\sigma}^\dagger d_{\sigma'} + d_{\sigma'}^\dagger c_{k\sigma} \right),$$

and summing over σ' gives the factor

$$\epsilon_d \left(c_{k\sigma}^\dagger d_\sigma + d_\sigma^\dagger c_{k\sigma} \right).$$

- iii. Interaction term

It is trivial that the commutator depends on the spin.

$\star \sigma = \uparrow$, $n_{d\bar{\sigma}} = n_{d\downarrow}$, the commutator gives a factor of

$$[n_{d\downarrow} (c_{k\uparrow}^\dagger d_\uparrow - d_\uparrow^\dagger c_{k\uparrow}), n_{d\uparrow} n_{d\downarrow}] = c_{k\uparrow}^\dagger d_\uparrow n_{d\downarrow} - (-d_\uparrow^\dagger n_{d\downarrow} c_{k\uparrow}) = n_{d\downarrow} \left(c_{k\uparrow}^\dagger d_\uparrow + d_\uparrow^\dagger c_{k\uparrow} \right).$$

$\star \sigma = \uparrow$, by symmetry, just reverse the sign of σ , we get a factor

$$n_{d\uparrow} \left(c_{k\downarrow}^\dagger d_\downarrow + d_\downarrow^\dagger c_{k\downarrow} \right).$$

To summarize, the interaction term will bring the factor

$$[c_{k\sigma}^\dagger d_\sigma - d_\sigma^\dagger c_{k\sigma}, U n_{d\uparrow} n_{d\downarrow}] = U n_{d\bar{\sigma}} \left(c_{k\sigma}^\dagger d_\sigma + d_\sigma^\dagger c_{k\sigma} \right).$$

Now, we arrive at the result of the commutator $[\mathcal{S}, \mathcal{H}_0]$

$$[\mathcal{S}, \mathcal{H}_0] = \sum_{k\sigma} (A_k + B_k n_{d\bar{\sigma}}) V_k (-\epsilon_k + \epsilon_d + U n_{d\bar{\sigma}}) (c_{k\sigma}^\dagger d_\sigma + d_\sigma^\dagger c_{k\sigma}) = -\mathcal{H}_v.$$

Compare the expression of $\mathcal{H}_v$, we have

$$(A_k + B_k n_{d\bar{\sigma}}) (-\epsilon_k + \epsilon_d + U n_{d\bar{\sigma}}) = -1.$$

Since the sign of σ can be $\uparrow$ or $\downarrow$, let $n_{d\bar{\sigma}} = 0$ and 1 , we obtain the coefficients

$$A_k = \frac{1}{\epsilon_k - \epsilon_d}, \quad \text{and} \quad B_k = \frac{U}{(\epsilon_k - \epsilon_d)(\epsilon_k - \epsilon_d - U)}.$$

2. Second term of the transformed Hamiltonian

Then, we need to get $[\mathcal{S}, \mathcal{H}_v]$, the second term of $\tilde{\mathcal{H}}$. Similar to the method of obtaining $[\mathcal{S}, \mathcal{H}_0]$, the result is

$$\begin{aligned} [\mathcal{S}, \mathcal{H}_v] &= \sum_{k\sigma} V_k^2 (A_k d_\sigma^\dagger d_\sigma + B_k n_{d\bar{\sigma}} d_\sigma^\dagger + \text{H.c.}) \\ &\quad + \sum_{kk'\sigma} V_k V_{k'} (A_k c_{k\sigma}^\dagger c_{k'\sigma'} + B_k d_{\bar{\sigma}}^\dagger c_{k'\bar{\sigma}} c_{k\sigma}^\dagger d_\sigma + B_k c_{k'\bar{\sigma}}^\dagger d_{\bar{\sigma}} c_{k\sigma}^\dagger d_\sigma + B_k c_{k\sigma}^\dagger c_{k'\sigma'} n_{d\bar{\sigma}} + \text{H.c.}), \end{aligned}$$

where A_k and B_k have been already determined.

Eventually, we finished the SWT for SIAM

$$\tilde{\mathcal{H}} = \mathcal{H}_0 + \frac{1}{2} [\mathcal{S}, \mathcal{H}_v] + \mathcal{O}(V^3),$$

where the corresponding terms/coefficients are obtained in the previous step.

CHAPTER 5 Methodology

5.1 Computation Gods

Q#1 We want to have an interacting Hamiltonian

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{V}$$

normally, we consider $\tilde{E}_0 > \langle V \rangle$.

$$\sum t_{ij} c_i^\dagger c_j + \epsilon_A - 2t(\cos kx + \cos ky)$$

$2Dt = W$. while V energy $un_\uparrow n_\downarrow$, $W > u$; or $u > W$ is for cookrate. $U \sim \delta t$.

- (a) The first thing we will discuss is the Ground State (zero temperature); and we also need some thermo equilibrium states (finite T), ρ : density matrix.
- (b) Observables $\Leftrightarrow$ Experiments. $\langle \hat{O} \rangle$.
 - Thermodynamic: specific heat C_v ; M (magnetisation), χ susceptibility.
 - Second type of measurement is spectroscopy (Can be transform to many body Green Funcs): it's energy dependent. ARPES means Angel Result . Another called Raman (low frequency). X-ray. Using potitros (μ -SR); neutron.

For MB: $L \rightarrow \infty$, $N \rightarrow \infty$. infinity - $\langle x|n \rangle$.

Q#1.1 $|\Psi\rangle$: how to represent efficiently? $\rightarrow$ Quantum Statistical.

For a system: n, T, P, V, μ, S . Macros $F(n, \mu, T, \dots)$ to get equation of states.

A "few" important (enough to determine physical properties) observables (parameter). For landau parameters.

5.2 How to compute

1. Represent eff.

- Perturbatim, and something beyond that: rariation (functional), mens to minimize sth, $(\delta F, \delta E)$ to some differential intense Equal group. In class method, EL - eq, δL .
 - unperturbed theory (H_0): pay attention to degeneracy quasi-degeneracy (FL), Gapless $\Delta E|_{L \rightarrow \infty} \rightarrow 0$.
 - generate perturbation sequences Feynman diagram, bookkeeping = wilsn RG.

5.3 QM: formulation

- Hamiltonian Mechanics
- Lagrangian Mechanins: path integral

Pictures (under Hamiltonian)

- Schrödinger picture
- Heisenberg picture
- interaction picture (Dirac picture)

means where u keep the TIME EVOLUTION.

In Schrödinger picture:

$$i \frac{\partial}{\partial t} |\Psi_s(t)\rangle = H |\Psi_s(t)\rangle \quad (5.1)$$

and operators are time-independent.

In Heisenberg picture, states are time-independent.

$$-i \frac{\partial}{\partial t} \hat{O}_H(t) = [H, O(t)]$$

and the relation

$$O_H(t) = e^{iHt/\hbar} \hat{O}_p e^{-iHt/\hbar}$$

For Dirac picture, states change slowly

$$i \partial_t |\Psi_I(t)\rangle = V_I(t) |\Psi_I(t)\rangle$$

and Evolution accroding to H

$$-i \partial_t O_I(t) = [H_o, O_I(t)]$$

In some sense, they are equivalent

$$O_H(t) = e^{iHt/\hbar} \hat{O}_p e^{-iHt/\hbar}, O_I(t) = e^{iHt/\hbar} \hat{O}_I e^{-iHt/\hbar}$$

5.3.1 Pictures

Schrödinger Picture:

$$i \hbar \partial_t |\psi_s(t)\rangle = H |\psi_s(t)\rangle \quad (5.2)$$

Left: mixed state; Right: ρ_s . $-i \hbar \partial_t \rho_s = [H, \rho_s]$

From the above, we derive

$$i \hbar \partial_t U_s(t, t_0) = H_t U_s(t, t_0)$$

while $u \sim e^{iHt/\hbar}$.

Time evolution operator.

$$|\Psi_s(t)\rangle = U_s(t, t_0)|\Psi_s(t_0)\rangle.$$

and $U^\dagger = U^{-1}$, $U(t_0, t_0) = \mathbb{1}$.

While $U(t, t_0) = U_s(t, t')U_s(t', t_0)$

$$U_s(t, t_0) = 1 - \frac{i}{\hbar} \int_{t_0}^t dt_1 H_t \times U_s(t_1, t_0)$$

So, we can do the following

$$U_s(t, t_0) = 1 - \frac{i}{\hbar} \int_{t_0}^t dt_1 H_t, \quad U_s(t_1, t_0) = 1 - \frac{i}{\hbar} \int_{t_0}^{t_1} dt_2 U(t_1, t_0)$$

then,

$$U_s(t, t_0) = 1 - \frac{i}{\hbar} \int_{t_0}^t dt_1 H_{t_1} \left(1 - \frac{i}{\hbar} \int_{t_0}^{t_1} dt_2 H_{t_2} U(t_0, t_1) \right) = \dots$$

equivalent to the equation

$$x^2 - x + 1 = 0, \quad x = \frac{-1}{1-x} = \frac{-1}{1 - \left(\frac{-1}{1-\dots}\right)} \text{ Continued fraction}$$

recursion (von Neumann's series)

$$U_s(t, t_0) = 1 + \sum_{n=1}^{\infty} U_s^{(n)}(t, t_0), \quad (5.3)$$

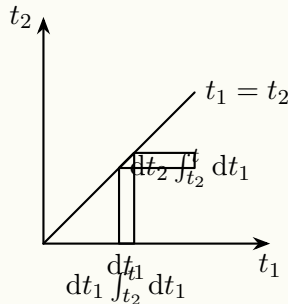
or

$$U_s^{(n)}(t, t_0) = \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \int_{t_0}^{t_2} dt_3 \cdots \int_{t_0}^{t_{n-1}} dt_n \times H_{t_1} H_{t_2} \cdots H_{t_n}.$$

while $t \geq t_1 \geq t_2 \cdots \geq t_n \geq t_0$.

5.3.2 Dyson's time-ordering operator

$$T_D(A(t_1)B(t_2)) = \begin{cases} A(t_1)B(t_2), & \text{for } t_1 > t_2 \\ B(t_2)A(t_1), & \text{for } t_2 > t_1 \end{cases}$$



$$dt_2 \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 H_{t_1} H_{t_2} = \int_{t_3}^t dt_2 \int_{t_2}^t dt_1 H_{t_1} H_{t_2}$$

then, $t_1 \Leftrightarrow t_2$

$$\begin{aligned} \int_{t_2}^t dt_1 \int_{t_2}^{t_1} dt_2 H_{t_1} H_{t_2} &= \int_{t_0}^t dt_1 \int_{t_1}^t dt_2 H_{t_2} H_{t_1} \\ &= \frac{1}{2} \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 (H_{t_1} H_{t_2} \theta(t_1 - t_2) + H_{t_2} H_{t_1} \theta(t_2 - t_1)) \\ &= \frac{1}{2!} \int_{t_0}^t dt_1 dt_2 T_D(H_{t_1} H_{t_2}) \end{aligned}$$

Generalize

$$U_s^{(n)}(t, t_0) = \frac{1}{n!} \left(-\frac{i}{\hbar} \right)^n \int_{t_0}^t \int_{t_0}^t dt_1 \cdots dt_n T_D(H_{t_1} H_{t_2} H_{t_3} \cdots H_{t_n}) \quad (5.4)$$

$T_D(H_{t_1}, H_{t_2}, H_{t_3})$: $\theta(t_1 - t_2), \theta(t_2 - t_1), \theta(t_2 - t_3), \theta(t_3 - t_2), \theta(t_3 - t_1), \theta(t_1 - t_3)$, like $2^3 = 8$. The formal expression

$$U_s(t, t_0) = T_D \exp \left(-\frac{i}{\hbar} \int_{t_0}^t dt' H_{t'} \right) \quad (5.5)$$

Now, let's try to

$$H = H_0 + V$$

5.3.3 Conventional pert

$$H_0 |\eta_n\rangle = \eta_n |\eta_n\rangle \quad (5.6)$$

$$H |E_0\rangle = E_0 |E_0\rangle \quad (5.7)$$

while

$$\Delta E_0 = E_0 - \eta_0 = \frac{\langle \eta_0 | V | E_0 \rangle}{\langle \eta_0 | E \rangle}$$

where $|E_0\rangle \approx |\eta_0\rangle$. define $P_0 = |\eta_0\rangle\langle\eta_0|$, $Q_0 = 1 - P_0 = \sum_{n=1}^{\infty} |\eta_n\rangle\langle\eta_n|$.

$$(D - H_0) |E_0\rangle = (D - \hat{H} + \hat{V}) |E_0\rangle = (D - E_0 + \hat{V}) |E_0\rangle$$

we get

$$|E_0\rangle = \frac{1}{D - H_0} (D - E_0 + V) |E_0\rangle = P_0 |E_0\rangle + D_0 |E_0\rangle = |\eta_0\rangle\langle\eta_0| E_0\rangle + Q_0 |E_0\rangle \quad (5.8)$$

and

$$|\tilde{E}_0\rangle = \frac{|E_0\rangle}{\langle\eta_0|E_0\rangle}$$

with this, we can write down the perturbation state

$$|\tilde{E}_0\rangle = |\eta_0\rangle + \frac{1}{D - H_0} Q_0 (D - E_0 + V) |\tilde{D}_0\rangle, \quad (5.9)$$

$$[H_0, P_0] = [Q_0, H_0] = 0, \quad (5.10)$$

while Q_0 stands for the projection for orthogonormal basis. Formally,

$$|\tilde{E}_0\rangle = \sum_{m=0}^{\infty} \left\{ \frac{1}{D - H_0} Q_0 (D - E_0 + V) \right\}^m |\eta_0\rangle \quad (5.11)$$

and $\Delta E_0 = \langle \eta_0 | V | \tilde{E}_0 \rangle$.

- Schrodinger pert: $D = \eta_0$, $\Delta E_0(m=0) = \langle \eta_0 | \lambda v | \eta_0 \rangle \sim \lambda$
- $\Delta E_0(m=1) = \frac{\sum |\langle \eta_0 | \lambda v | \eta_n \rangle|^2}{\eta_1 - \eta_n} \sim \lambda^2$, $\Delta E_0(m) \sim \lambda^{m+1}$.
- $\Delta E_0(m=2) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{\langle \eta_0 | V | \eta_m \rangle \langle \eta_m | V | \eta_n \rangle \langle \eta_n | V | \eta_0 \rangle}{(\eta_0 - \eta_m)(\eta_0 - \eta_n)} (\sim \lambda^3) - \Delta E_0 \sum \frac{|\langle \eta_0 | V | \eta_n \rangle|^2}{(\eta_0 - \eta_n)^2} (\sim \lambda^2)$ where $\Delta E_0 \sim (\lambda^0 + \lambda + \dots + \lambda^m)$.
 $f(\lambda, \lambda^2, \lambda^3)$ mixing $(\lambda^1, \lambda^2, \lambda^3)$.
 interaction problem: diff-integral.

5.4 Brillouin-Wigner Pert.

Here, $D = E_0$.

5.4.1 Lippmann-Schwinger Equation

$$\hat{G} = \frac{1}{Z - H_0}, \hat{G}(z) = \frac{1}{Z - \hat{H}}.$$

we can write

$$Z - H_0 = Z - H + V$$

and

$$(1 + V(Z - H)^{-1})(Z - H) = (1 + \hat{V}\hat{G})(Z - H) = (1 + \hat{V}\hat{G})\hat{G}^{-1} = (Z - H_0)G$$

and $G_0^{-1}(Z - H_0)G = 1 + VG$, $\hat{G} = \hat{G}_0 + \hat{G}\hat{V}\hat{G}$. To expand it,

$$\hat{G} = G_0 + G_0VG_0 + G_0VG_0VG_0 + \dots$$

5.5 Necessity of Gell-Mann-Lon (GML) theorem (No necessity ...)

Doing sth different

$$H_\alpha = H_0 + V e^{-\alpha|t|}, \alpha > 0 \quad (5.12)$$

scuicthing on the interactionL ADIABADIACALLY. there are two limits,

$$\lim_{t \rightarrow \pm\infty} H_\alpha = H_0; \quad (5.13)$$

$$\lim_{t \rightarrow 0} H_\alpha = H \quad (5.14)$$

in the first one, the eigenstate

$$|\psi_\alpha(t)\rangle|_{t \rightarrow \pm\infty} = \begin{cases} |\eta_0\rangle, & t \rightarrow -\infty \\ e^{i\varphi(\alpha)}|\eta_0\rangle & t \rightarrow \infty \end{cases}$$

for the second one

$$|\Psi_\alpha(t)\rangle'|_{t \rightarrow 0} \rightarrow |\Psi_D(t=0)\rangle, .$$

and $H|\Psi_D\rangle = E|\Psi_D\rangle$. as $\alpha > 0$, $\alpha \rightarrow 0$, $\frac{\partial}{\partial\alpha}|_{\alpha \rightarrow 0}$.

$$V_D(t) e^{-|t|\alpha} = e^{\frac{1}{\hbar}H_0 t} V e^{-\frac{1}{\hbar}H_0(t)} e^{-|t|\alpha}$$

and we want to compute this: time evolution of:

$$U_\alpha^D(t, t_0) = \mathcal{T}_D e^{-\frac{i}{\hbar} \int dt' V_D(t') e^{-|t|\alpha}} = \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{1}{\hbar}\right)^n \int_{t_0}^t \int dt_1 \cdots dt_n e^{-\alpha(|t_1|+\cdots)} \quad (5.15)$$

$$T_d\{V_D(t_1), \cdots V_D(t_n)\}$$

$$|\Psi_\alpha^D(t)\rangle = U_\alpha^D(t, t_0) |\Psi_\alpha^D(t_0)\rangle|_{t_0 \rightarrow \infty}$$

1. $|E_0^D\rangle = \lim_{\alpha \rightarrow 0} |\Psi_\alpha^D(0)\rangle_V$ Gell-Mann-Low Theorem

$$\lim_{\alpha \rightarrow 0} \frac{U_\alpha^D(0, -\infty)|\eta_0\rangle}{\langle\eta_0|U_\alpha^D(0, -\infty)|\eta_0\rangle} = \lim_{\alpha \rightarrow 0} \frac{|\Psi_\alpha^D(0)\rangle}{\langle\eta_0|\Psi_\alpha^D(0)\rangle}$$

so

$$\langle\eta_0|\Psi_\alpha^D(0)\rangle \xrightarrow{\alpha \rightarrow 0} \exp\left(-i \frac{f(\lambda)}{\alpha}\right).$$

Linked-Cluster Theorem

$$\begin{aligned} & \langle\Psi_\alpha^D(0)|\hat{A}^D(t)\hat{B}^D(t')|\Psi_\alpha^D(0)\rangle \\ & \Rightarrow \langle\eta_0|V^D(+\infty, 0)\hat{A}(t)\hat{B}(t')U_{(0, -\infty)}^D|\eta\rangle \\ & \Rightarrow \langle\eta_0|T_D(V^D, \cdots), \hat{A}(t)\hat{B}(t')T_D(V^D \cdots)|\eta_0\rangle, \\ & \Rightarrow F(G_D^{cansal}) \end{aligned}$$

From this, we have

$$G^c = \frac{1}{G_v - \epsilon_k^0 - \Sigma}$$

$$|\eta_0\rangle \longrightarrow G^c \longrightarrow |\Psi_\alpha^p(0)\rangle \longrightarrow |\eta_0\rangle \text{ Q:}$$

5.6 No-Necssity of GML

From $\rho_0: |\Psi_0\rangle, H_0 + \delta H$, to $\rho = \rho_0 + \delta\rho$. To describe MB system more efficiently we need operator basis (orthogonal, complete) set. $\{|\eta_n\rangle\}$: inner product. $\text{Tr}[u^\alpha, u^\beta] = \delta^{\alpha\beta}$. $\{u^\alpha\}$ is complete.

$$\hat{\rho} = \sum u^\alpha \langle u^\alpha \rangle$$

Fano regonance. and u find that

$$\rho^2 = \rho \Rightarrow \{\rho_0, \delta\rho\} = \delta\rho + \delta\rho^2$$

so called polynomial eq.

5.7 GML & proof

$$H = H_0 + V e^{-\alpha|t|} = H_0 + \lambda \delta H$$

where D means in Dirac picture

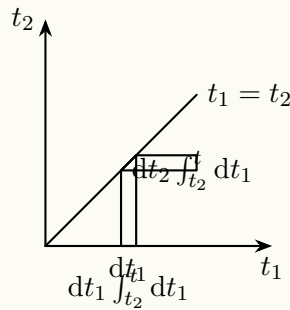
$$\lim_{\alpha \rightarrow 0} \frac{u_{\alpha}^D(0, -\infty)|\eta_0\rangle}{\langle \eta | u_{\alpha}^D(0, -\infty) | \eta_0 \rangle} = \lim_{\alpha \rightarrow 0} \frac{|\psi_{\alpha}^D(0)\rangle}{\langle \eta_0 | \psi_{\alpha}^D(0) \rangle}$$

Proof.

$$(H_0 + V) \frac{|\psi_{\alpha}^D(0)\rangle}{\langle \eta_0 | \psi_{\alpha}^D(0) \rangle} = \frac{\langle \eta_0 | H | \psi_0^D(0) \rangle}{\langle \eta_0 | \psi_0^D \rangle} \frac{|\psi_{\alpha}^D(0)\rangle}{\langle \eta_0 | \psi_{\alpha}^D(0) \rangle}$$

$$\begin{aligned} (H_0 - \eta_0) |\psi_{\alpha}^D(0)\rangle &= - \sum_n \frac{1}{n!} \left(-\frac{i}{\hbar} \right)^{n-1} \int_{-\infty}^0 \int_{-\infty}^0 dt_1 \cdots dt_n e^{-\alpha(|t_1| + \cdots + |t_n|)} \\ &= \left(\sum_{j=1}^n \right) \overset{n \frac{\partial}{\partial t_n}}{\mathcal{T}_D(V_D(t_1) V_D(t_2) \cdots V_D(t_n))} |\eta_0\rangle \end{aligned}$$

where $\left(\sum_{j=1}^n \frac{\partial}{\partial t_j} \right) \rightarrow n \frac{\partial}{\partial t_n}$.



$\mathcal{T}_D(V(t_1) \cdots V(t_n))$ symmetric of $(t_1 \cdots t_n)$ it will return corresponding factor due to the order of $t_1 t_2 t_3 + t_2 t_3 t_1 + \cdots$ then u can do integral by part

Causal GFs

We usually convert

$$\langle \psi_{\alpha}^D(0) | \hat{a}_i^{\dagger}(t) a_j(t') | \psi_{\alpha}^D(0) \rangle$$

into Casaul Green Functions

$$G = \frac{1}{\omega - \epsilon_0 - \Sigma}$$

5.7.1 No-Necessity of GML

Basically, we can also do perturbation & variational method. When given a H_0 , $\rho_0 + \delta H$ where $\delta\rho = \rho - \rho_0$, and energy δE .

Reasit adiabaticity.

route cause: $|\psi\rangle = \frac{1}{E_0}$ when write into Taylor series: $|\psi(\lambda)\rangle = |\psi_0\rangle + \lambda|\delta\psi\rangle + \dots$ and the energy $E_0(\lambda) = E_0 + \lambda\delta\epsilon^{(1)} + \frac{\lambda^2}{2}\delta\epsilon^{(2)}$ they should be true separate.

There's no need for E_0 .

In QM: Lippmann-Schwinger

$$\hat{G} = (G_0 - H)^{-1}$$

But the Lippmann-Schwinger equation we put down is

$$\hat{G}(\omega) = G_0 + G_0 V G$$

diagonal in ω .

As well,

$$G(\mathbf{r}; \mathbf{r}') = G_0(\mathbf{r}, \mathbf{r}') - \int d\mathbf{r}'' G_0(\mathbf{r}, \mathbf{r}'') \delta V(\mathbf{r}'') G(\mathbf{r}'', \mathbf{r}')$$

DOES NOT depend on E_0 .

5.8 Complete operator basis set

wave function $|\Psi\rangle = e^{i\alpha(\dots)} = \langle \hat{O} \rangle$

“Inner product” (orthogonal, completeness) $\text{Tr}[\hat{u}^\alpha, \hat{u}^\beta] = \delta_{\alpha,\beta} C$

$$\hat{O} = \sum \langle u^\alpha \rangle \delta \hat{U}^\alpha = \sum_\alpha (\mathbf{u}^\alpha \cdot \mathbf{O}) \mathbf{u}^\alpha$$

where $\langle \rangle_{\hat{O}} = \text{tr}[u^\alpha, \hat{O}]$

$$\hat{\rho} = \sum \langle \hat{u}^\alpha \rangle \hat{u}^\alpha$$

so we come to some oelgebraic description

$$[\rho, H] = 0$$

(⊗!!! SSB $[\rho, H] \neq 0$) and the project $\hat{\rho}^2 = \hat{\rho}$, with the eigenenergy for pure state condition

$$\{\rho, H\} = 2E\hat{\rho}$$

* Gerivile to thermal states

$\hat{\rho}^2 \neq \rho$, but

5.8.1 Perturbation theory

$$\rho(g) = \rho_0 + g\rho_1 + \frac{g^2}{2!}\rho_2 \quad (5.16)$$

$$E = E_0 + \delta E = E_0 + gE_1 + \frac{g^2}{2!}E_2 + \dots \quad (5.17)$$

$$\hat{\rho} = \hat{\rho}_0 + \delta\hat{\rho}$$

$$\rho^2 = \rho_0^2 + \delta\rho^2 + \{\hat{\rho}_0, \delta\hat{\rho}\} = \hat{\rho} = \hat{\rho}_0 + \delta\hat{\rho} = \rho_0 + \delta\rho^2 + \{\hat{\rho}_0, \delta\hat{\rho}\}$$

we obtain

$$\{\hat{\rho}_0, \delta\rho\} = \delta\hat{\rho} - \delta\hat{\rho}^2$$

For the Hamiltonian

$$H = H_0 + \delta H$$

the first order term becomes

$$O(g') : [\rho_1, H_0] + [\hat{\rho}_0, \delta H] \approx 0.$$

then

$$\{\hat{\rho}_0, \hat{\rho}_1\} = \hat{\rho}_1$$

and we have

$$[\{\hat{\rho}_0, \hat{\rho}_1\}, H_0] + [\hat{\rho}_0, \delta H] = 0$$

we expand $\rho_1 = \sum \langle \hat{u}^\alpha \rangle_1 \hat{u}^\alpha$ while $\hat{\rho}_0 = \sum_{\alpha_0} \langle u^{\alpha_0} \rangle u^{\alpha_0}$ and the trace $\text{Tr}[u^\alpha(\text{the equation})] = 0$. Finally, we get

$$\langle [\delta H, \hat{u}^{\alpha_0}] \rangle + \sum_{\alpha} \langle \{[H_0, \hat{u}^{\alpha_0}], \hat{u}^\alpha\} \rangle_0 \langle u^\alpha \rangle_1 = 0$$

and we can write it into linear equation

$$\hat{L} \langle u \rangle_1 + \langle [\delta H, u^{\alpha_0}] \rangle = 0$$

where The $\hat{L}$ is Liouver super op. for time evo of ρ :

$$[H, \delta\rho] = \frac{d}{dt}\rho = \hat{L}\rho$$

More formally,

$$[[L_0]][\rho_1] + [V]_0 = 0$$

so u can see

$$\{\rho_0, \delta\rho\} = \delta\hat{\rho} - \delta\hat{\rho}^2 = \delta\hat{\rho} - \delta\hat{\rho}(\{\rho_0, \delta\rho\} + \delta\hat{\rho}^2) = \delta\hat{\rho}(1 - \{\rho_0, \delta\rho\}) - \delta\hat{\rho}^3$$

concerning the 2nd order term

$$O = \langle [\delta H, u^{\alpha_0}] \rangle_0 + \sum \delta \langle u^\alpha \rangle \langle \{u^\alpha, [H, \hat{u}^{\alpha_0}]\} \rangle_0 + \sum_{\gamma\beta} \delta \langle u^\alpha \rangle \delta \langle u^\beta \rangle \langle \{u^\gamma, [H, \{u^{\alpha_0}, u^\beta\}]\} \rangle_0 + \dots$$

then we start from $\mathbb{1} \in coBasisset$. Eq of $\mathbb{1}$, $u^{\alpha_0} = \mathbb{1}$:

$$0 = 0 + \sum \dots$$

called linked-cluster-theorem. keep connected term.

In a global form, we can find

$$\begin{aligned} [V]_{0,\alpha_0} + [[L]] \cdot [\delta\hat{\rho}] + \text{Tr}[\hat{u}^{\alpha_0}[\delta\hat{\rho}, H]] &= 0 \\ [V] + [[L]][\delta\hat{\rho}] + [[L_2]][\delta\rho^2] + \text{Tr}[u^{\alpha_0}, [\delta\rho^3, H]] & \end{aligned}$$

then L_n can be defined as

$$L_n \cdot \delta\hat{\rho}^n = [H, \delta\rho^n]$$

then

$$[V]_0 + \sum_{n=1}^{\infty} [[L_n]][\delta\rho^n] = 0$$

then u can get δE from $\delta\rho$.

5.8.2 Example

Transversefield Ising model

$$H_{\text{TFIM}} = -J \sum_{ij} S_i^z S_j^z - h_z \sum_i S_i^z = -\delta H - H_0$$

$|\psi_0\rangle = |\rightarrow\rightarrow\rightarrow\cdots\rangle$ $\rho_0 : \langle S_i^x \rangle_0 = \frac{1}{2}$, $\langle SA^{\alpha}xx \rangle_0 = 0$, $\langle S_i^{\alpha} S_j^{\beta} \rangle_{C,0} = 0^{\dagger}$. Denote $u^{\alpha_0} = S_{i_0}^y S_{i_0+1}^z$

$$\langle [\delta H, n^{\alpha_0}] \rangle_0 \propto \langle S_i^x S_{i_0+1}^x \rangle_0$$

we often get

$$[H_0, u^{\alpha_0}] = S_{i_e}$$

$$\{u^{\alpha}, [H_0 u^{\alpha_0}]\} = s_{10}y + S_{i_0+1}^x Y + \langle S^x \rangle_x \langle S_x \rangle_z = 0$$

that is

$$\langle S_i^x, S_{i+1}^x \rangle \sim h/1g - g_c|^{1/8}$$

2D Heisenberg module

5.9 Perturbation

5.9.1 Review of Path Integral in QM

$$\psi(\mathbf{x}, t) = \int d^3x' K(\mathbf{x}, t; \mathbf{x}', t_0) \psi(\mathbf{x}', t_0)$$

¹Cumulant: $\langle S_i S_j \rangle_C = \langle (S_i - \langle S_i \rangle)(S_j - \langle S_j \rangle) \rangle$.

$$\langle x|\psi(t)\rangle \Rightarrow K(\mathbf{x}, t; \mathbf{x}', t_0) = \langle \mathbf{x} | e^{-iH(t-t_0)/\hbar} | \mathbf{x}' \rangle = \langle 0 | \varphi(\mathbf{x}, t) \varphi^\dagger(\mathbf{x}', t_0) | 0 \rangle$$

PI is based computed on GF directly.

$$\sum e^{i \int_{t,t}^{t,t} L/t}$$

$$e^{iHt} \rightarrow e^{N_i H t / N} = E^o$$

$$\left(-\frac{\hbar^2}{2m} (\nabla')^2 + V(\mathbf{x}) - i\hbar \frac{\partial}{\partial t} \right) K(\mathbf{x}, t; \mathbf{x}', t_0) = -i\hbar \delta^3(\mathbf{x} - \mathbf{x}') \delta(G - G_{aaa})$$

5.9.2 bass Precedures

$$L = L(x, \partial_t, xx, \dots t)$$

- lassical route
- Fluctuations around classic route

5.9.3 Auadratic Legrangian

$$\psi(x_N, t_\sim, |X_0 t_0\rangle) = \int d[x(t)] \exp \left[\frac{i}{\hbar} S[x, t] \right]$$

while consider the Lagrangian

$$\mathcal{L}(x, \dot{x}, t) = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} c(t) x^2 - c(t) x$$

and since $\delta L = 0$, Since Euler-Lagrangian eq., we have EOM

$$m \ddot{X}_{cl} + c(t) x_{cp} + e(t) = 0$$

then $x(t) = x_{cl}(t) + y(t)$, and $y(t_0) = y(t_N) = 0$. then $L(x_{cl} + y, \dot{x}_{cl} + \dot{y}, t) = L_{cl} + L'(y, \dot{y}(t), t) + \delta L(|\dot{y}\rangle)$

$$L_{cl} = \frac{1}{2} m \dot{x}_{cl}^2 - \frac{1}{2} c(t) x_{cl} - e(t) x_{cl}$$

$$L' = \frac{1}{2} m \dot{y}^2 - \frac{1}{2} c(t) y^2 \delta L = m \dot{x}_{cl} \dot{y}(t) - c(t) x_{cl} y - e(t) y$$

5.9.4 Perturbation Theory by path integral

$$\dot{x}_{cl} \dot{y} = \frac{d}{dt} (\dot{x}_{cl} y) - \ddot{x}_{cl} y$$

$$\int_{t_0}^{t_N} dt \delta L = m [\dot{x}_{cl} y] \Big|_{t_1}^{t_N} - \int_{t_0}^{t_N} (m \ddot{x}_{cl}(t) + c(t) x_{cl}(t) + e(t)) y(t) dt$$

then

$$\phi(x_N, t_N; x_0, t_0) = \exp \left\{ \frac{i}{\hbar} S[x_{cl}(t)] \right\} \tilde{\phi}(0, t_N; 0, t_0)$$

then

$$\tilde{\phi}(0, t_N, 0, t_0) = \int_{y(t_0)=d}^{y(t_N)=0} d[y(t_1)] \exp\left(\frac{i}{\hbar} \int dt L'\right)$$

For discretisation on t :

$$\epsilon_N = (t_N - t_0)/N, \text{ and } t_j = t_0 + \epsilon_N j$$

$$\tilde{\phi} = \lim_{N \rightarrow \infty} \left(\frac{m}{2\pi i \hbar \epsilon_0} \right)^{N/2} \times \int dy_1 dy_2 \cdots dy_{N-1} \exp \left[\frac{i}{\hbar} c_N \sum \frac{1}{2} M \frac{(y_{j+1} - y_j)^2}{\epsilon_N^2} - \frac{1}{2} c_j y_j^2 \right]$$

and the energy $E = i \sum y_i a_{jk} y_k$, we have a $(N-1) \times (N-1)$ matrix, where $N \rightarrow \infty$

$$(a_{jk}) = \frac{m}{2\hbar\epsilon_N} \begin{pmatrix} 2 & -1 & \dots & \\ -1 & 2 & -1 & \\ & -1 & 2 & \dots \end{pmatrix} - \frac{\epsilon_N}{2\hbar} \begin{pmatrix} c_1 & & & \\ & \ddots & & \\ & & c_j & \\ & & & \ddots \end{pmatrix} = \lim_{N \rightarrow \infty} \left(\frac{1}{\det[a]} \right)^{1/2}$$

In the first term of a_{jk} , only the diag and two subdiag is non-zero, other elements are zero.

let $D_{N-1} = \det[a]$, then

$$D_n = \left(2 - \frac{\epsilon_N^2}{m} c_N \right) D_{n-1} - D_{n-2}$$

and then u get

$$D_0 = 1, D_1 = 2 - \frac{\epsilon_N^2}{m} c_1$$

to the infinite limit, $\frac{D_{n+1} - 2D_n + D_{n-1}}{\epsilon_N^2} = -\frac{c_{n+1} D_n}{m}$ and when $N \rightarrow \infty, \epsilon_N \rightarrow 0$,

$$\frac{d^2 f(t_0, t)}{dt^2} = -\frac{c(t)}{m} f(t_1, t)$$

apply the bound condition, show that

$$f(t_1, t_0) = 0, \frac{d}{dt} f(t_0, t) = 1$$

$$\begin{cases} \phi(xt; x_0 t_0) = \left(\frac{m}{2\pi i \hbar f(t_0, t)} \right)^{1/2} \exp \left[\frac{i}{\hbar} S[X_{ct}(t)] \right], \\ \frac{d^2 f(t_0, t)}{dt^2} = -\frac{c(t)}{m} f(t_0, t) \end{cases}$$

For Harmonic Oscillator, $c(t) = m\omega^2$, then $\ddot{f} = -\omega^2 f$, so $f(t_0, t) = \sin(\omega(t - t_0))/\omega$.

If consider $\omega \sim \omega_0 + gc(t)$, ...

We are getting $H, \epsilon_n, |n\rangle$, then we get $\langle x|\psi(t)\rangle = \sum_n \langle x|n\rangle e^{i\epsilon_n t/\hbar}$.

To be specifitly,

$$\langle xt|x't'\rangle = \sum_n \varphi_n^*(t) \varphi_n(t') e^{-i\epsilon_n \Delta t/\hbar}$$

5.9.5 Enclidean Path Integral

means $t \rightarrow i\tau$. Relation to partition func, and spectral repr. decomposition of GFs.

$$\langle x_f, -i\tau/2 | x_i, i\tau/2 \rangle|_{\tau \rightarrow \infty} = \lim_{T \rightarrow c_0} e^{-E_0 T} \varphi_0(x_f) \varphi_0^*(x_i) = \langle x_f | e^{-TH} | x_i \rangle$$

5.9.6 Double well problem for PI

$$V(x) = \lambda(x-a)^2 = \lambda(x-a)^2(x+a)^2 \sim \lambda(x+a)^2(2a)^2 \sim \lambda(x-a)^2(2a)^2$$

eigenstates: $|n, +a\rangle, |n, -a\rangle, \langle n_1 + a | H | n'_1 - a \rangle$

$$H' = \begin{pmatrix} H_{a-} & \Delta H \\ \Delta H & H_{-aa} \end{pmatrix}$$

1. x_{cl} : $E = 0$.
2. $t \rightarrow i\tau$, the potential $V(x)$, the Lagrangian from $L = T - V$ to $T + V = T - (-V)$, a flipped potential. (Why? consider e^{idtL}), get $-\dot{x}^2 - V(x)$, but when $t \rightarrow i\tau$, u will get $e^{-\tau(\dot{x}^2 - (-V(x)))}$, under the view of Hamiltonian

For 1: $e^{\sum x_{cl}}$

For 2: $t \rightarrow \infty, e^{\sum_{EPI} inst}$

5.10 Apply P.I. for second Quantization

Begin with the coherent state of H.O.. For Bosons,

$$[a_i, a_j^\dagger] = \delta_{ij}$$

Apply H.O. to Bosons for second quantization. The eigenstate of the annihilation / creation operator is

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle, \quad \text{but it's wrong that } \hat{a}^\dagger|\alpha\rangle = \alpha|\alpha\rangle$$

where the coherent state can be expanded in terms of the quantum number state

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum \frac{\alpha^n}{\sqrt{n!}} |\varphi_n\rangle$$

It has some properties

(a) Non-orthogonal

$$\langle\alpha|\alpha'\rangle = e^{-|\alpha|^2/2} e^{-|\alpha'|^2/2} e^{\alpha^* \alpha'} \quad (5.18)$$

(b) In the quantum number state, we have the identity

$$\sum_n |\varphi_n\rangle \langle\varphi_n| = \mathbb{1},$$

While in the coherent state

$$\mathbb{1} = \frac{1}{\pi} \int |\alpha\rangle \langle\alpha| d\text{Re } \alpha d\text{Im } \alpha \quad (5.19)$$

which is overcompleteness.

5.10.1 Coherentstate of P.I.

We want to compute

$$\langle x|\mathcal{U}|x'\rangle = \langle x|\mathcal{T} e^{iH(t-t')}|x'\rangle = \langle x|e^{i\mathcal{H}\Delta_t} e^{i\mathcal{H}\Delta_t} \dots e^{i\mathcal{H}\Delta_t}|x'\rangle \quad (5.20)$$

where $\Delta t = \frac{t-t'}{N} \big|_{N \rightarrow \infty}$, and we can insert the identity $\mathbb{1}$ between the $e^{i\mathcal{H}\Delta_t}$ s. Assume for the coherent state $|z\rangle$, we have

$$\hat{a}|z\rangle = z|z\rangle, \quad \hat{a}^\dagger|z\rangle = \partial_z|z\rangle$$

also for the invert

$$\langle z|\hat{a} = z \partial_{\bar{z}} \langle z|, \quad \langle z|\hat{a}^\dagger = z \langle z|,$$

and we can define

$$\psi(\bar{z}) = \langle z|\psi\rangle, \quad \text{and} \quad \bar{\psi}(z) = \langle\psi|z\rangle \quad (5.21)$$

Then the integral can be written as

$$I = \int \frac{dz d\bar{z}}{2\pi i} e^{-z\bar{z}} |z\rangle \langle z| \quad (5.22)$$

then, after inserting the identity $\int \frac{dz(t_n)d\bar{z}(t_n)}{2\pi i}$, we have

$$\cdots |z_{t_{n-1}}\rangle \langle z_{t_{n-1}}| \exp(i\mathcal{H}(a^\dagger, a)\Delta t) |z_{t_n}\rangle \langle z_{t_n}| e^{i\mathcal{H}} \cdots$$

where $\langle z_{t_{n-1}}| \exp(i\mathcal{H}(a^\dagger, a)\Delta t) |z_{t_n}\rangle$ is normal ordering, we can rewrite it as

$$\langle z| f(a_i^\dagger a_j^\dagger \cdots a_i a_j) |z'\rangle = f(\bar{z}_i, \bar{z}_j, \cdots z_i, z_j \cdots)$$

i.e., $a^\dagger \rightarrow \bar{z}$, $a \rightarrow z$. For Δt , the normal ordering term can be expressed as

$$\begin{aligned} \langle z_{t_{n-1}}| 1 - \frac{i}{\hbar} \Delta t H(a^\dagger, a) |z_{t_n}\rangle &= \langle z_{t_{n-1}}| z_{t_n}\rangle \left(1 - \frac{i}{\hbar} \Delta \mathcal{H}(\bar{z}_{t_{n-1}}, z_{t_n}) \right) \\ &= e^{\bar{z}_{t_{n-1}} z_{t_n}} e^{-|z_{t_{n-1}}|^2/2} e^{-|z_{t_n}|^2/2} (1 - \cdots) \\ &= \int \mathcal{D}z \mathcal{D}\bar{z} \exp \left[\frac{i}{\hbar} \int_{t_i}^{t_f} \frac{\hbar}{2i} (z \partial_t \bar{z} - \bar{z} \partial_t z) - H(\bar{z}, z) \right] \\ &= \cdots \exp \left[\frac{1}{2} (|z_i|^2 + |z_f|^2) \right] \bar{\psi}_f(z_f) \psi_i(\bar{z}_i) \end{aligned} \quad (5.23)$$

5.10.2 Fermionic coherent P.I.

For the fermions, it satisfy the anti-commutation

$$\{c_i c_j^\dagger\} = \delta_{ij}, \{c_i, c_j\} = 0 \quad (5.24)$$

then for the coherent state, we have

$$c_i |\xi_i\rangle = \xi_i |\xi_i\rangle, \quad (5.25)$$

where $|\xi_i\rangle = a|0\rangle + b|1\rangle$. Then, the action can be written into the number state

$$c|\xi_i\rangle = b_0 \quad (5.26)$$

where $b = \xi_i$. We also have the anti-commutator

$$\{\xi_\alpha, \xi_\beta\} = 0 \quad (5.27)$$

We get the Grassmann-variables ξ_i , which satisfy

$$\xi_\alpha^2 = 0, \quad f(\xi_\alpha) = a + b\xi_\alpha, \quad e^{\xi_\alpha} = 1 + \xi_\alpha, \quad \int d\xi 1 = 0, \quad \int d\xi \xi = 1, \quad \frac{\partial}{\partial \xi}(\xi^*) \xi = \frac{\partial}{\partial \xi} - \xi \xi^* = -\xi^* \quad (5.28)$$

We can have the table of Gaussian $[a_\alpha, a_\beta^\dagger] - \xi = \delta_{\alpha\beta}$, $|\xi\rangle = e^\xi \sum_\alpha \xi_\alpha a_\alpha^\dagger |0\rangle$, $a|\xi\rangle = \xi|\xi\rangle$

$\mathbb{I} = \cdots$

$$\int \mathcal{D}[\xi_\alpha, \xi_\beta^*] \exp \left[- \sum_{\alpha\beta} \xi_\alpha^* \mathcal{H}_{\alpha\beta} \xi_\beta + \sum_\alpha (\eta_\alpha \xi_\alpha^* + \eta_\alpha^* \xi_\alpha) \right] = [\det, \mathcal{H}]^{-\xi} \sum_{\alpha\beta} \eta_\alpha^* \mathcal{H}_{\alpha\beta}^{-1} \eta_\beta \quad (5.29)$$

we will discuss later: by taking $\frac{\partial}{\partial \eta} \frac{\partial}{\partial \eta^*}$ with $\eta \rightarrow 0$ and $\eta^* \rightarrow 0$ to the expression above.

5.10.3 spin-coherent state

Example : Wess-Zumino-Witten term & Haldane conjecture with S^+ , S^- , and $|S, \mu\rangle$

We can start from the state

$$|0\rangle = |SS\rangle, \quad S_z|0\rangle = S|0\rangle, \quad \hat{S}^2|0\rangle = S(S+1)|0\rangle \quad (5.30)$$

where S is the magnetic number. The three-dimensional vector $\mathbf{n}$

$$|\mathbf{n}\rangle = e^{i\theta(\mathbf{n}_0 \times \mathbf{n}) \cdot \mathbf{S}} |SS\rangle \quad (5.31)$$

where $\mathbf{n} \cdot \mathbf{n}_0 = \cos \theta$, and $\mathbf{n}_0 = (0, 0, 1)$ is a constant vector point to north.

$$|\mathbf{n}\rangle = \sum_{M=-S}^{M=S} \mathcal{D}(\mathbf{n})_{MS} |SM\rangle \quad (5.32)$$

and the inner product

$$\langle \mathbf{n}_1 | \mathbf{n}_2 \rangle = \left[\frac{(1 + \mathbf{n}_1 \mathbf{n}_2)}{2} \right]^S e^{i\Phi(\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_0)S} \quad (5.33)$$

The vector basis $\{\mathbf{n}_i\}$ can be expressed in a Bloch-sphere, while the identity

$$\mathbb{1} = \int d\mu[\mathbf{n}] |\mathbf{n}\rangle \langle \mathbf{n}| \quad (5.34)$$

where

$$d\mu(\mathbf{n}) = \left(\frac{2S+1}{4\pi} \right) d^3n \delta(\mathbf{n}^2 - 1) \quad (5.35)$$

Now, start from

$$\langle \mathbf{n}(t) | e^{i\Delta t \mathcal{H}} | \mathbf{n}(t_{j+1}) \rangle = (\langle \mathbf{n}(t_j) | \mathbf{n}(t_{j+1}) \rangle + i\Delta t \langle \mathbf{n}(t_j) | \mathcal{H} | \mathbf{n}(t_{j+1}) \rangle)$$

and

$$\frac{\langle \mathbf{n}(t_j) | \mathcal{H} | \mathbf{n}(t_{j+1}) \rangle}{\langle \mathbf{n}(t_j) | \mathbf{n}(t_{j+1}) \rangle} \cong \langle \mathbf{n}(t_j) | \mathcal{H} | \mathbf{n}(t_j) \rangle + O(\Delta t)$$

combine them together

$$\langle \mathbf{n}(t_j) | \mathbf{n}(t_{j+1}) \rangle = e^{i\Phi S} \left(1 + \frac{\mathbf{n}(t_j) \cdot \mathbf{n}(t_{j+1})}{2} \right)^S \quad (5.36)$$

then we have

$$iS[\mathbf{n}] = iS \sum_{j=1}^{N_t} \Phi(\mathbf{n}_0, \mathbf{n}(t_j), \mathbf{n}(t_{j+1})) + S \sum \ln \left(1 + \frac{\mathbf{n} \cdot \mathbf{n}}{2} \right) - \langle \mathbf{n} | H | \mathbf{n} \rangle \quad (5.37)$$

For a series of magnetic momentum $\mathbf{n}(t)$ along a chain, P.I. to the Hamiltonian is

$$\int d\mathbf{n}_i(t) e^{iJ \sum_i \mathbf{S}_i \mathbf{S}_{i+1}}$$

and we can see the P.I. approach the phase term

$$\frac{S}{2} \underbrace{\int dt \mathbf{n}(t) \cdot (\partial_t \mathbf{n} \times \partial_x \mathbf{n})}_{4\pi} = S$$

For $S = \frac{2n+1}{2}$, it behaves as a “crossing” (X); when $S = n$, it behaves as a gapped $\frown$.

$$I_x \mathbf{S}_i^x \mathbf{S}_j^x + I_y \mathbf{S}_i^x \mathbf{S}_j^y \longrightarrow S_i^x S_j^- + S_i^- S_j^+ + S^+ S^+ + S^- S^-$$

5.10.4 Review of Methodology & Source field method

(a) H-Q.M. GML

$$|\Phi\rangle = \mathcal{T}_0 \underbrace{\hat{S}|\Phi_0\rangle}_{\langle\Phi_0|\mathcal{T}_0\hat{S}|\Phi_0\rangle} \quad (5.38)$$

Then the inner product

$$\langle\Phi|\mathcal{T}\psi(1)\psi^\dagger(2)|\Phi\rangle = \underbrace{\langle\Phi|\mathcal{T}_0\hat{S}\psi(1)\psi^\dagger(2)|\Phi_0\rangle}_{\langle\Phi_0|\mathcal{T}_0\hat{S}|\Phi_0\rangle} \quad (5.39)$$

The operator $\hat{S}$ can be expressed as

$$\mathcal{S} = \mathcal{T}_0 \exp\left[-i \int V(t') dt'\right] = \mathcal{T}_0 \sum_n (\sim \int dt_1 \cdots dt_n) \quad (5.40)$$

(b) P.I. compute the S -matrix scattering

$$\langle\Phi_0|S|\Phi_0\rangle = \int \mathcal{D}[\psi, \psi^\dagger] \exp\left(\frac{i}{\hbar} \int dt \mathcal{L}^{\text{Gaussian}} - V\right) \quad (5.41)$$

we can map $t \rightarrow i\tau$. The question is how to get

$$\langle\Phi_0|S\psi(1)\psi^\dagger(2)|\Phi_0\rangle \quad (5.42)$$

(c) $i \partial_t \langle\Psi|\mathcal{T}_0\psi(1)\psi^\dagger(2)|\Psi\rangle = \delta[1-2] + \langle[\psi(1), H]|\rangle$, Then we will get the derivative of Green function

$$i \partial_t G = \delta[\] + G_k^0 G + \underbrace{\langle[\psi(1), V], \psi^\dagger(2)\rangle}_{\text{Vertex-function}}$$

from which we can get *Schrödinger-Dyson-EOM*. Link through source field method

$$H^{\text{source}} = H + \sum_{t,j} \nu_j(t) \hat{\eta}|_{\nu_\eta(t) \rightarrow 0} \quad (5.43)$$

Consider add a source of c , and $\hat{\eta}_1 = c_i$, $\hat{\eta}_2 = c_j^\dagger$. Then we can computer the Green function by taking the functional derivative

$$\langle|c_i(t)c_j^\dagger(t')|\rangle = \frac{\delta}{\delta\nu_{\eta_1}(t), \nu_{\eta_2}(t')} \langle|\hat{S}|\rangle \Big|_{\text{source}} \Big|_{\nu \rightarrow 0} \quad (5.44)$$

we can define

$$\Gamma = \langle\hat{O}\psi(1); \psi^\dagger(2)\rangle = \frac{\delta}{\delta\nu_{\hat{O}}} \langle\psi(1); \psi^\dagger(2)\rangle \Big|_{\text{Source}} \Big|_{\nu \rightarrow 0} \quad (5.45)$$

For $H_n = \hat{n}_\uparrow \hat{n}_\downarrow$, $\Gamma = u \langle n_\downarrow(1) c_\uparrow(1); c_\uparrow^\dagger(2) \rangle$. Then the Fourier transformation $\mathcal{F}[\langle c_\uparrow(1), c_\uparrow^\dagger(2) \rangle]$.

For exact G_0 , We can using the skeleton diagram to generate $G_0 + G_0 G_\alpha$.

The Green basically contains

$$G[\] = \Delta E_{(n-m)} \langle\Psi|\psi(x)\psi^\dagger(y)|\psi\rangle, \quad (5.46)$$

and $|\Psi\rangle, (\rho), \Sigma$.

Homework #4 [2025-09-23]

Problem 4.1 (Exercise 5.1 of Coleman). A particle with $S = \frac{1}{2}$ is placed in a large magnetic field $\mathbf{B} = (B_1 \cos \omega t, B_1 \sin \omega t, B_0)$, where $B_0 \gg B_1$.

- Treating the oscillating part of the Hamiltonian as the interaction, write down the Schrödinger equation in the interaction representation.
- Find $\hat{U}(t) = \mathcal{T} \exp[-i \int_{-\infty}^t \mathcal{H}_{\text{int}}(t') dt']$ by whatever method proves most convenient.
- If the particle starts at time $t = 0$ in the state $S_z = -\frac{1}{2}$, what is the probability it is in this state at time t ?

Solution.

- The Hamiltonian is

$$\hat{H}(t) = -\gamma \mathbf{S} \cdot \mathbf{B}(t) = -\gamma B_0 S_z - \gamma B_1 (S_x \cos \omega t + S_y \sin \omega t),$$

with $\mathbf{S} = \frac{\hbar}{2} \boldsymbol{\sigma}$. Denote $\hat{H}_0 = -\gamma B_0 S_z$ and $\hat{H}_1(t) = -\gamma B_1 (S_x \cos \omega t + S_y \sin \omega t)$, we obtain

$$\hat{H}_1(t) = -\frac{\gamma B_1}{2} (S_+ e^{-i\omega t} + S_- e^{i\omega t}).$$

where $S_{\pm} = S_x \pm iS_y$. In the interaction picture, the interaction Hamiltonian is expressed as

$$\hat{H}_{1,I}(t) = e^{i\hat{H}_0 t/\hbar} \hat{H}_1(t) e^{-i\hat{H}_0 t/\hbar},$$

with the BCH theory, using $e^{i\hat{H}_0 t/\hbar} S_{\pm} e^{-i\hat{H}_0 t/\hbar} = S_{\pm} e^{\mp i\omega_0 t}$, where $\omega_0 = \gamma B_0$, we obtain

$$\hat{H}_{1,I}(t) = -\frac{\gamma B_1}{2} (S_+ e^{-i(\omega+\omega_0)t} + S_- e^{i(\omega+\omega_0)t}).$$

So, the Schrödinger equation in the interaction representation is

$$i\hbar \frac{d|\psi_I(t)\rangle}{dt} = -\frac{\gamma B_1}{2} (S_+ e^{-i(\omega+\omega_0)t} + S_- e^{i(\omega+\omega_0)t}) |\psi_I(t)\rangle,$$

- Let $\Delta = \omega + \omega_0$, $\Omega = -\frac{\gamma B_1}{2}$. Then

$$\hat{H}_{1,I}(t) = \Omega (S_+ e^{-i\Delta t} + S_- e^{i\Delta t}) = 2\Omega e^{-i\Delta t S_z} S_x e^{i\Delta t S_z}.$$

The time-evolution operator is

$$\hat{U}(t) = \mathcal{T} \exp \left[-\frac{i}{\hbar} \int_{-\infty}^t \hat{H}_{1,I}(t') dt' \right].$$

Using the identity for time-ordered exponentials of this form, we find

$$\hat{U}(t) = e^{-i\Delta t S_z} \exp \left[-\frac{i}{\hbar} (2\Omega S_x - \Delta S_z) t \right] = e^{-i(\omega+\omega_0)t S_z} \exp \left[\frac{i}{\hbar} t (\gamma B_1 S_x + (\omega + \omega_0) S_z) \right].$$

(c) Assume in the interaction picture, $|\psi_I(0)\rangle = |\downarrow\rangle$. The amplitude to remain in $|\downarrow\rangle$ is

$$A(t) = \langle\downarrow|\hat{U}(t)|\downarrow\rangle.$$

Let $\Omega_R = \gamma B_1$, $\tilde{\Delta} = \omega + \omega_0$, and define the effective Hamiltonian according to (b)

$$H_{\text{eff}} = -\Omega_R S_x - \tilde{\Delta} S_z,$$

then

$$\hat{U}(t) = e^{-i\tilde{\Delta}tS_z} e^{-iH_{\text{eff}}t/\hbar}.$$

Since $|\downarrow\rangle$ is an eigenstate of S_z with eigenvalue $-\hbar/2$,

$$e^{-i\tilde{\Delta}tS_z}|\downarrow\rangle = e^{i\tilde{\Delta}t/2}|\downarrow\rangle,$$

so

$$A(t) = e^{i\tilde{\Delta}t/2} \langle\downarrow|e^{-iH_{\text{eff}}t/\hbar}|\downarrow\rangle, \quad \text{and} \quad H_{\text{eff}} = -\frac{1}{2}(\Omega_R \sigma_x + \tilde{\Delta} \sigma_z).$$

The eigenvalues of H_{eff} are $\pm \frac{\lambda}{2}$, where $\lambda = \sqrt{\Omega_R^2 + \tilde{\Delta}^2}$. So

$$A(t) = e^{i\tilde{\Delta}t/2} \left[\cos\left(\frac{\lambda t}{2}\right) - i \frac{\tilde{\Delta}}{\lambda} \sin\left(\frac{\lambda t}{2}\right) \right],$$

and we obtain the survival probability is

$$P(t) = |A(t)|^2 = 1 - \frac{(\gamma B_1)^2}{(\gamma B_1)^2 + (\omega + \gamma B_0)^2} \sin^2\left(\frac{1}{2}\sqrt{(\gamma B_1)^2 + (\omega + \gamma B_0)^2}t\right).$$

Problem 4.2 (Exercise 3.1.1 of Nolting). For the non-interacting electron gas (H_e) and for the non-interacting phonon gas (H_p),

$$H_e = \sum_{k,\sigma} \epsilon(k) a_{k\sigma}^\dagger a_{k\sigma}; \quad \text{and} \quad H_p = \sum_{q,r} \hbar \omega_r(q) \left(b_{qr}^\dagger b_{qr} + \frac{1}{2} \right),$$

compute the time dependence of the annihilation operators $a_{k\sigma}(t)$, $b_{qr}(t)$ in the Heisenberg picture.

Solution. We just need to apply EOM to $a_{k\sigma}$ and b_{qr} .

(a) Annihilation operator $a_{k\sigma}(t)$

Calculate the commutator $[H_e, a_{k\sigma}]$ first

$$\begin{aligned} [H_e, a_{k\sigma}] &= \sum_{k',\sigma'} \epsilon(k') [a_{k'\sigma'}^\dagger a_{k'\sigma'}, a_{k\sigma}] \\ &= \sum_{k',\sigma'} \epsilon(k') \left(a_{k'\sigma'}^\dagger [a_{k'\sigma'}, a_{k\sigma}] + [a_{k'\sigma'}^\dagger, a_{k\sigma}] a_{k'\sigma'} \right) = -\epsilon(k) a_{k\sigma}. \end{aligned}$$

Then the EOM becomes

$$\frac{da_{k\sigma}(t)}{dt} = \frac{i}{\hbar} [-\epsilon(k) a_{k\sigma}(t)].$$

It's trivial that

$$a_{k\sigma}(t) = a_{k\sigma}(0) e^{-i\epsilon(k)t/\hbar}.$$

(b) Similarly, the commutator

$$[H_p, b_{qr}] = -\hbar \omega_r(q) b_{qr},$$

and the EOM becomes

$$\frac{db_{qr}(t)}{dt} = -i\omega_r(q) b_{qr}(t),$$

and it's trivial that

$$b_{qr}(t) = b_{qr}(0) e^{-i\omega_r(q)t}.$$

Problem 4.3. Consider an arbitrary **time-dependent** operator in Schrödinger picture $\hat{A}_S(t)$ then re-express $\hat{A}_S(t)$ in Heisenberg picture as $\hat{A}_H(t)$ and in Dirac (interaction) picture as $\hat{A}_D(t)$.

1. Show that $\langle \psi_H | \hat{A}_H(t) | \psi_H \rangle \stackrel{!}{=} \langle \psi_S(t) | \hat{A}_S | \psi_S(t) \rangle$;
2. Derive the Heisenberg equation-of-motion for $\hat{A}_H(t)$ which should read;
3. Similarly, derive the equation-of-motion in Dirac (interaction) picture for $\hat{A}_D(t)$.

Solution.

(a) In Heisenberg picture, $\psi_H = \psi_S(0)$, so we have

$$\langle \psi_H | \hat{A}_H(t) | \psi_H \rangle = \langle \psi_S(0) | \hat{U}^\dagger(t, 0) \hat{A}_S(t) \hat{U}(t, 0) | \psi_S(0) \rangle.$$

Since $|\psi_S(t)\rangle = \hat{U}(t, 0) |\psi_S(0)\rangle$, so

$$\langle \psi_S(t) | \hat{A}_S(t) | \psi_S(t) \rangle = \langle \psi_S(0) | \hat{U}^\dagger(t, 0) \hat{A}_S(t) \hat{U}(t, 0) | \psi_S(0) \rangle,$$

which is exactly $\langle \psi_H | \hat{A}_H(t) | \psi_H \rangle$. So

$$\langle \psi_H | A_H(t) | \psi_H \rangle = \langle \psi_S(t) | A_S | \psi_S(t) \rangle.$$

(b) Since the operator in the Heisenberg picture can be expressed as

$$\hat{A}_H = \hat{U}^\dagger(t, 0) \hat{A}_S(t) \hat{U}(t, 0),$$

and its differentiate

$$\frac{d}{dt} \hat{A}_H(t) = \frac{\partial \hat{U}^\dagger}{\partial t} \hat{A}_S(t) \hat{U} + \hat{U}^\dagger \frac{\partial \hat{A}_S(t)}{\partial t} \hat{U} + \hat{U}^\dagger \hat{A}_S(t) \frac{\partial \hat{U}}{\partial t}.$$

Substitute the equation that $\hat{U}$ satisfies under the Heisenberg picture

$$i\hbar \frac{\partial}{\partial t} \hat{U} = \hat{H}_S \hat{U},$$

we have

$$\begin{aligned} \frac{d}{dt} \hat{A}_H &= \left(-\frac{1}{i\hbar} \hat{U}^\dagger H_S \right) \hat{A}_S \hat{U} + \hat{U}^\dagger \frac{\partial \hat{A}_S(t)}{\partial t} \hat{U} + \hat{U}^\dagger \hat{A}_S \left(\frac{1}{i\hbar} H_S \hat{U} \right) \\ &= \frac{1}{i\hbar} [-\hat{U}^\dagger H_S \hat{A}_S \hat{U} + \hat{U}^\dagger \hat{A}_S H_S \hat{U}] + \hat{U}^\dagger \dot{\hat{A}}_S \hat{U}, \end{aligned}$$

where $-\hat{U}^\dagger H_S \hat{A}_S \hat{U} = -H_H \hat{A}_H$, $\hat{U}^\dagger \hat{A}_S H_S \hat{U} = \hat{A}_H H_H$, and $\hat{U}^\dagger \dot{\hat{A}}_S \hat{U} = \left(\frac{\partial \hat{A}_S}{\partial t} \right)_H$. So we obtain the Heisenberg equation of motion

$$\frac{d}{dt} A_H(t) = \frac{1}{i\hbar} [\hat{A}_H(t), H_H(t)] + \left(\frac{\partial A_S}{\partial t} \right)_H.$$

(c) Similarly from

$$A_I(t) = \hat{U}_0^\dagger(t, 0) \hat{A}_S(t) \hat{U}_0(t, 0),$$

we have the differentiate

$$\frac{d\hat{A}_I}{dt} = \frac{\partial \hat{U}_0^\dagger}{\partial t} \hat{A}_S \hat{U}_0 + \hat{U}_0^\dagger \frac{\partial \hat{A}_S}{\partial t} \hat{U}_0 + \hat{U}_0^\dagger \hat{A}_S \frac{\partial \hat{U}_0}{\partial t}.$$

Substitute the equation that $\hat{U}$ satisfies

$$i\hbar \frac{\partial \hat{U}_0}{\partial t} = H_{0,S} \hat{U}_0,$$

we have

$$\frac{d\hat{A}_I}{dt} = -\frac{1}{i\hbar} \left[\hat{U}_0^\dagger H_{0,S} \hat{A}_S \hat{U}_0 + \hat{U}_0^\dagger \dot{\hat{A}}_S \hat{U}_0 + \frac{1}{i\hbar} \hat{U}_0^\dagger \hat{A}_S H_{0,S} \hat{U}_0 \right].$$

Since $H_{0,I}(t) = \hat{U}_0^\dagger H_{0,S} \hat{U}_0$ and $\hat{A}_I = \hat{U}_0^\dagger \hat{A}_S \hat{U}_0$, we obtain the Dirac picture equation of motion

$$\frac{d}{dt} A_I(t) = \frac{1}{i\hbar} [A_I(t), H_{0,I}(t)] + \left(\frac{\partial \hat{A}_S}{\partial t} \right)_I.$$

Homework #5 [2025-10-09]

Problem 5.1. Verify the key step for proving the GML theorem:

$$\left(\sum_{j=1}^n \frac{\partial}{\partial t_j} \right) \longrightarrow n \frac{\partial}{\partial t_n}$$

explicitly. Consider $n = 2$, and do the following

- Carry out the integration-by-path for each t_j .
- Show that for all j 's the final results are the same, i.e., independent on j .

Solution.

- Denote

$$F(t_1, t_2) = \langle 0 | \mathcal{T}[\phi_I(t_1) \phi_I(t_2) U] | 0 \rangle,$$

where U is the interaction picture evolution operator and $\mathcal{T}$ is the time-ordering operator. To simplify, define the unordered pieces

$$A(t_1, t_2) = \langle 0 | \phi_I(t_1) \phi_I(t_2) U | 0 \rangle, \quad \text{and} \quad B(t_1, t_2) = \langle 0 | \phi_I(t_2) \phi_I(t_1) U | 0 \rangle.$$

Then, by definition of $\mathcal{T}$, we have

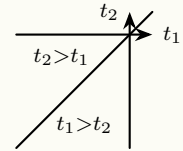
$$F(t_1, t_2) = \theta(t_1 - t_2) A(t_1, t_2) + \theta(t_2 - t_1) B(t_1, t_2),$$

where θ is the Heaviside function. Now act $(\partial_{t_1} + \partial_{t_1})$ on F

$$\begin{aligned} \frac{\partial F}{\partial t_1} + \frac{\partial F}{\partial t_2} &= \theta(t_1 - t_2) \left(\frac{\partial A}{\partial t_1} + \frac{\partial A}{\partial t_2} \right) + \theta(t_2 - t_1) \left(\frac{\partial B}{\partial t_1} + \frac{\partial B}{\partial t_2} \right) + \delta(t_1 - t_2)(A - B) \\ &= \theta(t_1 - t_2)(\partial_{t_1} A + \partial_{t_2} A) + \theta(t_2 - t_1)(\partial_{t_1} B + \partial_{t_2} B), \end{aligned}$$

where the last term vanishes since $(A - B)|_{t_1=t_2} = 0$. Then, consider the integration

$$\begin{aligned} I &= \int_{-\infty}^0 dt_1 \int_{-\infty}^0 dt_2 (\partial_{t_1} + \partial_{t_2}) F(t_1, t_2) \\ &= \underbrace{\int_{-\infty}^0 dt_2 \int_{t_2}^0 dt_1 (\partial_{t_1} A + \partial_{t_2} A)}_{\text{Region 1: } t_1 > t_2} + \underbrace{\int_{-\infty}^0 dt_1 \int_{t_1}^0 dt_2 (\partial_{t_1} B + \partial_{t_2} B)}_{\text{Region 2: } t_2 > t_1}. \end{aligned}$$



Consider Region I

$$\begin{aligned} I_1 &= \int_{-\infty}^0 dt_2 \int_{t_2}^0 dt_1 \partial_{t_1} A(t_1, t_2) + \int_{-\infty}^0 dt_2 \int_{t_2}^0 dt_1 \partial_{t_2} A(t_1, t_2) \\ &\stackrel{\text{Leibniz integral rule}}{=} \int_{-\infty}^0 dt_2 [A(0, t_2) - A(t_2, t_2)] + \int_{-\infty}^0 dt_2 \left[A(t_2, t_2) - \frac{d}{dt_2} \left(\int_0^{t_2} A(t_1, t_2) dt_1 \right) \right] \\ &= \int_{-\infty}^0 dt_2 A(0, t_2) - \left[\int_0^0 A(t_1, 0) dt_1 - \int_0^{-\infty} A(t_1, -\infty) dt_1 \right] = \int_{-\infty}^0 dt_2 A(0, t_2). \end{aligned}$$

The last term vanishes due to the boundary condition. Similarly, for Region II, the integration can be written as

$$I = \int_{-\infty}^0 dt_2 A(0, t_2) + \int_{-\infty}^0 dt_1 B(0, t_1) = \int_{-\infty}^0 dt_2 A(0, t_2) + \int_{-\infty}^0 dt_1 A(t_1, 0) = 2 \int_{-\infty}^0 dt_1 A(0, t_1).$$

In the last step, both the integral region and the variables are switched. Since

$$F(t_1, 0) = \theta(t_1 - 0)A(t_1, 0) + \theta(0 - t_1)B(t_1, 0) = A(0, t_1),$$

integral over t_2 to $\partial_{t_2} F(t_1, t_2)$

$$\int_{-\infty}^0 dt_2 \partial_{t_2} F(t_1, t_2) = F(t_1, 0) - F(t_1, -\infty) = F(t_1, 0) = A(0, t_1).$$

So, we have

$$I = 2 \int_{-\infty}^0 dt_1 \int_{-\infty}^0 dt_2 \partial_{t_2} F(t_1, t_2) = \int_{-\infty}^0 dt_1 \int_{-\infty}^0 dt_2 (\partial_{t_1} + \partial_{t_2}) F(t_1, t_2).$$

Then we obtain

$$\partial_{t_1} + \partial_{t_2} = 2 \partial_{t_2}, \quad \text{i.e.,} \quad \left(\sum_{j=1}^2 \partial_{t_j} \right) \longrightarrow 2 \partial_{t_2}.$$

(b) Denote $T = \frac{1}{2}(t_1 + t_2)$, $\tau = t_1 - t_2$. Holding τ fixed,

$$\partial_{t_1}|_{\tau} = \frac{\partial}{\partial T} \frac{\partial T}{\partial t_1} = \frac{1}{2} \frac{\partial}{\partial T}, \quad \text{and} \quad \partial_{t_2}|_{\tau} = \frac{\partial}{\partial T} \frac{\partial T}{\partial t_2} = \frac{1}{2} \frac{\partial}{\partial T}$$

So, we have $2 \frac{\partial}{\partial t_1}|_{\tau} = 2 \frac{\partial}{\partial t_2}|_{\tau} = \partial_T$, hence the result is independent of which t_j is chosen.

Problem 5.2 (The two-spin problem). Consider two spin-1/2's, $\hat{s}_a = \hat{s}_b = \frac{1}{2}$, which are coupled to each other via an exchange interaction J and are presumed to be acted upon by a homogeneous magnetic field. We describe them in terms of the correspondingly simplified Heisenberg model

$$\mathcal{H}_0 = \left(\frac{J_{\perp}}{2} (\hat{s}_a^{++} \hat{s}_b^{--} + \hat{s}_a^{--} \hat{s}_b^{++}) + J_z \hat{s}_a^z \hat{s}_b^z \right) - h_z (\hat{s}_a^z + \hat{s}_b^z).$$

The parameter choice is to ensure that when $J_{\perp} = J_z$ the first term reduces to the Heisenberg interaction. For this problem, we take $J_{\perp} = 0$, which is called the Ising limit, $b = 0$ and also assume $J_z > 0$ (for this problem, the sign does not make much difference, but it will be important when we turn on $J_{\perp}$). Next, consider a transverse field as the perturbation

$$\delta \mathcal{H} = -h_x (\hat{s}_a^x + \hat{s}_b^x).$$

Consider the two-spin COBS, $U_{g2} = \{\hat{I}, \hat{S}^{\alpha}, \hat{\eta}, \hat{B}_S^{\alpha}, \hat{B}_A^{\alpha}, \hat{D}^{\alpha}\}$ defined as

$$\begin{aligned} \hat{S}_{ab}^{\alpha} &= \hat{s}_a^{\alpha} + \hat{s}_b^{\alpha}, \quad \hat{\eta}_{ab}^{\alpha} = \hat{s}_a^{\alpha} - \hat{s}_b^{\alpha}, \\ \hat{B}_{A,ab}^{\gamma} &= 2\epsilon_{\alpha\beta\gamma} \hat{s}_a^{\alpha} \hat{s}_b^{\beta}, \\ \hat{B}_{S,ab}^{\gamma} &= 2\epsilon_{\alpha\beta\gamma}^2 \hat{s}_a^{\alpha} \hat{s}_b^{\beta}, \\ \hat{D}_{ab}^{\alpha} &= 2\hat{s}_a^{\alpha} \hat{s}_b^{\alpha}, \end{aligned}$$

where the Einstein summation notation over dummy indices is assumed.

- Determine the exact ground state(s) of $\mathcal{H}_0$, specify the degeneracy, if any, by diagonalization, i.e., rewrite $\mathcal{H}_0$ as a 4×4 matrix.
- Parameterize the degenerate ground state subspace; compute expectation values of COBS elements for a parameterized state; try to use symmetry as a guide, only 5 out of the 15 operators have none-zero values; for two-body operators, use the cumulant expression.
- Determine the exact ground state energy and the ground state wavefunction for $\mathcal{H} = \mathcal{H}_0 + \delta\mathcal{H}$, assuming $J_z > 4h_x > 0$; also compute expectation values for the COBS elements; carry out the Taylor expansion for $\langle \hat{S}^x \rangle$ to the first order, to determine the transverse field susceptibility $\chi_x = \frac{\partial \langle \hat{S}^x \rangle}{\partial h_x} \Big|_{h_x \rightarrow 0}$; determine the ground state in the same limit $|\Psi\rangle|_{h_x \rightarrow 0^+}$.
- Try to use the perturbative VES Eq. (5.6.10)

$$0 = \langle [\delta\mathcal{H}, \hat{u}^{\alpha 0}] \rangle_0 + \sum_{\alpha} \delta \langle \hat{u}^{\alpha} \rangle \langle \{ \hat{u}^{\alpha}, [\mathcal{H}, \hat{u}^{\alpha 0}] \} \rangle_0 + \sum_{\gamma\beta} \langle \delta \hat{u}^{\gamma} \rangle \langle \delta \hat{u}^{\beta} \rangle \langle \{ \hat{u}^{\gamma}, [\mathcal{H}, \{ \hat{u}^{\alpha 0}, \hat{u}^{\beta} \}] \} \rangle_0 + \dots, \quad (5.6.10)$$

to do the computation, starting from a parameterized state, and express the transverse field susceptibility χ_x as a function of COBS expectation values; determine the parameterized state with the largest χ_x and compare with the diagonalization results you obtained in the previous question.

- You should find that the fully entangled triplet state has the largest χ_x , try to discuss the reason based on your calculation results.

Solution. Since we take $J_{\perp} = 0$, and omit the perturbation term. The Hamiltonian becomes

$$\mathcal{H}_0 = J_z \hat{s}_a^z \hat{s}_b^z.$$

- Recall $\hat{s}^z = \frac{1}{2}\sigma^z$, then $\hat{s}_a^z \hat{s}_b^z = \frac{1}{4}\sigma_a^z \sigma_b^z$. Taking the basis

$$\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}.$$

Therefore, the eigenenergies, i.e., the diagonal matrix elements, are

$$E(\uparrow\uparrow) = \frac{J_z}{4}, \quad E(\uparrow\downarrow) = -\frac{J_z}{4}, \quad E(\downarrow\uparrow) = -\frac{J_z}{4}, \quad E(\downarrow\downarrow) = \frac{J_z}{4}.$$

So, $\mathcal{H}_0$ can be written into a 4×4 diagonal matrix

$$\mathcal{H}_0 = \begin{pmatrix} \frac{J_z}{4} & & & \\ & -\frac{J_z}{4} & & \\ & & -\frac{J_z}{4} & \\ & & & \frac{J_z}{4} \end{pmatrix}.$$

The ground states are $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$. Since the two lowest eigenvalues are equal, the degeneracy is 2.

(b) Parameterize a normalized general state in the ground subspace as

$$|\psi(\theta, \phi)\rangle = \cos \frac{\theta}{2} |\uparrow\downarrow\rangle + e^{i\phi} \sin \frac{\theta}{2} |\downarrow\uparrow\rangle.$$

The general is normalized. The non-zero values of the 5 COBS elements are listed as follows

$$\begin{aligned} \hat{\eta}^z &= \hat{s}_a^z - \hat{s}_b^z, & \langle \hat{\eta}^z \rangle &= |\cos(\theta/2)|^2 - |e^{i\phi} \sin(\theta/2)|^2 = \cos \theta, \\ \hat{D}^x &= 2\hat{s}_a^x \hat{s}_b^x, & \langle \hat{D}^x \rangle &= \text{Re}\{\cos(\theta/2)^* e^{i\phi} \sin(\theta/2)\} = \frac{1}{2} \sin \theta \cos \phi, \\ \hat{D}^y &= 2\hat{s}_a^y \hat{s}_b^y, & \langle \hat{D}^y \rangle &= \langle \hat{D}^x \rangle = \frac{1}{2} \sin \theta \cos \phi, \\ \hat{D}^z &= 2\hat{s}_a^z \hat{s}_b^z, & \langle \hat{D}^z \rangle &= -\frac{1}{2}, \\ \hat{B}_A^z &= 2(\hat{s}_a^x \hat{s}_b^y - \hat{s}_a^y \hat{s}_b^x), & \langle \hat{B}_A^z \rangle &= -2 \text{Im}\{\cos(\theta/2)^* e^{i\phi} \sin(\theta/2)\} = -\sin \theta \sin \phi. \end{aligned}$$

(c) The total Hamiltonian is

$$\begin{aligned} \mathcal{H} &= \mathcal{H}_0 + \delta\mathcal{H} = J_z \hat{s}_a^z \hat{s}_b^z - h_x (\hat{s}_a^x + \hat{s}_b^x) = J_z \hat{s}_a^z \hat{s}_b^z - h_x \hat{S}_x \\ &= \begin{pmatrix} \frac{J_z}{4} & -\frac{h_x}{2} & -\frac{h_x}{2} & 0 \\ -\frac{h_x}{2} & -\frac{J_z}{4} & 0 & -\frac{h_x}{2} \\ -\frac{h_x}{2} & 0 & -\frac{J_z}{4} & -\frac{h_x}{2} \\ 0 & -\frac{h_x}{2} & -\frac{h_x}{2} & \frac{J_z}{4} \end{pmatrix} \sim \begin{pmatrix} -\frac{1}{4}\sqrt{J_z^2 + 16h_x^2} & & & \\ & -\frac{J_z}{4} & & \\ & & +\frac{J_z}{4} & \\ & & & +\frac{1}{4}\sqrt{J_z^2 + 16h_x^2} \end{pmatrix}. \end{aligned}$$

Therefore, the exact ground state energy is $E_0 = -\frac{1}{4}\sqrt{J_z^2 + 16h_x^2}$. The exact ground eigenvector (wavefunction) can be expressed as

$$|\Psi_0\rangle = |\uparrow\uparrow\rangle + \left(\frac{J_z}{4h_x} - \frac{\sqrt{J_z^2 + 16h_x^2}}{4h_x} \right) (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) + |\downarrow\downarrow\rangle.$$

The expectation values are

$$\begin{aligned} \langle \hat{S}_x \rangle &= \frac{h_x}{\sqrt{(J_z/4)^2 + h_x^2}}, & \langle \hat{S}^y \rangle &= \langle \hat{S}^z \rangle = 0, & \langle \hat{\eta} \rangle &= 0, \\ \langle \hat{B}_A^\alpha \rangle &= \langle \hat{B}_s^\alpha \rangle = 0, & \langle \hat{D}^x \rangle &= \langle \hat{D}^y \rangle = \frac{1}{2}, & \langle \hat{D}^z \rangle &= \frac{2\lambda^2 - 1}{2(1 + 2\lambda^2)}, \end{aligned}$$

where $\lambda = \frac{\sqrt{(J_z/4)^2 + h_x^2} - J_z/4}{\sqrt{2}h_x}$. Under the limitation of $h_x \rightarrow 0$, the ground energy can be expanded as

$$E_1 \cong -\frac{J_z}{4} - \frac{2h_x^2}{J_z} + \mathcal{O}(h_x^4),$$

and the expectation of $\langle \hat{S}^x \rangle$ under this limitation becomes

$$\lim_{h_x \rightarrow 0} \langle \hat{S}^x \rangle = \frac{4h_x}{J_z},$$

and the transverse field susceptibility $\chi_x = 4/J_z$, the ground state

$$\lim_{h_x \rightarrow 0} |\Psi\rangle = \frac{1}{\sqrt{2}} |\uparrow\uparrow\rangle + \frac{1}{\sqrt{2}} |\downarrow\downarrow\rangle,$$

where the normalization factor is $\frac{1}{\sqrt{2}}$.

(d) We only consider the top 2 orders of the VES equation. Choose $\hat{u}^{\alpha 0} = \hat{S}^x$, then

$$\langle [\delta \mathcal{H}, \hat{S}^x] \rangle_0 = -h_x \langle [\hat{S}^x, \hat{S}^x] \rangle_0 = 0,$$

and the VES equation becomes

$$\sum_{\alpha} \delta \langle \hat{u}^{\alpha} \rangle \langle \{ \hat{u}^{\alpha}, [\mathcal{H}_0, \hat{u}^{\alpha 0}] \} \rangle_0 = 0.$$

i. Compute $[\mathcal{H}_0, \hat{S}^x]$.

$$[\mathcal{H}_0, \hat{S}^x] = J_z [\hat{S}_a^z \hat{S}_b^z, \hat{S}_a^x + \hat{S}_b^x] \frac{[\hat{S}_a^z, \hat{S}_a^x] = i\hat{S}_a^y}{[\hat{S}_b^z, \hat{S}_b^x] = i\hat{S}_b^y} i J_z (\hat{S}_a^y \hat{S}_b^z + \hat{S}_a^z \hat{S}_b^y).$$

ii. Choose the relevant Operator $\hat{u}^{\alpha} = \hat{B}_A^y$.

From symmetry analysis, the operator is expressed as $\hat{B}_A^y = 2(\hat{S}_a^z \hat{S}_b^x - \hat{S}_a^x \hat{S}_b^z)$. Then

$$\langle \{ \hat{B}_A^y, [\mathcal{H}_0, \hat{S}^x] \} \rangle_0 = -i J_z \langle \hat{D}^x \rangle_0, \quad \text{and} \quad \delta \langle \hat{B}_A^y \rangle = -2\delta \langle \hat{S}^x \rangle,$$

where $\hat{D}^x = 2\hat{S}_a^x \hat{S}_b^x$.

Substitute them into the VES equation

$$0 = 0 + \delta \langle \hat{B}_A^y \rangle \cdot (-i J_z \langle \hat{D}^x \rangle_0) = (-2\delta \langle \hat{S}^x \rangle) (-i J_z \langle \hat{D}^x \rangle_0), \quad \langle \hat{S}^x \rangle \cong \delta \langle \hat{S}^x \rangle = \frac{2h_x}{J_z} (1 + 2\langle \hat{D}^x \rangle_0).$$

Then, the transverse susceptibility

$$\chi_x = \frac{\partial \langle \hat{S}^x \rangle}{\partial h_x} = \frac{2}{J_z} (1 + 2\langle \hat{D}^x \rangle_0) = \frac{2}{J_z} \left(1 + 2 \times \frac{1}{2} \right) = \frac{4}{J_z},$$

which is the same as the previous question.

(e) The triplet state $|1, 0\rangle$ maximizes χ_x because it achieves the largest $\langle \hat{D}^x \rangle_0 = 1/2$ in the susceptibility formula $\chi_x = (2/J_z)(1 + 2\langle \hat{D}^x \rangle_0)$, yielding $\chi_x = 4/J_z$. Its symmetric structure provides optimal matrix elements for transverse field coupling.

Problem 5.3 (extra credit, I don't see the answer right way yet but I think both approaches should agree). Use the results of section 5.7.1.5 to compute the propagator of a driven harmonic oscillator discussed in section 5.8.1.1. You may change the creation / annihilation operator representation into the x/p representation if needed; i.e., consider the following Hamiltonian and perturbation,

$$\hat{H}_0 = \omega \left(b^\dagger b + \frac{1}{2} \right), \quad \hat{V}(t) = \bar{z}(t)b(t) + b^\dagger(t)z(t),$$

and prove that the propagator is

$$S[\bar{z}, z] = \left\langle 0 \left| \mathcal{T} \exp \left(-i \int_{-\infty}^{\infty} dt [\bar{z}(t)b(t) + b^\dagger(t)z(t)] \right) \right| 0 \right\rangle = \exp \left[i \int_{-\infty}^{\infty} dt \int_{-\infty}^{\infty} dt' \bar{z}(t) G(t-t') z(t') \right],$$

where $G(t-t') = -i\theta(t-t') e^{-i\omega(t-t')}$.

Remark. Seems all contributions directly step from the factor $\exp\{\frac{i}{\hbar}S[x_c l(t), \dot{x}(t), t]\}$. Connect the contribution to the $G(t - t')$ in the absence of the external drive $z(t)$.

Solution. The time evolution of the creation and annihilation operators in Dirac picture can be written as

$$b_I(t) = b e^{-i\omega t}, \quad \text{and} \quad b_I^\dagger(t) = b^\dagger e^{i\omega t}.$$

The propagator can be written as a vacuum-to-vacuum amplitude in the Dirac picture

$$S[\bar{z}, z] = \left\langle 0 \left| \mathcal{T} \exp \left(-i \int_{-\infty}^{\infty} \hat{V}_I(t) dt \right) \right| 0 \right\rangle,$$

where $\hat{V}_I(t)$ is the Dirac picture perturbation

$$\hat{V}_I(T) = e^{i\hat{H}_0 T} \hat{V}(t) e^{-i\hat{H}_0 T} = \bar{z}(t) b e^{-i\omega t} + b^\dagger z(t) e^{i\omega t}$$

the above expression used the following identities

$$e^{i\hat{H}_0 t} b e^{-i\hat{H}_0 t} \xrightarrow[\text{[i}\hat{H}_0 t, b] = -i\omega t b]{\text{BCH identity}} b e^{-i\omega t}, \quad \text{and} \quad e^{i\hat{H}_0 t} b^\dagger e^{-i\hat{H}_0 t} \xrightarrow[\text{[i}\hat{H}_0 t, b^\dagger] = i\omega t b^\dagger]{\text{BCH identity}} b^\dagger e^{i\omega t}.$$

Then, we compute the vacuum expectation

$$\begin{aligned} \langle 0 | \mathcal{T} [\hat{V}_I(t_1) \hat{V}_I(t_2)] | 0 \rangle &= \theta(t_1 - t_2) e^{-i\omega(t_1 - t_2)} \bar{z}(t_1) z(t_2) + \theta(t_2 - t_1) e^{-i\omega(t_2 - t_1)} \bar{z}(t_2) z(t_1) \\ &= 2\theta(t_1 - t_2) e^{-i\omega(t_1 - t_2)} \bar{z}(t_1) z(t_2). \end{aligned}$$

The $\mathcal{S}$ -matrix can be expanded as

$$\mathcal{S} = \mathcal{T} \exp \left(-i \int_{-\infty}^{\infty} \hat{V}_I(t) dt \right) = 1 + (-i) \int dt_1 \hat{V}_I(t_1) + \frac{(-i)^2}{2} \int dt_1 \int dt_2 \mathcal{T} [\hat{V}_I(t_1) \hat{V}_I(t_2)] + \dots$$

Then, the connected part of the vacuum expectation is

$$\log \langle 0 | \mathcal{S} | 0 \rangle = -\frac{1}{2} \int dt_1 \int dt_2 \langle 0 | \mathcal{T} [\hat{V}_I(t_1) \hat{V}_I(t_2)] | 0 \rangle + \dots = -\int dt_1 \int dt_2 \theta(t_1 - t_2) e^{-i\omega(t_1 - t_2)} \bar{z}(t_1) z(t_2),$$

where only the contractions between b and $b^\dagger$ are non-zero. Therefore

$$\langle 0 | \mathcal{S} | 0 \rangle = \exp \left(-\int dt \int dt' \theta(t - t') e^{-i\omega(t - t')} \bar{z}(t) z(t') \right).$$

We define the retarded Green function

$$G(t - t') = -i\theta(t - t') e^{-i\omega(t - t')}.$$

Then, the propagator can be expressed as

$$\mathcal{S}[\bar{z}, z] = \exp \left[i \int_{-\infty}^{\infty} dt \int_{-\infty}^{\infty} dt' \bar{z}(t) G(t - t') z(t') \right],$$

where $G(t - t') = -i\theta(t - t') e^{-i\omega(t - t')}.$

CHAPTER 6 More Green's Functions

6.1 Green's function

6.1.1 GFs differential Eqs

$$\frac{d^2}{dx^2}f + S(x)f(x) = 0, \quad \text{or} \quad \hat{\mathcal{L}}f = 0 \quad (6.1)$$

then we can construct $\frac{d^2}{dx^2}g = S(x)$, $f = \int dx g \cdot S(x)$.

6.1.2 GF as propagators in Q.M.

From the S.E.

$$i\hbar \partial_t \psi = H\psi$$

we get the G.F. $G = \langle x|U|x' \rangle$. To be specific

$$\left(i\hbar \partial_t + \frac{\hbar^2}{2m} \nabla^2 \right) G_{t,x;t',x'} = \delta(t - t')\delta(x - x') \quad (6.2)$$

which is the EOM, and $G_{t,x;t',x'}$ is the P.I.

6.1.3 One particle GF & real frequency/real time

Consider the indicate label λ , the GF

$$G_{\lambda\lambda'} = -i\langle \phi | \mathcal{T} \psi_\lambda(t) \psi_{\lambda'}^\dagger(t') | \phi \rangle \quad (6.3)$$

where

$$\mathcal{T} \hat{\psi}_\lambda(t) \hat{\psi}_{\lambda'}^\dagger = \begin{cases} \psi_\lambda(t) \psi_{\lambda'}^\dagger(t'), & t > t' \\ \pm \psi_{\lambda'}^\dagger(t') \psi_\lambda(t), & t < t' \end{cases} \quad (6.4)$$

and the Bosons for $+$ and Fermions for $-$. To combine,

$$\mathcal{T} \hat{\psi}_\lambda(t) \hat{\psi}_{\lambda'}^\dagger = \theta(t - t') \psi_\lambda(t) \psi_{\lambda'}^\dagger(t') \pm \theta(t' - t) \psi_{\lambda'}^\dagger(t') \psi_\lambda(t) \quad (6.5)$$

when we take the time derivative,

$$\begin{aligned} i \partial_t G &= \delta(t - t') (\psi_\lambda \psi_{\lambda'}^\dagger \mp \psi_{\lambda'}^\dagger \psi_\lambda) + \langle \mathcal{T}[\psi, it], \psi_{\lambda'} \rangle \\ &= \begin{cases} \delta(t - t') [\psi_\lambda, \psi_{\lambda'}^\dagger] = \delta(t - t') \delta_{\lambda\lambda'}, & \text{Bosons} \\ \delta(t - t') \{ \psi_\lambda, \psi_{\lambda'}^\dagger \} = \delta(t - t') \delta_{\lambda\lambda'}, & \text{Fermions} \end{cases} + \langle \mathcal{T}[\psi, it], \psi_{\lambda'} \rangle \end{aligned} \quad (6.6)$$

Classically, the commutator is the Poisson bracket $\{ \quad \} = \delta_{ij}$. For bosons and fermions, $\delta_{\lambda\lambda'}$ are all canonical operators, also exist for many-body operators.

6.1.4 Linear response problem: double time – GFs for arbitrary operators

$$G_{\hat{A},\hat{B}}^{\text{ret}} = -i\theta(t-t')\langle\langle[\hat{A}(t),\hat{B}(t')]_{\pm}\rangle\rangle \quad (6.7)$$

$$G_{\hat{A},\hat{B}}^{\text{adv}} = +i\theta(t'-t)\langle\langle[\hat{A}(t),\hat{B}(t')]_{\pm}\rangle\rangle, \quad G_{\hat{A},\hat{B}}^{\text{C}} = -i\langle\langle\mathcal{T}_{\pm}(\hat{A}(t)\hat{B}(t'))\rangle\rangle \quad (6.8)$$

where C = Causal = T -ordered, and

$$\langle\langle\hat{A}(t),\hat{B}(t')\rangle\rangle = \langle\Phi|(\hat{A}(t) - \langle\hat{A}(t)\rangle)(\hat{B}(t') - \langle\hat{B}(t')\rangle)|\Phi\rangle = \text{Tr}[\rho\cdots]$$

and $\langle A \rangle = \langle\Phi|A|\Phi\rangle$, also for B .

6.1.5 Spectral Density of GFs

$$S_{AB}(t,t') = \frac{1}{2\pi}\langle[A(t),B(t')]\rangle_{\xi} \quad (6.9)$$

If set $t \rightarrow t'$, $G[0] = S_{AB}(t=t')$, and we can prove the quantities should evolve to

$$\langle A(t)B(t') \rangle = \langle A(t-t')B(0) \rangle$$

6.1.6 Spectral representation of GF

$$H|E_n\rangle = E_n|E_n\rangle, \quad \sum_n |E_n\rangle\langle E_n| = \mathbb{1}, \quad \langle E_n|E_m\rangle = \delta_{nm} \quad (6.10)$$

Take the trace with the exact eigenstates

$$\begin{aligned} \text{Tr}[e^{-\beta H} A(t)B(t')] &= \sum_{nm} \langle E_n|e^{-\beta H} A(t)|E_n\rangle \langle E_m|B(t')|E_m\rangle \\ &= \sum_n e^{-\beta E_n} \langle E_n|A(t')|E_n\rangle \langle E_n|B(t')|E_n\rangle \\ &= \sum_n e^{-\beta E_n} e^{\frac{i}{\hbar}(E_n-E_n)(t-t')} \langle E_n|\hat{A}|E_n\rangle \langle E_n|\hat{B}|E_n\rangle \end{aligned} \quad (6.11)$$

Then we can have

$$S_{AB}[E] = \sum_{n,m} \delta(E - (E_n - E_m))(e^{\beta E} - \epsilon) \langle E_n|B|E_m\rangle \langle E_m|A|E_n\rangle e^{-\beta E_n} \quad (6.12)$$

For the fourier transformation of the step function

$$\begin{aligned} \theta(t-t') &= \frac{i}{2\pi} \int_{-\infty}^{+\infty} dE \frac{e^{-iE(t-t')}}{E + i0^+}, \\ \theta(t'-t) &= \frac{i}{2\pi} \int_{-\infty}^{+\infty} dE \frac{e^{-iE(t-t')}}{E + i0^-} \end{aligned}$$

Then

$$G[E] = \frac{f_e(E\cdots)}{f_2(E\cdots)} = \frac{f_1(E)}{\prod(E-E_i)} = \sum_i \frac{g_i(E)}{E-E_i+i\text{sgn}[E_i]} \quad (6.13)$$

$$f_2(E\cdots) = \sum_n (E^n a_n + E^{n-1} a_n \cdots) = \sum_{i=1}^n (E-E_i) \quad (6.14)$$

Then, the Hilbert transformation of the retard/advanced/canonical GF and S -matrix can be written as

$$G^{\text{ret}}(E) = \int_{-\infty}^{\infty} dE' \frac{S_{AB}(E')}{E - E' + i0^+}, \quad (6.15)$$

$$G^{\text{adv}}(E) = \int_{-\infty}^{\infty} dE' \frac{S_{AB}(E')}{E - E' + i0^-}, \quad (6.16)$$

$$G^{\text{C}}(E) = \int_{-\infty}^{\infty} dE' \frac{S_{AB}(E')}{E - E' + i \operatorname{sgn}[E']0^+}, \quad (6.17)$$

$$S_{AB}(E) = \frac{i}{2\pi} [G_{AB}(E + i0^+) - G_{AB}(E - i0^+)] = -\frac{1}{\pi} \operatorname{Im}[G_{AB}^{\text{ret}}(E)]. \quad (6.18)$$

6.2 GFs continent

For arbitrary operator A and B , which could be both Fermionic or Boson

$$iG_{AB}^{\text{C}} = -i[\theta(t - t')\langle A(t)B(t') \rangle \pm \theta(t' - t)\langle B(t')A(t) \rangle] \quad (6.19)$$

where $\langle A(t)B(t') \rangle$ and $\langle B(t')A(t) \rangle$ is so-called the greater and lesser Green functions: $G^>$, $G^<$, and $[c, c^\dagger]_\xi = 1$. The retarded/advanced Green function are defined as

$$G_{AB}^{\text{ret}} = -i\theta(t - t')\langle [A(t), B(t')]_\xi \rangle \quad (6.20)$$

$$G_{AB}^{\text{adv}} = +i\theta(t' - t)\langle [A(t), B(t')]_{-\xi} \rangle \quad (6.21)$$

Now the solution will like

$$\frac{1}{E - E_i + i \operatorname{sgn}(E)}$$

where $\operatorname{sgn}(E)$ can be 0^+ (retarded Green function) and 0^- (advanced green function). The Heaviside function satisfies the Fourier Transformation

$$\theta(t - t') = \frac{1}{2\pi} \int dx \frac{e^{ix(t-t')}}{x + i0^+}$$

If the state Φ_0 is applied to $G^<$, i.e.,

$$\langle \Phi_0 | B(t')A(t) | \Phi_0 \rangle \rightarrow \operatorname{Tr}[\rho A(t)B(t')]$$

where

$$G[t, t'] = G(t - t', 0) = G(0, t - t')$$

We define

$$S_{AB}(E) = \langle A(t)B(t') \rangle \quad (6.22)$$

Then the retarded Green function becomes

$$G_{AB}^{\text{ret}}(E) = \int dE' \frac{S_{AB}(E')}{E - E' + i0^+} \sim \delta(E - E') \quad (6.23)$$

and we can get $S_{AB}(E)$ from the Green function

$$S_{AB}(E) = \mp \frac{1}{\pi} \operatorname{Im} G^{\text{ret/adv}} \quad (6.24)$$

and $-\frac{1}{\pi} \text{Im}(G[c_k, c_k^\dagger]) = A_k(\omega)$. Typically,

$$S_{AB}(t, t') = \langle A(t)B(t') \rangle, \quad (6.25)$$

$$S_{BA}(t, t') = \langle B(t')A(t) \rangle, \quad (6.26)$$

If one is measuring $k > k_F$, then $\langle c_k^\dagger, c_k \rangle = 0$, $\langle c_k, c_k^\dagger \rangle = 1$; $k < k_F$. $\langle S^+(t)S^-(t') \rangle$, $\langle S^-(t')S^+(t) \rangle$.

6.2.1 Spectral density

The commutator and anticommutator

$$S_{AB}^{(-)} = \langle \{A(t), B(t')\} \rangle,$$

$$S_{AB}^{(+)} = \langle [A(t), B(t')] \rangle,$$

If we define $\tilde{A} = A - \langle A \rangle$, $\tilde{B} = B - \langle B \rangle$, then $S_{AB}^{(+)} = S_{\tilde{A}\tilde{B}}^{(+)}$,

$$S_{AB}^{(\epsilon)}(E) = S_{\tilde{A}\tilde{B}}^{(\epsilon)}(E) + (1 - \epsilon)DS(E) \quad (6.27)$$

which is so-called spectral theorem.

Expand the $\tilde{A}$, $\tilde{B}$ term

$$\langle \{(A - \langle A \rangle)(B - \langle B \rangle)\} \rangle = \langle \{A, B\} \rangle - 2\langle A \rangle \langle B \rangle + \langle A \rangle \langle B \rangle$$

where we can define

$$D = -2\langle A \rangle \langle B \rangle = \frac{1}{Z} \sum_{n,m}^{E_n=E_m} e^{-\beta E_n} \langle E_n | B | E_m \rangle \langle E_m | A | E_n \rangle \quad (6.28)$$

We can get the relation between S

$$S[A, B] = \frac{1}{2}(S^{(-)} + S^{(+)}), \quad S[B, A] = \frac{1}{2}(S^{(-)} - S^{(+)})$$

If taking the limit

$$\lim_{E \rightarrow 0} EG^{(\epsilon)}(E) = (1 - \epsilon)D$$

Keldysh-formula.

From the EOM of $\tilde{G}^\pm$

$$i \partial_t \tilde{G}^{+,c} = \delta(t) \langle [A, B] \rangle + \Gamma^+(ee) = \delta(t) \tilde{G}^{(-)}(t=0) + \Gamma^+ \quad (6.29)$$

$$i \partial_t \tilde{G}_{AB}^{-,c} = \delta(t) \langle \{A, B\} \rangle + \Gamma^- \quad (6.30)$$

The COBS are $\{\hat{u}^\alpha\}$, where $u^\alpha u^\beta = f_\gamma^{\alpha\beta} u^\gamma$. The commutator

$$\langle [u^\alpha, u^\beta] \rangle = a^{\alpha\beta}, \quad \langle \{u^\alpha, u^\beta\} \rangle = c_\gamma^{\alpha\beta} u^\gamma$$

Since $[\langle S^+, S^z \rangle] = \langle S^x \rangle$.

6.3 Finite- T Green function & Matsubara method

Denote $\langle \quad \rangle_{\text{thermal}}$, then acting

$$\mathcal{T} e^{i \int dt H} \xrightarrow{t \leftrightarrow i\tau} e^{-\int_0^\beta H \tau d\tau} \quad (6.31)$$

Starting from the thermal average

$$\langle A(t)B(t') \rangle = \langle A(t-t')B(0) \rangle \quad (6.32)$$

considering setting $t \rightarrow t - i\beta$

$$Z \text{tr}[e^{-\beta H} e^{iHt} A e^{-iHt} e^{+iHt'} B e^{-iHt'}]$$

The trace $\text{Tr}[AB \cdots C]$ is invariant under cycle. The thermal average

$$\langle A(t - i\beta)B(t') \rangle_{\text{thermal}} = \text{Tr}[e^{-\beta H} e^{iH(t-i\beta)} A e^{-i(i)}] = \text{Tr}[B(t') e^{iHt} A e^{-iHt} e^{-\beta H}] = \text{Tr}[e^{-\beta H} B(t') A(t)] \quad (6.33)$$

Som we fund that when $t = t' = 0$

$$\langle A(\tau = \beta)B(0) \rangle_\beta = \langle A(i\tau)B(0) \rangle$$

and we obtain the EOM and solution

$$-\frac{\partial}{\partial \tau} A(\tau) = [A(\tau), H], \quad A(\tau) = e^{H\tau} A(0) e^{-HE}$$

Sincee

$$\theta(\tau) = \begin{cases} 1, & \text{if } \tau > 0 \\ 0, & \text{if } \tau < 0 \end{cases}$$

Then $T_\tau\{A(\tau)B(\tau')\}$.

$$\langle A(t;)B(t') \rangle \Rightarrow \tau, G_{AB}^M(\tau, \tau') = -\langle \mathcal{T}_\tau(A(\tau), B(\tau')) \rangle = G_{AB}^M(\tau - \tau', 0)$$

$$G_{AB}^M(\tau - \tau', 0) = \epsilon G_{AB}^M(\tau - \tau' + n\beta, 0)$$

To proof that,

Proof.

$$G_{AB}^M(\tau - \tau', 0) = \mathcal{E} G_{AB}^M(\tau - \tau' + n\beta, 0) \quad (6.34)$$

then

$$\begin{aligned} G_{AB}^M(\tau - \tau' + \beta, 0) &= \text{Tr}[e^{-\beta H} e^{H(\tau-\tau'+\beta)} A e^{-H(\tau-\tau'+\beta)} B] \\ &= \text{Tr}[e^{-\beta H} B e^{H(\tau-\tau')} A e^{-H(\tau-\tau')}] = G_{AB}^M[B(0)A(\tau - \tau')] = \epsilon G_{AB}^M(\tau - \tau', 0) \end{aligned}$$

For boson: periodic; for fermion: antiperiodic. Then apply the Fourier transformon, we have Matsubara freq. Then let $E \rightarrow iE_n$, $E_n = 2m\pi/\beta$, or $E_n = (2n+1)\pi/\beta$, where $\tau \in (-\beta, \beta)$.

$$\int d\omega e^{i\omega t} G[\omega] \dots \Rightarrow \sum_{n=-\infty}^{+\infty} e^{iE_n \tau} G[iE_n](\dots) \Rightarrow \sum_{n=-\infty}^{+\infty} e^{iE_n \tau} G[E_n]$$

then

$$G[E_n] = \int dE' \frac{S_{AB}(E')}{iE_n - E' + i0^+} \quad (6.35)$$

along the contour

$$\sum_{n=-\infty}^{+\infty} f(iE_n) = \oint dz \frac{1}{e^{\beta z} \mp 1} f(z)$$

to calculate the residue, let $e^{\beta z} = \pm 1$, then $\tau \rightarrow \beta$, we have

$$\sum_n e^{iE_n t} G[iE_n] = \int dz dz' \frac{e^{\tau z} S_{AB}(E')}{(e^{\beta z} \pm 1)(z - E')} = \int dz dz' f^{\text{BF}}(z) \frac{S_{AB}(E')}{(z - E')} \Big|_{\tau \rightarrow \beta} \quad (6.36)$$

SPEOM: G .

6.4 Examples of GFs

Example 6.4.1. For fermions, the Hamiltonian

$$H = H_0 - \mu N = \sum \epsilon_k c_k^\dagger c_{k\sigma}$$

So the free fermion

$$|\Phi\rangle = \prod_{k < k_F, \epsilon_k < 0} c_{k\sigma}^\dagger |0\rangle$$

then

where $n_k = \theta(k_F - k)$. Then

$$\langle \Phi | c_{k\sigma}(t) c_{k'\sigma'}^\dagger(t') | \Phi \rangle = \delta_{\sigma\sigma'} \delta_{kk'} e^{-i\epsilon_k t} \langle \phi | c_{k\sigma} c_{k\sigma}^\dagger | \phi \rangle = \delta_{\sigma\sigma'} \delta_{kk'} e^{-i\epsilon_k(\tau - \tau')} (1 - n_k)$$

So, the Green function

$$G^c(k, t) = -i[(1 - n_k)\theta(t) - n_k\theta(-t)] e^{-i\epsilon_k t} = \begin{cases} -i\theta(|k| - k_F) e^{-i\epsilon_k t}, & t > 0 : \text{particle} \\ i\theta(k_F - |k|) e^{-i\epsilon_k t}, & t < 0 : \text{holes} \end{cases}$$

From the Fourier transform

$$G^c(k, \omega) = -i \left(\frac{\theta(k_1 - k_F)}{\delta - i(\omega - \epsilon_k)} - \frac{\theta(k_1 - |k|)}{\delta + i(\omega - \epsilon_k)} \right) \rightarrow \frac{1}{\omega - \epsilon_k + i\delta^+ \text{sgn}(|k| - k_F)}$$

where $\theta(k_1 - k_F) \sim 1 - n_k$, $\theta(k_f - |k|) \rightarrow n$. i.e., the valid for any eigenstate of H

$$i \partial_t G = \delta(t) + \epsilon G \quad (6.37)$$

Then,

$$G^c = \frac{1}{\omega - \epsilon_k + i\delta^+ \text{sgn}(\epsilon_k)}$$

$$\prod_{k \in (k_1, k_0)} c_k^\dagger |0\rangle$$

$$\frac{1 - n_k}{\omega - \epsilon_k} + \frac{n_k}{\omega - \epsilon_k} \Rightarrow \frac{1}{\omega - \epsilon_k}$$

From this, we have

$$E = \sum_k \epsilon_k \langle n_k \rangle \sim \int G \epsilon dk$$

and then, we can minimize the energy: $\delta E = 0$, for S_{AB} and S_{BA} .

Example 6.4.2 (Free boson capped). **No BEC** The Hamiltonian

$$H = \sum \omega_q b_q^\dagger b_q$$

and the Green functions

$$G[b_q, b_q^\dagger](k, \omega) = \frac{1}{\omega - \omega_q}$$

$G[\phi_q, \Phi_q]$, where $\Phi_q = b_q + b_q^\dagger$.

$$H_0 = \sum_k \epsilon_k b_k^\dagger b_k$$

BEC The ground state is

$$|\Psi\rangle = (a_0^\dagger)^N |0\rangle$$

that is called the BEC. when $\epsilon_k \rightarrow 0$, $N/V = n_0|_{L \rightarrow \infty, N \rightarrow \infty}$. The Green function

$$G_0(k, \omega) = G(a_{k: \epsilon_k=0}, a_k^\dagger)(k, \omega) = \underbrace{-i(2\pi)^d n_0 \delta(k) \delta(\omega)}_{\text{Coherent Part}} + \underbrace{\frac{1}{\omega - \epsilon_k + i0^-}}_{\text{Incoherent Part}}$$

The interaction Hamiltonian

$$H_{\text{int}} = g \sum_{k, k'} a_k^\dagger a_{-k}^\dagger a_{k'} a_{-k'}$$

When acting on $|\Psi\rangle$, $k' = 0$

$$H_{\text{int}}|\Psi\rangle \sim H_{\text{eff, int}} = \sum_{k \neq 0} g n_0 \left(a_k^\dagger a_k + \frac{1}{2} a_k^\dagger a_{-k}^\dagger + \frac{1}{2} a_{-k} a_k \right), \quad H_{B,E} = \sum_{k \neq 0} \epsilon_k a_k^\dagger a_k$$

When acting on $\langle \Psi |$, $k = 0$

$$\langle \Psi | H_{\text{int}}$$

then $a_0|\psi_0\rangle \sim \sqrt{n_0}|\Psi_0\rangle$ In total, $H_B = H_{\text{eff, int}} + H_{B,E}$. Since Bogolinbov transformation

$$b_k = u_k a_k - v_k a_{-k}^\dagger$$

Then

$$E_k = \sqrt{\epsilon_k(c\epsilon_k + 2gn_0)} \cong \sqrt{\frac{k^2}{2\pi} \left(\frac{k^2}{2\pi} + 2gn_0 \right)} \cong |k| \sqrt{gn_0}/m$$

which is so-called Goldstone mode. The Green function

$$i \partial_t G = \delta + \epsilon_{k_0} G + \langle [a_{k_0'}, H_{\text{int}}], a_{k'}^\dagger \rangle \rightarrow \langle g \sum_{k, k'} a_k^\dagger a_k a_{k'} \delta_{k, k'} \rangle \dots$$

... Bascially we have $H_0|\psi_0\rangle$, by adding δH , we have

$$G_0 + \delta G \propto \delta H / \propto H_0 \delta \psi$$

6.5 Free spin wave / magnos

* Most accurate to know the Hamiltonian of magnetic field: Applied a strong field

$$H_0 = -B \sum_j S_j^z$$

The ground state

$$|\Psi\rangle = |\uparrow\uparrow \cdots\rangle$$

Then

$$\delta H = \sum T_{ij}^+ (S_i^+ S_j^- + S_i^- S_j^+) + \sum J_{ij}^z S_i^z S_j^z$$

Since $[S_i^+, S_j^-] = \delta_{ij} 2\langle S_i^z \rangle$, $\{S_i^+, S_i^- = 1\}$, $\{S_i^+, S_j^-\} \neq 0$, $[S_i^+, S_i^z] = S_i^+$ The correlation function

$$\begin{aligned} i \partial_x [S^+, S_-] &= \delta_{ij} 2\langle S_i^z \rangle \delta(t) + \underbrace{\langle [S^+, H_0], S_f^- \rangle}_{B\langle S_i^+, S_f^- \rangle} + \langle [S^+, \delta H^+], S^- \rangle \\ &+ J^2 \langle S_j^z \rangle \langle S_i^+, S_f^- \rangle + J^z \langle S_i^z \rangle \langle S_j^+, S_f^- \rangle + \langle [S^+, \delta H^+], S^- \rangle + J_{ij}^+ \langle 2S_i^z; S_j^+; S_f^- \rangle \end{aligned}$$

Then do the Foruier Transform

$$\omega G[S_i^+, S_f^-] \cong 2\langle S^z \rangle + B\langle S_i^+, S_f^- \rangle + \sum_{\langle ij \rangle} J_{ij} 2\langle S_i^z \rangle \langle S_j^+, S_f^- \rangle$$

Acting ∂_t , summing over possible spin moments. Finally, this can be

$$\omega G[S^+, S^-](k, \omega) = 2\langle S^z \rangle + BG + 2\langle S^z \rangle J \tilde{\epsilon}_k$$

$$G \cong \frac{2\langle S^z \rangle}{\omega - B - 2J^+ \epsilon_k}$$

Another:

$$H_0 = \sum_{\langle ij \rangle} J_{ij}^z S_i^z S_j^z, \quad \delta H = \sum J_{ij}^+ (S_i^+ S_j^- + \text{H.c.})$$

If $J^z < 0$, then FM; If $J^z > 0$: then AFM.

$$\left| \begin{smallmatrix} \uparrow \uparrow \\ \uparrow \uparrow \end{smallmatrix} \right\rangle, \quad \left| \begin{smallmatrix} \uparrow \downarrow \\ \uparrow \uparrow \end{smallmatrix} \right\rangle$$

are all the classic type: SSB.

For Ruanfum SSB

$$H = \sum J \mathbf{S} \cdot \mathbf{S}, \quad [\rho, H] \neq 0$$

$$\omega G = 2\langle S^z \rangle + \sum_{\langle ij \rangle} J_{ij}^z \langle S_j^z \rangle G[S_i^+, S_f^-] + \sum_{\langle ij \rangle} J_{ij}^z \langle S_i^z \rangle G[S_j^+, S_f^-] + \sum J^+ 2\langle S_i^z \rangle G[S_j^+, S_f^-]$$

For FM:

$$\omega G = 2\langle S^z \rangle + 4J^z \langle S^z \rangle G + \langle S^z \rangle J^z \epsilon_k G + \langle S^z \rangle J^2 \epsilon_k G$$

then

$$\omega G = 2\langle S^z \rangle + (4J^z \langle S^z \rangle + (\langle S^z \rangle J^z + \langle S^z \rangle J^z) \epsilon_k) F$$

then the Green function is

$$G = \frac{2\langle S^z \rangle}{\omega - 4J^+ \langle S^z \rangle} + \epsilon_k J^z (1 + J^z / J^+)$$

so-called renormalization of (effective) mass of magnon.

For the anti-ferromagnet $\left| \begin{smallmatrix} \uparrow \downarrow \\ \downarrow \uparrow \end{smallmatrix} \right\rangle$

$$\begin{pmatrix} G[S_A^+ S_A^-] & G[S_B^+ S_A^-] \\ G[S_A^+ S_B^-] & G[S_B^+ S_B^-] \end{pmatrix}$$

Assume $k - k' = nQ$, $Q = (\pi, \pi)$.

$$\begin{pmatrix} G[k, k] & G[k + Q, k] \\ G[k, k + Q] & G[k + Q, k + Q] \end{pmatrix}$$

$$\omega[G] = \begin{pmatrix} 2\langle S^z \rangle_A & \\ & 2\langle S^z \rangle_B \end{pmatrix} + \begin{pmatrix} J^z \epsilon_k & J^z \epsilon_k \\ -J^z \epsilon_k & -J^z \epsilon_k \end{pmatrix}$$

where $\epsilon_{k+Q} = -\epsilon_k$. Eventually we will find that

$$\sqrt{\epsilon_k(\epsilon_k + J^z)} \propto k$$

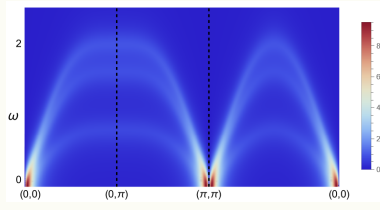
AFM: Spin-wave. In the k -space, the $\cos k$ will be flipped. For FM: $\epsilon_k \propto k^2$.

Only consider the very simple ground state $|\Psi\rangle_0$.

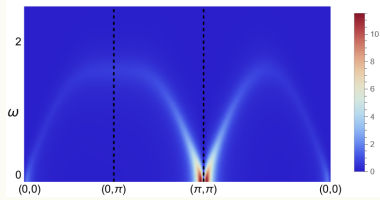
$G_{kk} \sim G_{AA}$, $k \in \text{First Brillouin Zone}$

6.8 EXAMPLES OF GFS · 277 ·

1. Gapless points: The dispersion relation Eq.(6.8.1) becomes gapless when $J_{xy} \geq J_z$. The gapless points are $Q = (0, 0)$ and (π, π) .



FIGURES 6.3: AFM spin wave DSF from a single sublattice G_{AA} at different magnetizations, from top to bottom: $m_z = 0.4, 0.2, 0.1$. For $m_z = 0.1$, the strength is amplified by a factor of 2.

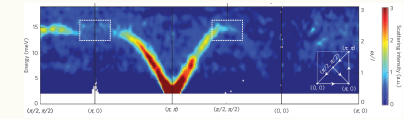


FIGURES 6.4: AFM spin wave DSF summed over all sublattices $S_{AA} + S_{BB} - S_{AB} - S_{BA}$. (note: it should all be plus sign, for some reason, I find it only fit experiment with minus sign for $S_{AB} = S_{BA}$.)

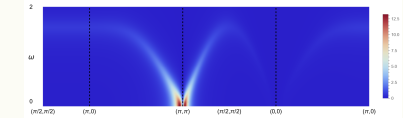
6.8.5 The two-spin problem by GFs

As a third example of an application, we wish to treat a model system with interactions, whose partition function can still be calculated exactly, so that all the interesting correlation functions are known in principle. This thus opens up the possibility of comparing the results of the Green's function method with the exact solutions.

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FIGURES 6.5: experimental data from Ref. Piazza et al. [2014]



FIGURES 6.6: AFM spin wave DSF summed over all sublattices $S_{AA} + S_{BB} - S_{AB} - S_{BA}$ for direct comparison with 6.4.

The model system in question consists of two spins of magnitudes

$$S_1 = S_2 = \frac{1}{2},$$

which are coupled to each other via an exchange interaction J and are presumed to be acted upon by a homogeneous magnetic field. We describe them in terms of the correspondingly simplified Heisenberg model (2.221):

$$H = -J(S_1^+ S_2^- + S_1^- S_2^+ + 2S_1^z S_2^z) - b(S_1^z + S_2^z),$$

where

$$b = \frac{1}{2} g \mu_B B_0.$$

The limitation to $S_1 = S_2 = 1/2$ allows some simplifications:

$$\begin{aligned} S_i^+ S_i^+ &= \frac{\hbar^2}{2} + \hbar S_i^z \\ S_i^+ S_i^- &= -S_i^z S_i^z = \mp \frac{\hbar}{2} S_i^z \\ (S_i^+)^2 &= 0; \quad (S_i^z)^2 = \frac{\hbar^2}{4} \end{aligned}$$

For our further discussion, we require several commutators:

$$\begin{aligned} [S_i^-, H]_- &= -J[S_i^-, S_1^+ S_2^- + S_1^- S_2^+ - 2J[S_i^-, S_1^z S_2^z] - b[S_i^-, S_1^z + S_2^z]_- \\ &= 2\hbar J(S_1^z S_2^- - S_1^- S_2^z) - \hbar b S_1^z \end{aligned}$$

Roton: liquid He. Roton get deeper during the salivation process.

6.6 Quasi-Particle Concept

$$i \partial_t G_{AB} = \delta(t) \langle [A, B] \rangle_\xi(t=0) + \langle [A, H], B \rangle = \delta(t) \cdots + \epsilon_k G_0 + \Gamma \quad (6.38)$$

where $\theta(t-t')AB = \theta(t'-t)BA$. $\langle [A, H], B \rangle$ is the so-called vertex function $\Gamma = \sum G$, $H = H_0 + V$. Then, we have the Dyson equation

$$G_{k\sigma}(E) = G_{k\sigma}^0 + G_k(E)^0 \Sigma(k, E) G_{k\sigma}(E) \quad (6.39)$$

and we can solve the Green function formally as

$$G_{k\sigma}(E) = \frac{1}{E - \epsilon_k + \Sigma}, \quad \text{where} \quad G_k^0 = \frac{1}{E - \epsilon_k} \quad (6.40)$$

6.6.1 Self energy

We assume that Σ is the self energy.

$$\Sigma = \text{Re}(k, E) + i \text{Im}(k, E)$$

we consider the relation between the advanced and retard Green function

$$(G^{\text{adv}})^* = G^{\text{ret}}.$$

In particular,

$$G_{k\sigma}^{\text{ret}}(E) = \frac{(E - \epsilon_k + R) + i\mathbb{1}}{(E - \epsilon_k + R)^2 + \mathbb{1}^2} \quad (6.41)$$

and the equivalent one-electron spectral density

$$S_{k\sigma}(E) = -\frac{1}{\pi} \frac{\mathbb{1}}{(E - \epsilon_k + R)^2 + I^2} \quad (6.42)$$

Case A $I = 0$ Denote $I \rightarrow -0^+$. The δ -function becomes

$$\delta(E - E_0) = \frac{1}{\pi} \lim_{x \rightarrow 0} \frac{x}{(E - E_0)^2 + x^2}$$

Then the equivalent one-electron spectral density becomes

$$S_k(E) = \delta(E - \epsilon_k + R) \quad (6.43)$$

To get the solution, let the δ -function to be 1

$$E - \epsilon_k + R = 0, \Rightarrow E_i(k)$$

then, we have

$$\delta(E - \epsilon_k + R) = S(E) = \sum \alpha_i(k) \delta(E - E_i(k)), \quad \text{and} \quad \alpha_i(k) = \left| 1 - \frac{\partial}{\partial E} k(E) \right|^{-1}$$

where make use of

$$\delta[f(x)] = \sum_i \frac{1}{f'(x_i)} \delta(x - x_i)$$

Case B $I \neq 0$

$$\langle S^+ | S^- \rangle \rightarrow \langle \psi(t') | \psi(t) \rangle \xrightarrow{t-t' \rightarrow \infty} \text{Const} \rightarrow 0;$$

$$\langle c | c^\dagger \rangle$$

means $|\psi\rangle = S^- |\Psi_0\rangle$ is no longer eigenstate. Particle will decay.

$$|I| \ll |\epsilon_k + R|$$

Since

$$F = \epsilon_k + R = F(E_i) + (E - E_i) \frac{\partial F}{\partial E} \Big|_{E=E_i} + \dots = E_i + (E - E_i) \frac{\partial F}{\partial E} \Big|_{E=E_i} + \dots$$

Then, the element of the density satisfies

$$S^{(i)} \cong \alpha_i e^{-iE_i(t-t')} \quad (6.44)$$

and we call the lifetime of quasi-particles $e^{-|\alpha| |t-t'|} \rightarrow e^{-|t-t'|/\tau}$, where the lifetime $\tau = 1/|\alpha I|$. The Real part $\text{Re}[\Sigma]$ and the imaginary part $\text{Im}[\Sigma]$ are *not independent* from each other.

$$\epsilon_k \cong T_0 + \frac{k^2}{2m} \quad (6.45)$$

$$E_i = T_0 + \frac{k^2}{2m^*} \quad (6.46)$$

then, we have

$$E_i = T_0 + \frac{m}{m^*} (\epsilon_k - T_0),$$

and the fraction of m and m^*

$$\frac{m}{m^*} = \frac{\partial E_i(k)}{\partial \epsilon_k} = 1 + \frac{\partial R}{\partial \epsilon_k} = 1 + \frac{\partial R}{\partial E_i} \frac{\partial E_i}{\partial k} = \frac{1 - \left(\frac{\partial R}{\partial E_i} \right)_{\epsilon_k}}{1 + \left(\frac{\partial R}{\partial \epsilon_k} \right)_{E_i}}$$

Homework #6 [2025-10-16]

Problem 6.1. Derive the completeness relation Eq. (1.123)

$$\int \prod_{\alpha} \frac{d\phi_{\alpha}^* d\phi_{\alpha}}{2i\pi} e^{-\sum_{\alpha} \phi_{\alpha}^* \phi_{\alpha}} |\phi\rangle \langle \phi| = \mathbb{1} \quad (1.123)$$

by integration. First consider one single-particle state α and let $|n\rangle$ denote the state with n particles in α . Show that by writing in polar form $\phi = \rho e^{i\theta}$ one obtains

$$\int \frac{d\phi^* d\phi}{2\pi i} e^{-\phi^* \phi} |\phi\rangle \langle \phi| = \int \rho \frac{d\rho d\theta}{\pi} e^{-\rho^2} \sum_m \frac{(\rho e^{i\theta})^m}{\sqrt{m!}} |m\rangle \sum_n \frac{(\rho e^{-i\theta})^n}{\sqrt{n!}} \langle n| = \sum_n |n\rangle \langle n|.$$

Now generalize to a set of single-particle states $\{|\alpha\rangle\}$ noting that the closure relation Eq. (1.89) may be written

$$\sum_{\{n_{\alpha}\}} |n_{\alpha_1} n_{\alpha_2} \dots\rangle \langle n_{\alpha_1} n_{\alpha_2} \dots| = \mathbb{1},$$

where $\{n_{\alpha}\}$ denotes a complete set of occupation numbers.

Solution. Expand the coherent state $|\phi\rangle$ in terms of number state $|n\rangle$

$$|\phi\rangle = \sum_{n=0}^{\infty} c_n |n\rangle.$$

Here, we merge the normalization factor into c_n to simplify, and it will be released at the end. Act $\hat{a}$ on the expand expression of $|\phi\rangle$, due to the property of annihilation operator

$$\hat{a}|\phi\rangle = \sum_{n=0}^{\infty} c_n \hat{a}|n\rangle = \sum_{n=1}^{\infty} c_n \sqrt{n} |n-1\rangle = \phi \sum_{n=0}^{\infty} c_n |n\rangle,$$

the third term summing from $n=1$ is due to $\hat{a}|0\rangle = 0$, which $n=0$ make no sense. To uniform the lower limit of n , shift the summation index

$$\hat{a}|\phi\rangle = \sum_{n=0}^{\infty} c_{n+1} \sqrt{n+1} |n\rangle = \phi \sum_{n=0}^{\infty} c_n |n\rangle,$$

then we obtain the expression of the expansion coefficient

$$c_{n+1} \sqrt{n+1} = \phi c_n, \quad \text{or} \quad c_n = \frac{\phi}{\sqrt{n}} c_{n-1} = \frac{\phi}{\sqrt{n}} \frac{\phi}{\sqrt{n-1}} c_{n-2} = \dots = \frac{\phi^n}{\sqrt{n!}} c_0,$$

thus we obtain the expansion of $|\phi\rangle$ in terms of number states

$$|\phi\rangle = c_0 \sum_{n=0}^{\infty} \frac{\phi^n}{\sqrt{n!}} |n\rangle \xrightarrow{\text{normalization}} e^{-|\phi|^2/2} \sum_{n=0}^{\infty} \frac{\phi^n}{\sqrt{n!}} |n\rangle.$$

Writing ϕ in polar form $\phi = \rho e^{i\theta}$, we expand $d\phi^* d\phi$ via Wedge product

$$d\phi^* \wedge d\phi = (e^{-i\theta} d\rho - i\rho e^{-i\theta} d\theta) \wedge (e^{i\theta} d\rho + i\rho e^{i\theta} d\theta) = 2i\rho d\rho \wedge d\theta,$$

then $d\phi^* d\phi = 2i\rho d\rho d\theta$. Substitute the above expressions into the identity

$$\mathbb{1} = \int \frac{d\phi^* d\phi}{2\pi i} e^{-\phi^* \phi} |\phi\rangle\langle\phi| = \int_0^{2\pi} d\theta \int_0^\infty \frac{\rho d\rho}{\pi} e^{-\rho^2} \sum_{m,n=0}^\infty \frac{\rho^{m+n} e^{(m-n)i\theta}}{\sqrt{m!n!}} |m\rangle\langle n|.$$

Evaluate the angular integral

$$\int_0^{2\pi} d\theta e^{(m-n)i\theta} = 2\pi \delta_{mn}.$$

Evaluate the radial integral for $m = n$ (since there will be a δ_{mn} factor)

$$\int_0^\infty d\rho e^{-\rho^2} \rho^{2n+1} = \frac{1}{2} \Gamma(n+1) = \frac{1}{2} n!.$$

Substitute them into the integral

$$I = \sum_{m,n=0} \frac{1}{\pi} \frac{1}{\sqrt{m!n!}} (2\pi \delta_{mn}) \left(\frac{1}{2} n! \right) |m\rangle\langle n| \xrightarrow{\text{only } m=n \text{ survives}} \sum_n |n\rangle\langle n|.$$

When generalize to a set of complete single-particle basis $\{|\alpha\rangle\}$ and build the full Fock space with occupation numbers $\{n_\alpha\}$, i.e.,

$$|\{\phi_\alpha\}\rangle = \prod_\alpha |\phi_\alpha\rangle_\alpha,$$

the product measure over modes gives the multi-mode resolution of identity. Concretely, for each mode α introduce a complex coordinate ϕ_α and form the product integral

$$\prod_\alpha \int \frac{d\phi_\alpha^* d\phi_\alpha}{2\pi i} e^{-\phi_\alpha^* \phi_\alpha} \otimes_\alpha |\phi_\alpha\rangle\langle\phi_\alpha|,$$

expanded into a Fock basis, each mode's angular integral enforces equality of occupation numbers in bra $\langle|$ and ket $| \rangle$, each radial integral produces the factorial that cancels the $1/\sqrt{n!}$ factors, so the identity ends up with the sum over all occupation-number configurations

$$\sum_{\{n_\alpha\}} |n_{\alpha_1} n_{\alpha_2} \dots\rangle \langle n_{\alpha_1} n_{\alpha_2} \dots| = \mathbb{1}_{\text{Fock}}.$$

In other words, the modes are independent, and for each that has the completeness shown above.

Problem 6.2. Generalize the properties of Grassmann variables demonstrated in Section 1.5 for the pair of generators ξ, ξ^* to the case $2p$ generators $\{\xi_1 \dots \xi_p \xi_1^* \dots \xi_p^*\}$. In particular, determine the general form of a function $f(\xi_\alpha)$ and an operator $A(\xi_\alpha^*, \xi_\alpha)$, show that $\frac{\partial}{\partial \xi_\alpha}, \frac{\partial}{\partial \xi_\beta}, \xi_\gamma$, and ξ_δ^* anticommute, determine if an analogous property holds for integration, find and verify an expression for the p -dimensional δ -function $\delta^P(\xi - \xi')$, and generalise Eq. (1.157)

$$\begin{aligned} \langle f|g \rangle &= \int d\xi^* d\xi (1 - \xi^* \xi) (f_0^* + f_1^* \xi) (g_0 + g_1 \xi^*) \\ &= - \int d\xi^* d\xi \xi^* \xi f_0^* g_0 + \int d\xi^* d\xi \xi \xi^* f_1^* g_1 = f_0^* g_0 + f_1^* g_1 \end{aligned} \quad (1.157)$$

for $\langle f|g \rangle$.

Solution.

- (a) Function $f(\xi) \equiv f(\xi_1, \xi_1, \dots, \xi_p)$ is a general element of the Grassmann algebra generated by $\{\xi_1, \xi_1, \dots, \xi_p\}$. Since $\xi_\alpha^2 = 0$, any function must be “at most” linear in each variable ξ_α , and the function is a *finite polynomial*. Apply Taylor expansion to $f(\xi)$, we can obtain its general form

$$f(\xi) = f_0 + \sum_{i=1}^p f_i \xi_i + \sum_{i < j} f_{ij} \xi_i \xi_j + \dots + f_{1\dots p} \xi_1 \dots \xi_p = \sum_{n=0}^p \frac{1}{n!} \sum_{\alpha_1 < \alpha_2 < \dots < \alpha_n} f_{\alpha_1 \dots \alpha_n} \xi_{\alpha_1} \dots \xi_{\alpha_n},$$

where the coefficients $f_{\{\alpha_i\}}$ are antisymmetric under interchange of any two indices. The Operator depends on $2p$ generators. Apply the same logic to the function, the operator is also a finite linear combination of monomials

$$A(\xi^*, \xi) = \sum_{m=0}^p \sum_{n=0}^p \sum_{\substack{\alpha_1 < \alpha_2 < \dots < \alpha_m \\ \beta_1 < \beta_2 < \dots < \beta_n}} \frac{1}{m!n!} A_{(\beta_1, \beta_2, \dots, \beta_n)}^{(\alpha_1, \alpha_2, \dots, \alpha_m)} \xi_{\alpha_1}^* \xi_{\alpha_2}^* \dots \xi_{\alpha_m}^* \xi_{\beta_1} \xi_{\beta_2} \dots \xi_{\beta_n},$$

with coefficients $A_{\{\beta_j\}}^{\{\alpha_i\}}$ antisymmetric in the upper indices and antisymmetric in the lower indices.

- (b) Act the partial derivatives with respect to Grassmann variables on a Grassmann variable, or a multiple of Grassmann functions, can be expressed as

$$\partial_{\xi_\alpha}(\xi_\beta) = \delta_{\alpha\beta}, \quad \text{and} \quad \partial_{\xi_\alpha}(AB) = (\partial_{\xi_\alpha} A)B + (-1)^{\epsilon_A} A(\partial_{\xi_\alpha} B),$$

where $\epsilon_A = 0$ (even) or 1 (odd) is the Grassmann parity of A . Then, we can obtain the canonical anticommutators

$$[\xi_\alpha, \xi_\beta]_+ = 0, \quad \text{and} \quad [\partial_{\xi_\alpha}, \partial_{\xi_\beta}]_+ = 0,$$

and the mixed relation

$$[\partial_{\xi_\alpha}, \xi_\beta]_+ = \partial_{\xi_\alpha} \xi_\beta + \xi_\beta \partial_{\xi_\alpha} = \delta_{\alpha\beta}.$$

With respect to the integration properties

$$\int d\xi_\alpha = 0, \quad \int d\xi_\alpha \xi_\alpha = 1, \quad d\xi_\alpha^{(*)} d\xi_\beta^{(*)} = -d\xi_\beta^{(*)} d\xi_\alpha^{(*)}.$$

With the chosen ordering the standard normalization is

$$d^p \xi \equiv d\xi_p d\xi_{p-1} \dots d\xi_1, \quad \int d^p \xi_{\alpha_1} \xi_{\alpha_2} \dots \xi_{\alpha_p} = (-1)^{\epsilon_{\xi_i}} \delta_{np}.$$

- (c) Due to the integration properties and the *Dirac measure*, the p -dimensional δ -function satisfies

$$\int d^p \xi f(\xi) \delta^p(\xi - \xi') = f(\xi'),$$

then, the p -dimensional δ -function can be expressed as

$$\delta^p(\xi - \xi') \equiv \prod_{\alpha=1}^p (\xi_{\alpha} - \xi'_{\alpha}),$$

and it can be trivially verified by denoting $f(\xi') = \sum_i f_i \xi'_i$.

$$\int d\xi_1 f(\xi_1, \xi_2, \dots, \xi_p) (\xi_1 - \xi'_1) \cdots (\xi_p - \xi'_p) = f(\xi'_1, \xi_2, \dots, \xi_p) (\xi_2 - \xi'_2) \cdots (\xi_p - \xi'_p),$$

which it is similar for $\xi_2, \dots, \xi_p$. And after p integrations, the RHS becomes $f(\xi'_1, \xi'_2, \dots, \xi'_p) \equiv f(\xi')$.

- (d) Expand $f(\xi)$ and $g(\xi^*)$ as

$$f(\xi) = \sum_i f_i \xi_i, \quad \text{and} \quad g(\xi^*) = \sum_j g_j \xi_j^*,$$

then, the inner product can be expressed as

$$\langle f | g \rangle = \int \left(\prod_{\alpha=1}^p d\xi_{\alpha}^* d\xi_{\alpha} \right) \left[\prod_{\alpha=1}^p (1 - \xi_{\alpha}^* \xi_{\alpha}) \right] f^*(\xi) g(\xi^*) \equiv \sum_i f_i^* g_i.$$

If $p = 2$, then we will generate Eq. (1.157), i.e., $\langle f | g \rangle = f_0^* g_0 + f_1^* g_1$.

Problem 6.3. Prove the closure relation Eq. (1.123)

$$\int \prod_{\alpha} \frac{d\phi_{\alpha}^* d\phi_{\alpha}}{2i\pi} e^{-\sum_{\alpha} \phi_{\alpha}^* \phi_{\alpha}} |\phi\rangle\langle\phi| = \mathbb{1} \quad (1.123)$$

to its fermionic form

$$\mathbb{1} = \int d\mu(\xi) e^{-\sum_{\alpha} \xi_{\alpha}^* \xi_{\alpha}} |\xi\rangle\langle\xi|$$

for Fermions using Schur's lemma. As in the Boson case, $[a_{\alpha}, |\phi\rangle\langle\phi|] = (\xi_{\alpha} - \frac{\partial}{\partial \xi_{\alpha}^*}) |\phi\rangle\langle\phi|$, so one must show

$$\int \prod_{\alpha} d\xi_{\alpha}^* d\xi_{\alpha} e^{-\sum_{\alpha} \xi_{\alpha}^* \xi_{\alpha}} \left(\xi_{\alpha} - \frac{\partial}{\partial \xi_{\alpha}^*} \right) A(\xi_{\alpha}, \xi_{\alpha}^*) = 0 \quad (P6.1)$$

for any A . First, prove Eq. (P6.1) as it stands, establishing that the left-hand side of Eq. (1.126)

$$\left[a_{\alpha}, \int \prod_{\alpha'} \frac{d\phi_{\alpha'}^* d\phi_{\alpha'}}{2\pi i} e^{-\sum_{\alpha'} \phi_{\alpha'}^* \phi_{\alpha'}} |\phi\rangle\langle\phi| \right] = \int \prod_{\alpha'} \frac{d\phi_{\alpha'}^* d\phi_{\alpha'}}{2\pi i} e^{-\sum_{\alpha'} \phi_{\alpha'}^* \phi_{\alpha'}} |\phi\rangle\langle\phi| = 0 \quad (1.126)$$

must be proportional to unity, and evaluate the constant of proportionality by calculating the matrix element in the zero-particle state.

Then note that Eq. (P6.1) is a special case of integration by parts for Grassmann variables. Show that the general rule for integration by parts is

$$\int \prod_{\alpha} d\xi_{\alpha}^* d\xi_{\alpha} A(\xi_{\alpha}^*, \xi_{\alpha}) \frac{\overrightarrow{\partial}}{\partial \xi_{\alpha}} B(\xi_{\alpha}^*, \xi_{\alpha}) = \int \prod_{\alpha} d\xi_{\alpha}^* d\xi_{\alpha} A(\xi_{\alpha}^*, \xi_{\alpha}) \frac{\overleftarrow{\partial}}{\partial \xi_{\alpha}} B(\xi_{\alpha}^*, \xi_{\alpha}) \quad (P6.2)$$

where $\frac{\overleftarrow{\partial}}{\partial \xi_{\alpha}}$ acts to the left and the variable ξ_{α} must be anticommutated to the right to be adjacent to the derivative. Note in particular that the sign in Eq. (P6.2) is reversed from the usual relation for complex variables, and that expressions like $\int \prod_{\alpha} d\xi_{\alpha}^* d\xi_{\alpha} \frac{\partial}{\partial \xi_{\alpha}} [A(\xi_{\alpha}^*, \xi_{\alpha})] B(\xi_{\alpha}^*, \xi_{\alpha})$ do not reproduce the right-hand side.

Solution. We need to list some identities of the Grassmann variables.

$$1. \xi^2 = 0 \quad 2. \xi = \xi^* \quad 3. \xi^* \xi = -\xi \xi^* \quad 4. \int 1 d\xi = 0 \quad 5. \int \xi d\xi = 1 \quad 6. \int \left[\frac{\partial f(\theta)}{\partial \theta} \right] d\theta = 0$$

Then the exponential term and $A(\xi^*, \xi)$ can be expanded as

$$e^{-\xi^* \xi} = 1 - \xi^* \xi, \quad e^{-\xi \xi^*} = 1 + \xi \xi^*, \quad \text{and} \quad A(\xi^*, \xi) = A_{00} + A_{01} \xi + A_{10} \xi^* + A_{11} \xi \xi^*,$$

where the other terms consist ξ^2 , ξ^{*2} , or $(\xi^* \xi)^2$ vanish. To prove Eq. (P6.1), consider

$$\int d\xi^* d\xi e^{-\xi^* \xi} (\xi - \partial_{\xi^*}) A(\xi^*, \xi).$$

The first term of the integral can be expanded as

$$\int d\xi^* \int d\xi e^{-\xi^* \xi} \xi A(\xi, \xi^*) = \int d\xi^* \int d\xi \xi (A_{00} + A_{01} \xi + A_{10} \xi^* + A_{11} \xi \xi^*) = A_{10},$$

where only the term containing $\xi \xi^*$ survives. Since the derivative of two Grassmann terms satisfies

$$\partial_{\xi^*} [e^{-\xi^* \xi} A(\xi, \xi^*)] = \partial_{\xi^*} (e^{-\xi^* \xi}) A(\xi, \xi^*) + e^{-\xi^* \xi} \partial_{\xi^*} A(\xi, \xi^*).$$

Then the second term of the integral can be expanded as

$$\begin{aligned} \int d\xi^* \int d\xi e^{-\xi^* \xi} \partial_{\xi^*} A(\xi, \xi^*) &= \int d\xi^* \int d\xi \partial_{\xi^*} [e^{-\xi^* \xi} A(\xi, \xi^*)] - \int d\xi^* \int d\xi \partial_{\xi^*} (e^{-\xi^* \xi}) A(\xi, \xi^*) \\ &= - \int d\xi^* \int d\xi \partial_{\xi^*} (1 - \xi^* \xi) A(\xi, \xi^*) = \int d\xi^* \int d\xi \xi (A_{00} + A_{01} \xi + A_{10} \xi^* + A_{11} \xi \xi^*) = A_{10}. \end{aligned}$$

Combining the two terms, we have

$$\int d\xi^* d\xi e^{-\xi^* \xi} (\xi - \partial_{\xi^*}) A(\xi^*, \xi) = A_{10} - A_{10} = 0.$$

Concerning multi-mode: for a fixed label β , consider the total derivative $\partial_{\xi_\beta^*}$ to $e^{-\sum_\alpha \xi_\alpha^* \xi_\alpha} A(\xi_\alpha, \xi_\alpha^*)$. The same algebra gives

$$\frac{\partial}{\partial \xi_\beta^*} e^{-\sum_\alpha \xi_\alpha^* \xi_\alpha} A(\xi_\alpha, \xi_\alpha^*) = -\xi_\beta e^{-\sum_\alpha \xi_\alpha^* \xi_\alpha} + e^{-\sum_\alpha \xi_\alpha^* \xi_\alpha} \partial_{\xi_\beta^*} A(\xi, \xi^*),$$

and integrating over $\prod_\alpha d\xi_\alpha^* d\xi_\alpha$ yields zero. Thus, the multi-mode identity Eq. (P6.1) follows. Then, we shall generate Eq. (1.126). Consider the single mode first, we calculate the “kernel” commutator $[\hat{a}, |\xi\rangle\langle\xi|]$.

(a) $a|\xi\rangle$. Since $|\xi\rangle$ can be expanded as

$$|\xi\rangle = e^{a^\dagger \xi} |0\rangle = (1 + a^\dagger \xi) |0\rangle.$$

Left act a on it, then we have

$$a|\xi\rangle = a|0\rangle + aa^\dagger \xi |0\rangle = 0 + (1 + a^\dagger a) \xi |0\rangle = \xi |0\rangle.$$

If we multiply ξ to $|\xi\rangle$

$$\xi|\xi\rangle = \xi(1 + a^\dagger \xi) |0\rangle = \xi |0\rangle,$$

the ξ^2 term vanishes. So, we get the expression $a|\xi\rangle\langle\xi| = \xi|\xi\rangle\langle\xi|$.

(b) $\langle\xi|a$. Since $\langle\xi|$ can be expanded as

$$\langle\xi| = e^{a \xi^*} \langle 0| = \langle 0| (1 + a \xi^*).$$

Right act a on it, then we have

$$\langle\xi|a = \langle 0| (1 + a \xi^*) a = \langle 0|a + \langle 0|a \xi^* a = \langle 0|a,$$

where $\langle 0|a = (a^\dagger |0\rangle)^\dagger = \langle 1|$, and aa forms a zero operator, i.e., $aa = 0$ (for Fermions). If we right act ∂_{ξ^*} on $\langle\xi|$

$$\langle\xi| \bar{\partial}_{\xi^*} = (1 + a \xi^*) \langle 0| \bar{\partial}_{\xi^*} = \langle 0|a,$$

So, we get the expression $\langle\xi|a = \langle\xi| \bar{\partial}_{\xi^*}$, and we have

$$(|\xi\rangle\langle\xi|) \bar{\partial}_{\xi^*} = (|\xi\rangle \bar{\partial}_{\xi^*}) \langle\xi| + |\xi\rangle (\langle\xi| \bar{\partial}_{\xi^*}) = ((1 + a^\dagger \xi) |0\rangle \bar{\partial}_{\xi^*}) \langle\xi| + |\xi\rangle (\langle\xi| \bar{\partial}_{\xi^*}) = |\xi\rangle (\langle\xi| \bar{\partial}_{\xi^*}).$$

(c) Prove the relation between left and right derivatives. Denote

$$d\mu(\xi) \equiv \prod_{\alpha} d\xi_{\alpha}^* \xi_{\alpha}.$$

Due to the identities of the derivative of Grassmann variables

$$\frac{\partial}{\partial \xi} (AB) = \left(\frac{\partial A}{\partial \xi} \right) B + (-1)^{\epsilon_A} A \left(\frac{\partial B}{\partial \xi} \right),$$

where ϵ_A is the parity of A . Integral over the total derivative, and using the partial integration formula of Grassmann variables

$$\begin{aligned} 0 &\equiv \int d\mu \vec{\partial}_{\xi} (AB) \xi = \int d\mu (\vec{\partial}_{\xi} A) B + (-1)^{\epsilon_A} \int d\mu A (\vec{\partial}_{\xi} B), \\ \int d\mu A (\vec{\partial}_{\xi} B) &= -(-1)^{\epsilon_A} \int d\mu (\vec{\partial}_{\xi} A) B. \end{aligned}$$

Due to the property of the Grassmann variable, the left-acting derivative to A can be expanded as

$$A \overleftarrow{\partial}_{\xi} = -(-1)^{\epsilon_A} \vec{\partial}_{\xi} A,$$

by combining the two expressions above, we can obtain

$$\int d\mu A (\vec{\partial}_{\xi} B) = \int d\mu A \overleftarrow{\partial}_{\xi} B.$$

Also for multi-mode, which we proved Eq. (P6.2).

Then, the “kernel” commutator becomes

$$[\hat{a}, |\xi\rangle\langle\xi|] = a|\xi\rangle\langle\xi| - |\xi\rangle\langle\xi|a = \xi|\xi\rangle\langle\xi| - (|\xi\rangle\langle\xi|) \overleftarrow{\partial}_{\xi^*} = (\xi - \overrightarrow{\partial}_{\xi^*})|\xi\rangle\langle\xi|,$$

and the commutator in Eq. (1.126) can be written as

$$\left[a, \int d\mu e^{\xi^* \xi} |\xi\rangle\langle\xi| \right] = \int d\mu e^{\xi^* \xi} (\xi - \overrightarrow{\partial}_{\xi^*}) |\xi\rangle\langle\xi| \stackrel{\text{Eq. (P6.1)}}{=} 0,$$

and it also satisfies multi-mode, which we proved Eq. (1.126). Due to Schur's lemma, the operator $\mathcal{O} \equiv \int d\mu e^{\xi^* \xi} |\xi\rangle\langle\xi|$ commutes to a , then it is proportional to unity, i.e., $\mathcal{O} = C\mathbb{1}$. To determine C , calculate the vacuum expectation of $\mathcal{O}$

$$C \equiv \langle 0 | \mathcal{O} | 0 \rangle = \int d\mu(\xi) e^{-\sum_{\alpha} \xi_{\alpha}^* \xi_{\alpha}} \langle 0 | \xi \rangle \langle \xi | 0 \rangle.$$

Since the expansion of $|\xi\rangle$, $\langle\xi|$, and $e^{-\xi^* \xi}$, we have

$$\langle 0 | \xi \rangle = \langle \xi | 0 \rangle = 1, \quad \text{and} \quad \int d\xi^* d\xi e^{-\xi^* \xi} = \int d\xi^* d\xi (1 - \xi^* \xi) = 1.$$

Therefore

$$C = 1, \quad \mathcal{O} = \mathbb{1}, \quad \text{and} \quad \int d\mu(\xi) e^{-\sum_{\alpha} \xi_{\alpha}^* \xi_{\alpha}} |\xi\rangle\langle\xi| = \mathbb{1},$$

which we proved the fermionic form for Eq. (1.123).

Homework #7 [2025-10-22]

Problem 7.1. Investigate a simple model Hamiltonian for BCS superconductivity,

$$\mathcal{H}_k = t(k)(a_{k\uparrow}^\dagger a_{k\uparrow} + a_{-k\downarrow}^\dagger a_{-k\downarrow}) - \Delta(a_{k\uparrow}^\dagger a_{-k\downarrow}^\dagger + a_{-k\downarrow} a_{k\uparrow}).$$

- Find the energy eigenvalues of $\mathcal{H}_k$.
- Give the corresponding eigenstates.
- Give the possible excitation energies of the system.
- Identify the excitation gap, i.e. the minimum of the excitation energies. Consider the simplest 1D case: assuming $t(k) = -2t \cos k - \mu$.

Solution.

- We consider the two big terms of the Hamiltonian, respectively. Working in the 4D Fock space, there are 4 identities

$$1. |0\rangle \equiv |0_{k\uparrow}, 0_{-k\downarrow}\rangle \quad 2. |k\uparrow\rangle \equiv a_{k\uparrow}^\dagger |0\rangle \quad 3. |-k\downarrow\rangle \equiv a_{-k\downarrow}^\dagger |0\rangle \quad 4. |k\uparrow, -k\downarrow\rangle \equiv a_{k\uparrow}^\dagger a_{-k\downarrow}^\dagger |0\rangle$$

Now, we shall compute how each of the 4 operators in $\mathcal{H}_k$ acts on the basis.

- Number term

$$\begin{aligned} a_{k\uparrow}^\dagger a_{k\uparrow} |k\uparrow\rangle &= |k\uparrow\rangle, & a_{k\uparrow}^\dagger a_{k\uparrow} \{|0\rangle, |-k\downarrow\rangle, |k\uparrow, -k\downarrow\rangle\} &= \{0\}, \\ a_{-k\downarrow}^\dagger a_{-k\downarrow} |k\uparrow\rangle &= |-k\downarrow\rangle, & a_{-k\downarrow}^\dagger a_{-k\downarrow} \{|0\rangle, |-k\downarrow\rangle, |k\uparrow, -k\downarrow\rangle\} &= \{0\}. \end{aligned}$$

- Pairing term

$$\begin{aligned} a_{k\uparrow}^\dagger a_{-k\downarrow}^\dagger |0\rangle &= |k\uparrow, -k\downarrow\rangle, & a_{k\uparrow}^\dagger a_{-k\downarrow}^\dagger \{|k\uparrow\rangle, |-k\downarrow\rangle, |k\uparrow, -k\downarrow\rangle\} &= \{0\}, \\ a_{-k\downarrow} a_{k\uparrow} |k\uparrow, -k\downarrow\rangle &= |0\rangle, & a_{-k\downarrow} a_{k\uparrow} \{|0\rangle, |k\uparrow\rangle, |k\downarrow\rangle\} &= \{0\}. \end{aligned}$$

The number term acts diagonally, giving

Pairing term couples only $|0\rangle$ and $|k\uparrow, -k\downarrow\rangle$

State	$ 0\rangle$	$ k\uparrow\rangle$	$ -k\downarrow\rangle$	$ k\uparrow, -k\downarrow\rangle$	
Energy	0	$t(k)$	$t(k)$	$2t(k)$	$-\Delta(a_{k\uparrow}^\dagger a_{-k\downarrow}^\dagger + a_{-k\downarrow} a_{k\uparrow}) 0\rangle = -\Delta k\uparrow, -k\downarrow\rangle,$ $-\Delta(a_{k\uparrow}^\dagger a_{-k\downarrow}^\dagger + a_{-k\downarrow} a_{k\uparrow})\Delta k\uparrow, -k\downarrow\rangle = - 0\rangle.$

Since some of the actions act on, or create / annihilate the number of particles are odd, and other actions are even, so we can separate the Hamiltonian into two independent blocks

$$\mathcal{H}_k = \begin{cases} \{|k\uparrow\rangle, |-k\downarrow\rangle\}, & \text{Odd sector,} \\ \{|0\rangle, |k\uparrow, -k\downarrow\rangle\}, & \text{Even sector,} \end{cases}$$

where

$$\begin{aligned} \mathcal{H}_{\text{odd}} &= \begin{pmatrix} \langle k\uparrow | \mathcal{H}_k | k\uparrow \rangle & \langle k\uparrow | \mathcal{H}_k | -k\downarrow \rangle \\ \langle -k\downarrow | \mathcal{H}_k | k\uparrow \rangle & \langle -k\downarrow | \mathcal{H}_k | -k\downarrow \rangle \end{pmatrix} = \begin{pmatrix} t(k) & 0 \\ 0 & t(k) \end{pmatrix}, \\ \mathcal{H}_{\text{even}} &= \begin{pmatrix} \langle 0 | \mathcal{H}_k | 0 \rangle & \langle 0 | \mathcal{H}_k | k\uparrow, -k\downarrow \rangle \\ \langle k\uparrow, -k\downarrow | \mathcal{H}_k | 0 \rangle & \langle k\uparrow, -k\downarrow | \mathcal{H}_k | k\uparrow, -k\downarrow \rangle \end{pmatrix} = \begin{pmatrix} 0 & -\Delta \\ -\Delta & 2t(k) \end{pmatrix}. \end{aligned}$$

It is trivial that in the odd sector, both states have energy $E_{\text{odd}} = t(k)$, corresponding to 2nd degeneracy. In the even sector, we need to solve the eigenvalue equation

$$\det |\mathcal{H}_{\text{even}} - \lambda| = 0.$$

Hence, we have the eigenvalues for the even sector

$$\lambda_1 = t(k) + \sqrt{t^2(k) + \Delta^2}, \quad \lambda_2 = t(k) - \sqrt{t^2(k) + \Delta^2}.$$

In summary, the eigenvalues of $\mathcal{H}_k$ are

$$E_{\text{odd}} = t(k), \quad E_{\text{ground}} = t(k) - \sqrt{t^2(k) + \Delta^2}, \quad (\text{Degeneracy} = 2) \quad E_{\text{excited}} = t(k) + \sqrt{t^2(k) + \Delta^2}.$$

- (b) Since in the odd sector, the Hamiltonian is diagonal, so it is trivial that the eigenstates in the odd sector are

$$\psi_{\text{odd } 1} = |k \uparrow\rangle, \quad \psi_{\text{odd } 2} = |-k \downarrow\rangle.$$

Concerning the even sector. Denote $E_k = \sqrt{t^2(k) + \Delta^2}$ for simplification. Since in the 4D Fock space, the basis should be orthonormal. So, assume the two eigenstates in the even sector to be

$$\psi_{\text{ground}} = a_1|0\rangle + b_1|k \uparrow, -k \downarrow\rangle, \quad \psi_{\text{excited}} = a_2|0\rangle + b_2|k \uparrow, -k \downarrow\rangle.$$

Since the orthonormality $\langle \psi_{\text{even } 1} | \psi_{\text{even } 2} \rangle = 0$, then we can derive

$$\frac{a_1}{b_1} = -\frac{b_2}{a_2}.$$

Let $a_1 = a_k, b_1 = b_k$. Then, the two eigenstates in the even sector can be expressed as

$$\psi_{\text{ground}} = a_k|0\rangle + b_k|k \uparrow, -k \downarrow\rangle, \quad \psi_{\text{excited}} = b_k|0\rangle - a_k|k \uparrow, -k \downarrow\rangle.$$

where a_k and b_k should satisfies the normalization $a_k^2 + b_k^2 = 1$. To obtain a_k and b_k , substitute ψ_{ground} into the eigenequation

$$\begin{pmatrix} 0 & -\Delta \\ -\Delta & 2t(k) \end{pmatrix} \begin{pmatrix} a_k \\ b_k \end{pmatrix} = E_{\text{ground}} \begin{pmatrix} a_k \\ b_k \end{pmatrix}.$$

Then we will obtain

$$a_k^2 = \frac{1}{2} \left(1 + \frac{t(k)}{E_k} \right), \quad b_k^2 = \frac{1}{2} \left(1 - \frac{t(k)}{E_k} \right).$$

So, the eigenstates in the even sector are

$$\begin{aligned} \psi_{\text{ground}} &= \sqrt{\frac{1}{2} \left(1 + \frac{t(k)}{E_k} \right)} |0\rangle + \sqrt{\frac{1}{2} \left(1 - \frac{t(k)}{E_k} \right)} |k \uparrow, -k \downarrow\rangle, \\ \psi_{\text{excited}} &= \sqrt{\frac{1}{2} \left(1 - \frac{t(k)}{E_k} \right)} |0\rangle - \sqrt{\frac{1}{2} \left(1 + \frac{t(k)}{E_k} \right)} |k \uparrow, -k \downarrow\rangle. \end{aligned}$$

- (c) Since the energy eigenvalues of $\mathcal{H}_k$ are

$$\{t(k), t(k), t(k) - E_k, t(k) + E_k\}.$$

It is trivial that the ground energy is $E_0 = t(k) - E_k$, and other there energies corresponding states are all excited states. So, the excitation energies of the system are

$$\Delta E_{1qp} = t(k) - (t(k) - E_k) = E_k, \quad \Delta E_{2qp} = (t(k) + E_k) - (t(k) - E_k) = 2E_k.$$

(Note: “qp” means “quasi-particle”).

- (d) It is trivial that the minimum of the excitation energies is

$$\Delta E_{\min} = \Delta E_{1qp} = E_k = \sqrt{t^2(k) + \Delta^2}.$$

Since $t(k) = -2t \cos k - \mu$, so the minimum occurs when $t(k) = 0$, i.e.,

$$k = \cos^{-1}\left(-\frac{\mu}{2t}\right) = \pi - \cos^{-1}\left(\frac{\mu}{2t}\right).$$

Then, the minimum excitation energy in the simplest 1D case is

$$\Delta E_{\min} = \Delta.$$

Problem 7.2.

- (a) Show that the excitations of a BCS superconductor are generated from Exercise 7.1 by the operators

$$\rho_{k\uparrow}^\dagger = u_k a_{k\uparrow}^\dagger - v_k a_{-k\downarrow}; \quad \rho_{-k\downarrow}^\dagger = u_k a_{-k\downarrow}^\dagger + v_k a_{k\uparrow}.$$

The coefficients u_k and v_k are defined as

$$u_k^2 = \frac{1}{2} \left\{ 1 + \frac{t(k)}{[t^2(k) + \Delta^2]^{1/2}} \right\}, \quad v_k^2 = 1 - u_k^2.$$

- (b) Prove that these operators are purely Fermionic operators.
 (c) Compute the commutator

$$[\mathcal{H}^*, \rho_{k\uparrow}^\dagger]_-.$$

How is this result to be interpreted?

- (d) Formulate and solve the equation of motion of the retarded Green's function

$$\hat{G}_{k\uparrow}^{\text{ret}}(E) = \langle \langle \rho_{k\uparrow}; \rho_{k\uparrow}^\dagger \rangle \rangle_E^{\text{ret}}.$$

- (e) Compute Green's functions of the original fermions a_k s and $a_k^\dagger$ s, consider all non-zero combinations. You can express them via linear transformations between ρ_k s and a_k s.

Solution.

- (a) *Proof.* Obviously, the given coefficients u_k and v_k is just a_k and b_k that obtained in Exercise 7.1. List $\rho_{k\uparrow}$, $\rho_{-k\downarrow}$ and their conjugates first

$$\begin{aligned} \rho_{k\uparrow}^\dagger &= u_k a_{k\uparrow}^\dagger - v_k a_{-k\downarrow}, & \rho_{-k\downarrow}^\dagger &= u_k a_{-k\downarrow}^\dagger + v_k a_{k\uparrow}, \\ \rho_{k\uparrow} &= u_k a_{k\uparrow} - v_k a_{-k\downarrow}^\dagger, & \rho_{-k\downarrow} &= u_k a_{-k\downarrow} + v_k a_{k\uparrow}^\dagger. \end{aligned}$$

from which we can obtain the inverse transformations

$$\begin{aligned} a_{k\uparrow}^\dagger &= u_k \rho_{k\uparrow}^\dagger + v_k \rho_{-k\downarrow}, & a_{-k\downarrow}^\dagger &= u_k \rho_{-k\downarrow}^\dagger - v_k \rho_{k\uparrow}, \\ a_{k\uparrow} &= u_k \rho_{k\uparrow} + v_k \rho_{-k\downarrow}^\dagger, & a_{-k\downarrow} &= u_k \rho_{-k\downarrow} - v_k \rho_{k\uparrow}^\dagger. \end{aligned}$$

Then, using the identities that can be easily obtained

$$u_k^2 - v_k^2 = \frac{t(k)}{E_k}, \quad 2u_k v_k = \frac{\Delta}{E_k}, \quad u_k^2 + v_k^2 = 1, \quad E_k = \sqrt{t^2(k) + \Delta^2},$$

and the operator identities of the fermions $a_{k\uparrow}$, $a_{-k\uparrow}$ and their conjugates to substitute them into $\mathcal{H}_k$, we have

$$\mathcal{H}_k = E_k(\rho_{k\uparrow}^\dagger \rho_{k\uparrow} + \rho_{-k\uparrow}^\dagger \rho_{-k\uparrow}) + (t(k) - E_k) \equiv \mathcal{H}^*,$$

where the constant term $t(k) - E_k$ is just an energy shift and doesn't affect the excitation spectrum. Thus, the operators $\rho_{k\uparrow}^\dagger$ and $\rho_{k\downarrow}^\dagger$ (and their conjugates) create excitations with energy E_k above the BCS ground state.

- (b) *Proof.* Since the fact that Fermionic operators $a_{k\uparrow}, a_{-k\downarrow}$ satisfy the anticommutation relations of, then calculate the same anticommutation relations between the operators $\rho_{k\uparrow}, \rho_{-k\downarrow}$, and their conjugates

i. $\{\rho_{k\uparrow}, \rho_{k\uparrow}^\dagger\}$

$$\begin{aligned}\rho_{k\uparrow}\rho_{k\uparrow}^\dagger &= u_k^2 a_{k\uparrow} a_{k\uparrow}^\dagger - u_k v_k a_{k\uparrow} a_{-k\downarrow} - u_k v_k a_{-k\downarrow}^\dagger a_{k\uparrow}^\dagger + v_k^2 a_{-k\downarrow}^\dagger a_{-k\downarrow}, \\ \rho_{k\uparrow}^\dagger \rho_{k\uparrow} &= u_k^2 a_{k\uparrow}^\dagger a_{k\uparrow} - u_k v_k a_{k\uparrow}^\dagger a_{-k\downarrow}^\dagger - u_k v_k a_{-k\downarrow} a_{k\uparrow} + v_k^2 a_{-k\downarrow} a_{-k\downarrow}^\dagger.\end{aligned}$$

Adding and using fermionic anticommutators

$$\{a_{k\uparrow}, a_{k\uparrow}^\dagger\} = 1, \quad \{a_{-k\downarrow}, a_{-k\downarrow}^\dagger\} = 1,$$

where the cross terms vanish. We get

$$\{\rho_{k\uparrow}, \rho_{k\uparrow}^\dagger\} = u_k^2 + v_k^2 = 1.$$

ii. $\{\rho_{-k\downarrow}, \rho_{-k\downarrow}^\dagger\}$

$$\begin{aligned}\rho_{-k\downarrow}\rho_{-k\downarrow}^\dagger &= u_k^2 a_{-k\downarrow} a_{-k\downarrow}^\dagger + u_k v_k a_{-k\downarrow} a_{k\uparrow} + u_k v_k a_{k\uparrow}^\dagger a_{-k\downarrow}^\dagger + v_k^2 a_{k\uparrow}^\dagger a_{k\uparrow}, \\ \rho_{-k\downarrow}^\dagger \rho_{-k\downarrow} &= u_k^2 a_{-k\downarrow}^\dagger a_{-k\downarrow} + u_k v_k a_{-k\downarrow}^\dagger a_{k\uparrow}^\dagger + u_k v_k a_{k\uparrow} a_{-k\downarrow} + v_k^2 a_{k\uparrow} a_{k\uparrow}^\dagger.\end{aligned}$$

Adding and using anticommutators

$$\{\rho_{-k\downarrow}, \rho_{-k\downarrow}^\dagger\} = u_k^2 + v_k^2 = 1.$$

iii. $\{\rho_{k\uparrow}, \rho_{-k\downarrow}^\dagger\}$

$$\begin{aligned}\rho_{k\uparrow}\rho_{-k\downarrow}^\dagger &= u_k^2 a_{k\uparrow} a_{-k\downarrow}^\dagger + u_k v_k a_{k\uparrow} a_{k\uparrow} - u_k v_k a_{-k\downarrow}^\dagger a_{-k\downarrow}^\dagger - v_k^2 a_{-k\downarrow}^\dagger a_{k\uparrow}, \\ \rho_{-k\downarrow}^\dagger \rho_{k\uparrow} &= u_k^2 a_{-k\downarrow}^\dagger a_{k\uparrow} + u_k v_k a_{k\uparrow} a_{k\uparrow} - u_k v_k a_{-k\downarrow}^\dagger a_{-k\downarrow}^\dagger - v_k^2 a_{k\uparrow} a_{-k\downarrow}^\dagger.\end{aligned}$$

All terms vanish due to anticommutation

$$\{\rho_{k\uparrow}, \rho_{-k\downarrow}^\dagger\} = 0.$$

iv. $\{\rho_{k\uparrow}, \rho_{-k\downarrow}\}$

$$\begin{aligned}\rho_{k\uparrow}\rho_{-k\downarrow} &= u_k^2 a_{k\uparrow} a_{-k\downarrow} + u_k v_k a_{k\uparrow} a_{k\uparrow}^\dagger - u_k v_k a_{-k\downarrow}^\dagger a_{-k\downarrow} - v_k^2 a_{-k\downarrow}^\dagger a_{k\uparrow}^\dagger, \\ \rho_{-k\downarrow}\rho_{k\uparrow} &= u_k^2 a_{-k\downarrow} a_{k\uparrow} + u_k v_k a_{k\uparrow}^\dagger a_{k\uparrow} - u_k v_k a_{-k\downarrow} a_{-k\downarrow}^\dagger - v_k^2 a_{k\uparrow}^\dagger a_{-k\downarrow}^\dagger.\end{aligned}$$

Adding and simplifying

$$\{\rho_{k\uparrow}, \rho_{-k\downarrow}\} = u_k v_k - u_k v_k = 0.$$

- v. For other anticommutators, by similar computations or Hermitian conjugation

$$\{\rho_{k\uparrow}^\dagger, \rho_{-k\downarrow}^\dagger\} = 0, \quad \{\rho_{k\uparrow}, \rho_{k\uparrow}\} = 0, \quad \text{etc.}$$

In summary, the operators $\rho_{k\sigma}, \rho_{k\sigma}^\dagger$, and their conjugates satisfy

$$\{\rho_{k\sigma}, \rho_{k'\sigma'}^\dagger\} = \delta_{kk'} \delta_{\sigma\sigma'}, \quad \{\rho_{k\sigma}, \rho_{k'\sigma'}\} = 0, \quad \{\rho_{k\sigma}^\dagger, \rho_{k'\sigma'}^\dagger\} = 0.$$

Thus, they are purely Fermionic operators. $\square$

(c) i. The first term

$$[\rho_{k'\uparrow}^\dagger \rho_{k'\uparrow}, \rho_{k\uparrow}^\dagger] = \rho_{k'\uparrow}^\dagger \{\rho_{k'\uparrow}, \rho_{k\uparrow}^\dagger\} - \{\rho_{k'\uparrow}^\dagger, \rho_{k\uparrow}^\dagger\} \rho_{k\uparrow}^\dagger = \delta_{kk'} \rho_{k'\uparrow}^\dagger \cdot 1 - 0 = \delta_{kk'} \rho_{k\uparrow}^\dagger.$$

ii. The second term

$$[\rho_{-k'\downarrow}^\dagger \rho_{-k'\downarrow}, \rho_{k\uparrow}^\dagger] = \rho_{-k'\downarrow}^\dagger \{\rho_{-k'\downarrow}, \rho_{k\uparrow}^\dagger\} - \{\rho_{-k'\downarrow}^\dagger, \rho_{k\uparrow}^\dagger\} \rho_{-k'\downarrow} = 0.$$

Thus, only the term with $k' = k$ contributes

$$[\mathcal{H}, \rho_{k\uparrow}^\dagger] = E_k \rho_{k\uparrow}^\dagger.$$

This result interrupts that $\rho_{k\uparrow}^\dagger$ is an eigenoperator of the Hamiltonian $\mathcal{H}^*$ with eigenvalue E_k . Equivalently, $\rho_{k\uparrow}^\dagger$ to an eigenstate of $\mathcal{H}$ creates an excitation of energy E_k measured from the BCS ground state, confirming that $\rho_{k\uparrow}^\dagger$ is the quasiparticle creation operator.

(d) The retarded Green's function for the quasiparticle is formulated as

$$\hat{G}_{k\uparrow}^{\text{ret}}(E) = \langle \langle \rho_{k\uparrow}; \rho_{k\uparrow}^\dagger \rangle \rangle_E = -i \int_{-\infty}^{\infty} dt e^{iEt} \theta(t) \langle \{ \rho_{k\uparrow}(t), \rho_{k\uparrow}^\dagger(0) \} \rangle.$$

where we used the “God-Given” unit to omit the $\hbar$, and use the Heaviside step function to complete the integral range to the whole time-axis. Apply the equation of motion for the retarded Green's function

$$E \hat{G}_{k\uparrow}^{\text{ret}}(E) = \langle \{ \rho_{k\uparrow}, \rho_{k\uparrow}^\dagger \} \rangle + \langle \langle [\rho_{k\uparrow}, \mathcal{H}^*]; \rho_{k\uparrow}^\dagger \rangle \rangle = 1 + E_k \langle \langle \rho_{k\uparrow}; \rho_{k\uparrow}^\dagger \rangle \rangle_E = 1 + E_k \hat{G}_{k\uparrow}^{\text{ret}}(E),$$

where $\{ \rho_{k\uparrow}, \rho_{k\uparrow}^\dagger \} = 1$. Rearrange the equation of motion, we have

$$(E - E_k) \hat{G}_{k\uparrow}^{\text{ret}}(E) = \mathbb{1}, \quad \hat{G}_{k\uparrow}^{\text{ret}}(E) = \frac{1}{E - E_k + i0^+}$$

where the $i0^+$ enforces retarded boundary conditions.

(e) We have already calculated the retarded Green's function

$$\hat{G}_{k\uparrow}^{\text{ret}}(E) = \langle \langle \rho_{k\uparrow}; \rho_{k\uparrow}^\dagger \rangle \rangle_E^{\text{ret}}.$$

So, similarly, we can replace $\rho_{k\uparrow}$ and $\rho_{k\uparrow}^\dagger$ with $a_{k\uparrow}$, $a_{-k\downarrow}$ and their conjugates. By substituting the inverse transformations obtained in (a)

$$\begin{aligned} a_{k\uparrow}^\dagger &= u_k \rho_{k\uparrow}^\dagger + v_k \rho_{-k\downarrow}, & a_{-k\downarrow}^\dagger &= u_k \rho_{-k\downarrow}^\dagger - v_k \rho_{k\uparrow}, \\ a_{k\uparrow} &= u_k \rho_{k\uparrow} + v_k \rho_{-k\downarrow}^\dagger, & a_{-k\downarrow} &= u_k \rho_{-k\downarrow} - v_k \rho_{k\uparrow}^\dagger. \end{aligned}$$

we can obtain the following Green functions

$$\begin{aligned} G_{\uparrow\uparrow}^{\text{ret}}(k, E) &= \langle \langle a_{k\uparrow}; a_{k\uparrow}^\dagger \rangle \rangle = \frac{u_k^2}{E - E_k + i0^+} + \frac{v_k^2}{E + E_k + i0^+}, & \text{Normal Green's function,} \\ G_{\downarrow\downarrow}^{\text{ret}}(k, E) &= \langle \langle a_{k\downarrow}; a_{k\downarrow}^\dagger \rangle \rangle = \frac{u_k^2}{E - E_k + i0^+} + \frac{v_k^2}{E + E_k + i0^+}, & \text{Normal Green's function,} \\ G_{\uparrow\downarrow}^{\text{ret}}(k, E) &= \langle \langle a_{k\uparrow}; a_{-k\downarrow} \rangle \rangle = -\frac{\Delta}{E^2 - E_k^2 + i0^+}, & \text{Anomalous Green's function,} \\ \tilde{G}_{\downarrow\uparrow}^{\text{ret}}(k, E) &= \langle \langle a_{k\uparrow}^\dagger; a_{-k\downarrow}^\dagger \rangle \rangle = \frac{\Delta}{E^2 - E_k^2 + i0^+}, & \text{Anomalous Green's function,} \end{aligned}$$

and other combinations vanish due to symmetry.

Homework #8 [2025-10-28]

Problem 8.1 (Read material Sec. 6.10 and finish the following exercise.). Consider the Haldane model. The Bloch Hamiltonian is given by

$$\mathcal{H}(\mathbf{k}) = \hat{d}_x(\mathbf{k})\hat{\sigma}_x + \hat{d}_y(\mathbf{k})\hat{\sigma}_y + \hat{d}_z(\mathbf{k})\hat{\sigma}_z,$$

where $\sigma_{x,y,z}$ are Pauli matrices, and the vector $\hat{\mathbf{d}}(\mathbf{k})$ has components

$$d_x(\mathbf{k}) = t_1 \sum_{i=1}^3 \cos(\mathbf{k} \cdot \mathbf{a}_i), \quad d_y(\mathbf{k}) = t_1 \sum_{i=1}^3 \sin(\mathbf{k} \cdot \mathbf{a}_i), \quad d_z(\mathbf{k}) = M - 2t_2 \sin \phi \sum_{i=1}^3 \sin(\mathbf{k} \cdot \mathbf{b}_i),$$

where t_1 is the real NN hopping amplitude, t_2 is the NNN hopping amplitude with complex phase ϕ , M is the sublattice mass (Semenoff mass), and $\mathbf{a}_i, \mathbf{b}_i$ are lattice vectors for NN and NNN hoppings, respectively. The parameter space is spanned by (M, ϕ) . We formally obtain the Green's function as

$$\mathbf{G}[\omega, k] = [\omega \mathbb{1} - \mathcal{H}(k)]^{-1} = \text{adj}(\omega \mathbb{1} - \mathcal{H}(k)) \det[\omega \mathbb{1} - \mathcal{H}(k)]^{-1} = \begin{pmatrix} \frac{\omega + d_z}{\omega^2 - \mathbf{d}^2} & \frac{d_x + i d_y}{\omega^2 - \mathbf{d}^2} \\ \frac{d_x - i d_y}{\omega^2 - \mathbf{d}^2} & \frac{\omega - d_z}{\omega^2 - \mathbf{d}^2} \end{pmatrix}.$$

Try to write the above Green function into self-energy form in the following different ways.

(a) Focusing on a diagonal component, say $\frac{\omega + d_z}{\omega^2 - \mathbf{d}^2}$, write it into a self-energy form.

Remark. Consider $G^0 = (\omega - d_z)^{-1}$ and note $\mathbf{d}^2 = d_x^2 + d_y^2 + d_z^2$. Discuss the “mass renormalization” and “lifetime”. Discuss how to interpret them.

Solution. Since $\mathbf{d}^2 = d_x^2 + d_y^2 + d_z^2$, then, the denominator of the diagonal element of the Green function can be expressed as

$$\omega^2 - \mathbf{d}^2 = (\omega - d_z)(\omega + d_z) - (d_x^2 + d_y^2).$$

So, the numerator and denominator can be divided by $(\omega + d_z)$. Yields,

$$G_{11}(\omega, \mathbf{k}) = \frac{\omega + d_z}{\omega^2 - \mathbf{d}^2} = \frac{1}{(\omega - d_z) - \frac{d_x^2 + d_y^2}{\omega + d_z}}.$$

It can be written as the usual “bare inverse minus self-energy” form

$$G_{11}(\omega, \mathbf{k}) = \frac{1}{G_{11}^0(\omega, \mathbf{k})^{-1} - \Sigma_{11}(\omega, \mathbf{k})},$$

where

$$G_{11}^0(\omega, \mathbf{k})^{-1} \equiv \omega - d_z(\mathbf{k}), \quad \text{and} \quad \Sigma_{11}(\omega, \mathbf{k}) = \frac{d_x(\mathbf{k})^2 + d_y(\mathbf{k})^2}{\omega + d_z(\mathbf{k})}.$$

show that the off-diagonal kinetic terms d_x and d_y appear as an energy-dependent self-energy for the diagonal component.

Physics interpretation This exact representation shows how inter-sublattice hopping (d_x, d_y) modifies the bare propagator $(\omega - d_z)^{-1}$ through an energy-dependent self-energy.

- (a) **Mass Renormalization.** The real part of Σ shifts the quasiparticle pole from d_z to the band energy $|\mathbf{d}| = \sqrt{d_x^2 + d_y^2 + d_z^2}$. Near $\omega \approx d_z$, we have the approximations

$$\Sigma \approx \frac{d_x^2 + d_y^2}{2d_z}, \quad \text{and} \quad d_z^{\text{eff}} \approx d_z + \frac{d_x^2 + d_y^2}{2d_z},$$

showing how inter-sublattice hopping renormalizes the sublattice mass.

- (b) **Lifetime.** The imaginary part $-\text{Im } \Sigma = \pi(d_x^2 + d_y^2)\delta(\omega + d_z)$ is non-zero only at $\omega = -d_z$. Since the band energies are $\pm|\mathbf{d}|$, this causes no broadening at the quasiparticle poles, which interprets that the clean, noninteracting system exhibits infinite lifetime.

CHAPTER 7 Landau Fermi Liquid Theory

To begin with [Superfluid Helium](#) on YouTube.

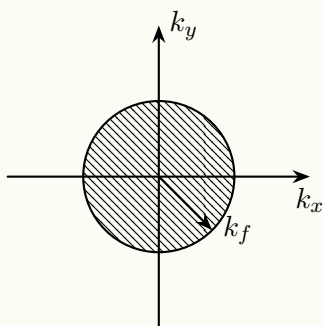
Stick stand straightly in a cup of liquid He $p \propto \frac{\partial S}{\partial n} \frac{\partial n}{\partial T}$, i.e., entropy per particle, $\frac{\partial n}{\partial T} \equiv \nabla n$ is the density gradient. Ref: Volovik, Universe in a drop of He.

LFLT paradigm of many-theory. 1900, Onnes.

- (a) Adiabatic: quasi-particle concept weight, effective mass
- (b) Variational $\frac{\langle n \rangle \rightarrow \delta \langle n \rangle}{\rightarrow}$ functional
- (c) Quantum-statistical: Landau parameters
- (d) Effective theory
- (e) Collective: mode/excitation, zero sound; first sound $\partial n / \partial \mu$.
- (f) $J^{-1}(\epsilon) \propto (\epsilon^2 + \pi^2 T^2)$
- (g) Degenerate Fermi gas of Sommerfeld

$$\epsilon_k = \frac{k^2}{2m} - \mu$$

$$\text{let } k_f^2/2m = \mu, \text{ then } n(\mathbf{k}) = \begin{cases} 1, & |\mathbf{k}| < k_f \\ 0, & |\mathbf{k}| > k_f \end{cases}$$



The density of states at Fermi energy (or Fermi surface along the circle in the Figure above)

$$N(0) = 2 \frac{(4\pi)p^2}{(2\pi\hbar)^3} \frac{dp}{d\epsilon_p} \Big|_{p=p_f} = \frac{mp_f}{\pi^2\hbar^3} \quad (7.1)$$

$$N(E) = \int_{\epsilon_k=0}^{\epsilon_k=E} \frac{d\mathbf{k}}{(2\pi)^2} n(\mathbf{k}) = \int d\epsilon \rho(\epsilon) n(\epsilon) \quad (7.2)$$

where $\rho(\epsilon)$ is the density of state. The total energy

$$E = \int d\epsilon \rho(\epsilon) \epsilon n(\epsilon) \quad (7.3)$$

Similarly, the heat capacity

$$C_v = \frac{dE}{dT} = \frac{\pi^3}{3} N(0) k_0^2 \Big|_{C_v, T \rightarrow 0} = \gamma \quad (7.4)$$

$\partial C_v / \partial T \Big|_{T \rightarrow 0}$. The coefficient $\chi_c = \partial n / \partial \mu$.

(h) magnetic susceptibility.

When a magnetic field $\mathbf{B}$ is applied, we will get an energy difference from spin-up and spin-down $\mu(N_\uparrow - N_\downarrow)\mu_B B$. The magnetic susceptibility

$$\chi = \mu_F(N_\uparrow - N_\downarrow)/B = \mu_F^2 N(0) \quad (7.5)$$

$\mu_F = \mu_B = \frac{e\hbar}{2m} \cdot \mathcal{E}(\epsilon, k, \dots)$, $\mu_B(\epsilon, k, \dots)$. The Wilson ratio

$$W = \frac{\chi}{\gamma} = 3 \left(\frac{\mu_F}{\pi k_B} \right)^2 \quad (7.6)$$

$v_i = \frac{k^2}{2m} \approx \frac{2k_F \delta k}{2m}$, $m^* = (2.8)m_{(3\text{He})}$, and the effective magnetic moment $(g^*)^2 = 3.3(g^2)_{3\text{He}}$. $\epsilon(k) \sim m^*$.

The low energy excitation of unperturbed Hamiltonian.

(a) Quasi-particle $c_{p_0\sigma_0}^\dagger$:

$$n_{p\sigma} = \begin{cases} 1, & (p < p_f), \text{ or } p = p_0, \sigma = \sigma_0 \\ 0, & \text{otherwise} \end{cases}$$

(b) Quasi-hole $c_{k_0\sigma_0}$:

$$n_{p\sigma} = \begin{cases} 1, & p < p_f \text{ except } p = p_0, \sigma = \sigma_0 \\ 0, & \text{otherwise} \end{cases}$$

(c) Particle-hole pair (create one and annihilate one at the same time).

On the Fermi surface,

$$[H, n_{p_f, \sigma}] = 0 \quad (p_f \in \text{FS})$$

If $p - p_f = \delta p$, then the function will be linear to δp . When taking the limit

$$[H, n_{p_f, \sigma}] \propto \delta p \Big|_{\delta p \rightarrow 0} \rightarrow 0$$

Then, the expectation

$$\langle \Psi_0 | [H, n_p, \sigma] | \Psi_0 \rangle \propto \delta p \Big|_{\delta \rightarrow p}$$

The residual scattering remains on the Fermi surface, which is forward scattering. The two particles

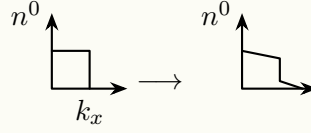
$$(\mathbf{p}_1, \mathbf{p}_2) \rightarrow (\mathbf{p}_1 - \mathbf{q}, \mathbf{p}_2 + \mathbf{q}), \quad \text{and} \quad \mathbf{q} = 0$$

form the restricted scattering phase space, $\text{Mod}(\mathbf{q}) = 0$. It is a narrow window, i.e., thin shell, near the Fermi surface.

7.1 Change-neutral Fermi Liquid (With short range interaction: $\sim 1/r^6$)

$$\delta n_{p\sigma} = n_{p\sigma} - n_{p\sigma}^{(0)}$$

from equilibrium static



Then the functional equation

$$E = E_0 + \sum_{p\sigma} (E_{p\sigma}^{(0)-\mu}) \delta n_{p\sigma} + \frac{1}{2} \sum_{pp'\sigma\sigma'} f_{p\sigma,p'\sigma'} \delta n_{p\sigma} \delta n_{p'\sigma'} + \dots \quad (7.7)$$

means that $\delta n_{p\sigma}(E_{p\sigma}^{(0)}, \mu) E_{p\sigma}^{(0)}(\delta n_{p\sigma}, \dots) f(\delta n_p, E_p^{(0)}, \dots)$.

$$\left. \frac{\delta E}{\delta n_{p\sigma}} \right|_{\delta n_{p'\sigma'}=0} = \epsilon_{p\sigma}^{(0)} = E_{p\sigma}^{(0)} - \mu, \quad v_f = \left. \frac{d\epsilon_p}{dp} \right|_{p=p_f} = \frac{p_f}{m^*}, \quad \text{and} \quad N^*(0) = \frac{m^* p_F}{\pi^2 \hbar^3} \quad (7.8)$$

Then, we can also the landau parameters as the second order perturbation

$$f_{p\sigma,p'\sigma'} = \left. \frac{\delta^2 E}{\delta n_{p\sigma} \delta n_{p'\sigma'}} \right|_{\delta n_{p''\sigma''}=0} \quad (7.9)$$

From the functional equation, we can have

$$\frac{\delta E}{\delta n_{p\sigma}} = \epsilon_{p\sigma} = \epsilon_{p\sigma}^{(0)} + \sum_{p'\sigma'} f_{p\sigma,p'\sigma'} \delta n_{p'\sigma'}$$

then the entropy

$$S \cong -k_B \sum_{p,\sigma} [n_{p\sigma} \ln n_{p\sigma} + (1 - n_{p\sigma}) \ln(1 - n_{p\sigma})]$$

Just mathematically, no coherent term

$$\langle n_p n_{p'} \rangle_C = \langle n_p n_{p'} \rangle - \langle n_p \rangle \langle n_{p'} \rangle$$

the really defiened $S = -\rho \ln \rho = f(\langle n \rangle, \langle n^2 \rangle, \langle n^3 \rangle, \dots)$.

Combine the two quantities

$$F(\{n_{p\sigma}\}) = E_0(\mu) + \sum_{p\sigma} \epsilon_{p\sigma}^{(0)} \delta n_{p\sigma} + \frac{1}{2} \sum_{pp'\sigma\sigma'} f_{pp'\sigma\sigma'} \delta n_{p\sigma} \delta n_{p'\sigma'} + k_B T \sum [n_{p\sigma} \ln n_{p\sigma} + (1 - n_{p\sigma}) \ln(1 - n_{p\sigma})], \quad (7.10)$$

and $\delta F = 0$. The (Helmholtz) Free energy is

$$F = E - ST$$

where

$$E = \int dk n_k \epsilon_k, \quad N = \int dk n_k, \quad \text{and} \quad E_0 = \int dk \theta(|k - k_F|) \epsilon_k$$

Since $\delta n = n_k - \theta(k - k_f)$, and $n^0 = \theta$, then

$$\delta E = \int dk \epsilon_k \delta n_k$$

to do the variation δn_k

$$\delta n = \underbrace{\nu^{(0)}(k)}_{\rightarrow \delta p_f} + \sum_{\delta(k-k_F) \sim \partial_k \theta} \underbrace{\frac{\partial n^0}{\partial k}}_{\nu_k^{(1)}} + \frac{1}{2} \frac{\partial^2 n^0}{\partial k_i \partial k_j} \nu_{k_i, k_j}^{(2)} + \dots$$

7.2 Landau Fermi Liquid Theory

The energy can be written as

$$E = E_0 + \sum \epsilon_{p\sigma}^{(0)} \delta n_{p\sigma} + \frac{1}{2} \sum_{pp'\sigma\sigma'} f_{pp'\sigma'\sigma} \delta n_{p\sigma} \delta n_{p'\sigma'} \quad (7.11)$$

where $\epsilon_{p\sigma}^{(0)}$ is the functional derivation

$$\epsilon_{p\sigma}^{(0)} = \left. \frac{\delta E}{\delta n_{p\sigma}} \right|_{\text{1st order}}, \quad (7.12)$$

with $|p| = |p_F|$ and the heat capacity $C_v \propto N^*(i)$. The Fermi velocity is defined of the Fermi surface

$$v_F = \left. \frac{d\epsilon^{(0)}}{dp} \right|_{P_F} = \frac{p_F}{m^*} \quad (7.13)$$

In 2nd order theory, we get the actual physical particle energy

$$\left. \frac{\delta E}{\delta n_{p\sigma}} \right|_{\text{2st order}} = \epsilon_{p\sigma} = \delta \epsilon_{p\sigma}^{(0)} + \underbrace{\sum_{p\sigma} f_{pp'\sigma'\sigma} \delta n_{p'\sigma'}}_{\text{Renormalization part}} \quad (7.14)$$

The free energy for the Fermi liquid

$$F = E - TS[\{\delta n_{p\sigma}\}] \quad (7.15)$$

Example 7.2.1. Consider the Hamiltonian

$$\mathcal{H} = \sum E_p n_{p\sigma} + \frac{1}{2} \sum V(q) c_{p-q\sigma}^\dagger c_{p'+q\sigma'}^\dagger c_{p'\sigma'} c_{p\sigma}$$

Denote $\langle \Psi | \mathcal{H} | \Psi \rangle$, where $|\Psi\rangle = |n_{p_1\sigma_1} n_{p_2\sigma_2} \dots\rangle$. Then, the energy

$$E = \sum E_p n_{p\sigma} + \frac{\lambda}{2} \sum V(q) \langle \Psi | \dots | \Psi \rangle$$

we can pair

- (a) $c_{p'+q\sigma'}^\dagger$ with $c_{p'\sigma'}$, then $p' + q$ with $p': q = 0$. (c) $c_{p'+q\sigma'}^\dagger$ with $c_{p\sigma}$, then $p' + q = p$.
 (b) $c_{p-q\sigma}^\dagger$ with $c_{p\sigma}$, then $p - q$ with $p: q = 0$. (d) $c_{p-q\sigma}^\dagger$ with $c_{p'\sigma'}$, then $p - q = p'$.

Then we will get

$$(\delta_{q=0} - \delta_{p-q,p'}\delta_{\sigma\sigma'})n_{p\sigma}n_{p'\sigma'}$$

the energy will become

$$E = \sum E_p n_{p\sigma} + \frac{\lambda}{2} \sum (V(0) - V(p-p')\delta_{\sigma\sigma'})n_{p\sigma}n_{p'\sigma'}$$

with

$$f_{p\sigma p'\sigma'} = \lambda(V(0) - V(p-p')\delta_{\sigma\sigma'})$$

which is so-called the Fermion-Renormalization group. Ref: Shanker, RMP; or Polchinski on arXiv, EFT & Fermi Surface.

7.2.1 Feedback effects of interaction

$$\delta\epsilon_{p\sigma}^0 = \begin{cases} -\delta\mu, & \text{isotropic enlarge / compression of Fermi surface} \\ -\sigma\mu_F B, & \text{magnetic field,} \\ \frac{1}{2}m(\mathbf{v} - \mathbf{u})^2, & \text{shift in } \mathbf{v}, \text{ in } \mathbf{u}, \text{ the Galilean boost.} \end{cases} \quad (7.16)$$

The actual physical particle energy

$$\delta\epsilon_{p\sigma} = \delta\epsilon_{p\sigma}^0 + \sum f_{p\sigma p'\sigma'} \delta n_{p'\sigma'} \quad (7.17)$$

where

$$n_{p\sigma} = f(\epsilon_{p\sigma}^0 + \delta\epsilon_{p\sigma}) \underset{T \rightarrow \sigma}{\approx} f(\epsilon_p^0) + f'(\epsilon_p^{(0)})\delta\epsilon_{p\sigma} = \underbrace{\theta(-\epsilon_p^0)}_{n_{p\sigma}^0} + \underbrace{[-\delta(\epsilon_p^0), \delta\epsilon_{p\sigma}]}_{\delta n_{p\sigma}}$$

i.e., $n_{p\sigma} \rightarrow n_{p\sigma}^0$, and $\delta n_{p\sigma} \rightarrow \delta n_{p'\sigma'}$. So, the actual physical particle energy can be written as

$$\delta\epsilon_{p\sigma} = \delta\epsilon_{p\sigma}^0 + \sum_{p'} f_{pp'}(-\delta(\epsilon_{p'}^0)) \cdot \delta\epsilon_{p'}$$

where

$$f_{pp'} = \sum_l f_l P_l(\cos \theta), \quad \delta\epsilon_p^{(0)} = v_l Y_{lm}(\theta, \phi), \quad \delta_{p\sigma} = \sum_l t_l Y_{lm}$$

then, we obtain

$$t_l = v_l - F_l^s t_l = \frac{v_l}{1 + F_l^s}$$

We shall only consider the leading term, i.e. effective mass per coleman

$$v_1 \propto \delta\epsilon_p^0 \approx -m\mathbf{v}_F \cdot \mathbf{u} = -m \cos \theta p_F \propto P_1(\cos \theta)$$

Then, we have

$$t_1 = \frac{v_1}{1 + F_1^s}$$

There will have a displacement of the fermi surface.

The effective mass satisfies

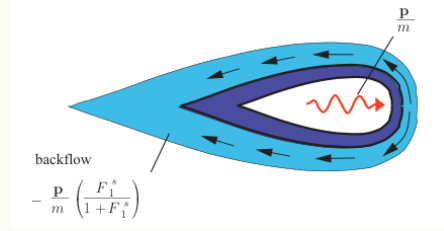
$$\frac{d\epsilon}{dp} = \frac{p_F}{m^*} = \frac{p_F}{m} + \frac{F_1^s p_F}{m^*}$$

then, it can be expressed as

$$m^* = m(1 + F_1^s)$$

V.s. the band effect mass,

$$m^* = m \left(1 + \frac{1}{3} F_1^s \right)$$



7.2.2 Equilibrium Properties

We specific head

$$\delta S = \frac{1}{TV} \sum (\epsilon_p - \mu) \delta n_p, \quad (7.18)$$

$$C_V = S_1 = \frac{m^* p_F}{3\hbar^2} k_B^2 T. \quad (7.19)$$

E.Fradin. We can write

$$\delta n_\sigma(T, \mu, \epsilon_p) = \frac{\partial n}{\partial \epsilon} [-(\epsilon_p - \mu) \delta T + \delta \epsilon_p \delta \mu] \frac{\delta S}{\delta T} \quad (7.20)$$

Then, we have

$$\begin{aligned} \delta S &= -\frac{1}{V} \sum_p \frac{\partial n}{\partial \epsilon} (\epsilon_p - \mu^2) \delta T = -\frac{1}{V} \underbrace{\int p^2 \frac{dp}{d\epsilon}}_{3D} d\epsilon_p \left(\frac{\partial n}{\partial \epsilon} \right)_p (\epsilon_p - \mu)^2 \delta T \\ &= -k_B^2 N^*(0) \int dx \frac{\partial}{\partial x} \left(\frac{1}{e^x + 1} \right) x^2 \delta T \end{aligned} \quad (7.21)$$

The Fermi energy for Landau Fermi liquid $E_F \propto T_F$, then

$$T_F = \frac{p_F^2}{2m^* k_B} = \frac{\epsilon_F}{k_B} \quad (7.22)$$

Then,

$$\mu'(n, T) = -\left(\frac{\partial F}{\partial n} \right)_T = \mu(n, 0) - \frac{T_1^2}{4} k_s \left(\frac{1}{3} + \frac{n}{m^*} \frac{\partial m^*}{\partial n} \right) \frac{T^2}{T_F} \quad (7.23)$$

The compressibility, $\delta \epsilon_p^{(0)=-\delta \mu}$. Then,

$$\kappa = -\frac{1}{V} \frac{\partial V}{\partial p} = \frac{1}{n^2} \frac{\partial n}{\partial \mu} \quad (7.24)$$

which is so-called the charge susceptibility. Then,

$$\delta n_p = \frac{\partial n}{\partial \epsilon_p} (\delta \epsilon_p - \delta \mu),$$

where $\delta E_p = F_0^S \delta n_{pF}$. Substitute it

$$\delta n_p = -N(0)(F_D^S \delta n - \delta \mu) \quad (7.25)$$

Then, we can obtain the relation between δn and $\delta \mu$

$$\frac{\delta n}{\delta \mu} = \frac{\partial n}{\partial \mu} = \frac{N(0)}{1 + F_0^S}. \quad (7.26)$$

The Fermi sphere get smaller

$$\delta \epsilon_p^{(0)} = -\sigma \mu_F \beta \begin{cases} \delta \mu_h = +\frac{B}{2}, \\ \delta \mu_l = -\frac{B}{2}, \end{cases}, \quad \frac{\partial S_Z}{\partial B} = \frac{\delta n_{\uparrow} - \delta n_{\downarrow}}{B}, \quad \chi = \frac{\hbar^2}{2} \gamma^2 N(0) \frac{1}{1 + p_0^s}.$$

7.2.3 Thermodynamic stability of Fermi liquid

$$n_{p\sigma}^0 = \theta(p_F(0) - |\mathbf{p}|) \quad (7.27)$$

Since the Gibbs free energy

$$G = E - \mu n \quad (7.28)$$

then

$$(E - \mu n) - (E - \mu n)_0 = \frac{1}{V} \sum (\epsilon_p^0 - \mu) \delta n_p + \frac{1}{2V^2} \sum \delta p_F(\theta) f_{p\gamma'} \delta n_p \delta n_{p'} \quad (7.29)$$

and

$$\delta n_p = n_p - n_p^0 = \delta p_F \delta(p_F - |\mathbf{p}|) - \frac{1}{2} (\delta p_F)^2 \frac{\partial}{\partial p} ((p_F - p)) \quad (7.30)$$

where

$$\delta p_F(\theta) = p_F(\theta) - p_F^0(\theta) \longrightarrow v_F \delta p_F(\theta) = \sum_{l=0}^{\infty} v_l P_l(\cos \theta)$$

The 1st term

$$\delta - \mu \delta n = \sum \frac{N(0)}{(2l+1)\delta} \left[(v_{l\uparrow} + v_{l\downarrow})^2 \left(1 + \frac{F_l^S}{2l+1} \right) + (v_{l\uparrow} - v_{l\downarrow})^2 \left(1 + \frac{F_l^A}{2l+1} \right) \right].$$

We should find the Fermi-stability condition

$$\delta E - \mu \delta n > 0, \quad \text{and} \quad 1 + \frac{F_l^{S,A}}{2l+1} \geq 0, \quad (7.31)$$

and the Pometanchuk inequility (transition)

$$F_l^{S,A} \geq -(2l+1). \quad (7.32)$$

If p -inequility, violated, then $\delta p_F = v_{l_0}$, which will change the Fermi surface shape.

7.2.4 Non-Equilibrium Properties

Steady state

7.2.5 Changed Fermi liquid: Landau-Silin theory

$\delta n_p(x)$:

$$\nabla^2 \phi_p = \frac{e}{\epsilon_0} \sum_{p'} \delta n_{p'}(x) = \frac{e}{\epsilon_0} \delta n(x) \quad (7.33)$$

We can have

$$\epsilon_{p\sigma}(x) = \epsilon_p^{(0)} + e\phi_p + \sum_{p'\sigma'} f_{p\sigma p'\sigma'} \delta n_{p'}(x) \quad (7.34)$$

Then, take the Fourier transform

$$\delta \epsilon_{p\sigma}(q) = e\phi_p(q) + \sum_{p'\sigma'} f_{p\sigma p'\sigma'} \underbrace{\delta n_{p'\sigma'}(\epsilon)}_{\delta n(\delta \epsilon_p)} = \sum_{p'\sigma'} \left(\frac{e^2}{\epsilon_0 q^2} + f_{p\sigma p'\sigma'} \right) \delta n_{p'\sigma'}(q) \quad (7.35)$$

where

$$\tilde{f}_{p\sigma p'\sigma'}(q) = \frac{e^2}{\epsilon_0 q^2} + f_{p\sigma p'\sigma'} = \frac{N^*(0)}{1 + \left(\frac{e^2 N^*(0)}{\epsilon_0 q^2} + F_0^s \right)} \equiv \chi_c(q) \propto \frac{\partial n}{\partial \mu}$$

where we can approximately write

$$1 + \left(\frac{e^2 N^*(0)}{\epsilon_0 q^2} + F_0^s \right) \propto 1 + \frac{\kappa^2}{q^2}$$

with $q \rightarrow 0$. Then,

$$\delta \mu = \delta \mu \cdot \delta(x) 9, \quad \text{the Fourier Transform} \quad \delta n(x - x_0) = \mathcal{F}(\delta n(q)) \rightarrow \delta n(x) \delta n(x_0) \sim -\kappa e^{-\kappa|x-x_0|}$$

where

$$\kappa \cong \frac{e^2 N^*(0)}{\epsilon_0} \frac{1}{2} (1 + F_0^s)$$

The above is the so-called Thomas-Fermi screening.

Consider the quasi-particle length scale

$$l \gg \frac{\hbar}{\Delta p} \sim \frac{\hbar v_F}{k_B T} \quad (7.36)$$

$\delta n_p(x)$ is the Wigner distribution function, where $p \rightarrow \langle p \rangle$ and $x \rightarrow \langle x \rangle$. The wavepacket of δn_p

$$W(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2) = \int d\mathbf{p}_1^3 d\mathbf{p}_2^3 e^{i/\hbar(\mathbf{p}_1 \mathbf{r}_1 - \mathbf{p}_2 \mathbf{r}_2)} \langle a_{p_2 \sigma_2}^\dagger(t) a_{p_1 \sigma_1}(t) \rangle \quad (7.37)$$

The wave function of q and p : $U^\dagger c_{p\sigma}^\dagger U$. Ref: Fradkin's.

7.2.6 Kinetic equation

$$\frac{d}{dt} \delta n_{p\sigma}(\mathbf{r}, t) = 0 \quad (7.38)$$

where

$$[\delta n_{p=p_F}, H] = 0, \quad \text{and} \quad [\delta n_{p \neq p_F}, H] \neq 0$$

Then, q.p. will decay $\rightarrow$ q.p. collision, cause the quasi-particle lifetime

$$p_1 \leftrightarrow p_2 : p_3, p_4.$$

If we take the collisionless limit, then only achievable of $T = 0$. Then term $I(\delta n_{p'})$, the collision integral, will describe the process of quasi-particle lifetime.

The derivatives

$$\begin{aligned}\frac{d}{dt}\delta n_{\mathbf{p}}(\mathbf{r}, t) &= \frac{\partial}{\partial t}\delta n + \frac{\partial}{\partial \mathbf{r}}\left(\frac{d\mathbf{r}}{dt}\delta n_{\mathbf{p}}(\mathbf{r}, t)\right) + \frac{\partial}{\partial \mathbf{p}}\frac{d\mathbf{p}}{dt}\delta n_{\mathbf{p}}(\mathbf{r}, t), \\ \frac{d\mathbf{r}_p}{dt} &= \mathbf{v}_{\mathbf{p}} = \frac{\partial \epsilon_{\mathbf{p}}(\mathbf{r}, t)}{\partial \mathbf{p}}, \\ \frac{d\mathbf{p}}{dt} &= \mathbf{f}_{\mathbf{p}}(\mathbf{r}, t) = -\frac{\partial}{\partial \mathbf{r}}\epsilon_{\mathbf{p}}(\mathbf{r}, t)\end{aligned}$$

Then, the Landau's kinetic equation becomes

$$\frac{\partial}{\partial t}\delta n_{\mathbf{p}}(\mathbf{r}, t) - \{\epsilon_{\mathbf{p}}(\mathbf{r}, t), \delta n_{\mathbf{p}}(\mathbf{r}, t)\} = \begin{cases} 0, & T \rightarrow 0, \\ T(\delta n_{p'}), & T \text{ is finite} \end{cases} \quad (7.39)$$

where the Poisson brackets

$$\{\epsilon_p, \delta n_p\}_{\text{PB}} = \frac{\partial}{\partial \mathbf{r}}\epsilon_p \cdot \frac{\partial}{\partial \mathbf{p}}\delta n_p - \frac{\partial \epsilon_p}{\partial \mathbf{p}} \cdot \frac{\partial \delta n_p}{\partial \mathbf{r}}$$

The external potential $U(\mathbf{r}, t)$ satisfies

$$\int d\mathbf{r} U(\mathbf{r}, t) \delta n_{\mathbf{p}}(\mathbf{r}, t)$$

Then, the derivative becomes

$$\frac{\partial \epsilon_{\mathbf{r}}}{\partial \mathbf{r}} = \frac{\partial U}{\partial \mathbf{r}} + \int \frac{d^3 \mathbf{p}'}{(2\pi\hbar)^3} f_{pp'} \frac{\partial}{\partial \mathbf{r}} \delta n_{\mathbf{p}}(\mathbf{r}, t)$$

To linearize this equation: keep to the 1st order derivative in $\mathbf{r}$

$$\frac{d}{dt}\delta n = \frac{\partial}{\partial t}\delta n + \frac{\partial}{\partial \mathbf{r}}(\mathbf{v}_p \cdot \delta n) + \frac{\partial}{\partial \mathbf{p}} - \frac{\partial \epsilon}{\partial \mathbf{r}} \cdot \delta n = 0$$

where $\epsilon = \epsilon^0 + \delta\epsilon$, and the term

$$\frac{\partial \epsilon}{\partial \mathbf{r}} \cdot \delta n \sim \frac{\partial \delta \epsilon}{\partial \mathbf{r}} \cdot \frac{\partial}{\partial \mathbf{p}} \delta n$$

where $\frac{\partial}{\partial \epsilon} \frac{\partial \epsilon}{\partial \mathbf{p}} \delta$ the first derivative act on δn . Then $\frac{d}{dt}\delta n$ satisfies

$$\frac{\partial}{\partial t}\delta n + \mathbf{v}_p \frac{\partial}{\partial \mathbf{r}} \delta n - \frac{\partial}{\partial \epsilon} n_p^0 \delta \epsilon_p = \begin{cases} 0, \\ T(\delta n) \end{cases}$$

Apply the Fourier transformation to $t \rightarrow \omega$, $\mathbf{r} \rightarrow \mathbf{q}$

$$(\omega - \mathbf{q} \cdot \mathbf{v}_p) \delta n_{\mathbf{p}}(\mathbf{q}, \omega) - \mathbf{q} \cdot \mathbf{v}_p \frac{\partial n_p^0}{\partial \epsilon_p} \delta \epsilon_p(\mathbf{q}, \omega) = \begin{cases} 0, \end{cases}$$

The conservation laws

$$n(\mathbf{r}, t) = \int d^3\mathbf{p} n_p(\mathbf{r}, t)$$

Since $\int d^3\mathbf{p} [t, n_{p'}] = 0$, $\int d\mathbf{p} \frac{\partial}{\partial p} [f_p, n_p] = 0$, and the continuity equation

$$\frac{\partial}{\partial t} n(\mathbf{r}, t) + \nabla \cdot \mathbf{j} = 0$$

then we have

$$\mathbf{j} = \sum_{p, \sigma} \mathbf{v}_p \cdot n_{p, \sigma} = \sum \nabla_p \epsilon_p \cdot n_p \quad (7.40)$$

The energy conservation law (thermal transport $\Delta U(\mathbf{r}, t)$)

$$\frac{\partial}{\partial t} (E - Un) - \nabla \mathbf{j}_E = -\mathbf{j} \cdot \nabla \mathbf{u} \quad (7.41)$$

Then, we have

$$\mathbf{j}_E = \int d\mathbf{p} \nabla_r \epsilon_p(\mathbf{r}, t) (\epsilon_p - U) n_{p\sigma}$$

and also the gradient energy.

7.3 Collective modes: zero sound

$$\delta n_p(\mathbf{r}, t) = \sum_q \delta n_p(q, \omega) e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)}$$

and the potential

$$U(\mathbf{r}, t) = U e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)}$$

The frequency satisfies

$$(\omega - \mathbf{q} \cdot \mathbf{v}_p) \delta n_p(q, \omega) + \frac{\partial n_p^0}{\partial \epsilon_p} \mathbf{q} \cdot \mathbf{v}_p (u + \int d\mathbf{p}' f_{pp'} \delta n_{p'}(q, \omega)) = 0 \quad (7.42)$$

where

$$\delta n_p = -\frac{\partial n_p^0}{\partial \epsilon_p} v_p, \quad \text{and} \quad v_p = \sum_{l=0}^{\infty} v_l P_l(\cos \theta)$$

Substitute them

$$v_p + \frac{\mathbf{q} \cdot \mathbf{v}_p}{\omega - \mathbf{q} \cdot \mathbf{v}_p} \int d\mathbf{p}' f_{pp'} \frac{\partial n^0}{\partial \epsilon_p} = \frac{\mathbf{q} \cdot \mathbf{v}_p}{\omega - \mathbf{q} \cdot \mathbf{U}_p} U$$

Then, we have the single frequency

$$S = \frac{\omega}{qv_F}, \quad \text{and} \quad \Omega_{ll'}(S) = \frac{1}{2} \int_{-1}^1 dx P_l(x) \frac{x}{x - S} P_{l'}(x)$$

To diagonal l

$$\frac{v_l}{2l+1} + \sum_{l'=0}^{\infty} \Omega_{ll'} F_{l'}^s \frac{v_{l'}}{2l'+1} = -\Omega - l_0 U$$

take $l' = l = 0$

$$\Omega_{00} = 1 + \frac{S}{2} \ln \left| \frac{S-1}{S+1} \right| + i \frac{\pi}{2} S \theta(1 - |S|).$$

When S is small, we will omit the imaginary part

$$v_0(s) = -\frac{\Omega_{00}(s)U}{1 + F_0^S \Omega_{00}(S)}$$

Take $0 \leq S < 1$ (1t sound damped), then

$$V_0(S) \rightarrow \delta n_p(x, t) \sim \delta n_p^{S_0}(x, t) \propto e^{-S},$$

which is the so-called $p - n$ -continuity. When $S > 1$,

$$1 + F_0^{(S)} \Omega_{00}(S_0) = 0, \quad \text{zero sound}$$

means that the Fermi liquid self is resonating with external driving force, corresponding to an collective eigenmode, and there is none dissipative in the collisionless limit.

The frequency $S_0 = \frac{\omega_0}{qv_F}$, i.e., the dispersion relation of the collective eigenmode.

7.4 Non-equilibrium properties: linear response

Consider the time derivative of the number variable

$$\frac{d}{dt} \delta n_p(\mathbf{r}, t) = 0 \quad \text{or} \quad \mathbb{I}[\delta n_p] \quad (7.43)$$

- (a) $\frac{d}{dt} \delta n_p(\mathbf{r}, t) = 0$: Collisionless limit, $T \rightarrow 0$.
- (b) $\frac{d}{dt} \delta n_p(\mathbf{r}, t) = \mathbb{I}[\delta n_p]$: Collision integral: Inelastic CMP scattering (decay), or interaction V.s. collision

7.4.1 Inelastic quasi-particle scattering

For fermi liquid

$$\Gamma = \frac{1}{\tau} \propto \epsilon^2 + \pi^2 T^2 k_B^2 \quad (7.44)$$

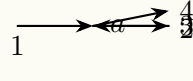
in Landau Fermi Temperature cd, $T \rightarrow 0$, $\epsilon \rightarrow 0$, then $\frac{1}{\tau} \rightarrow 0$, $\tau \rightarrow \infty$.

The finite lifetime: at finite T , we have finite energy,

$$\sigma(a) = \frac{1}{\omega^2}.$$

This decay happens is due to the “particle-particle interaction”. We consider the case one particle traveling, then goes into multi particles, i.e.,

Decay


 $\hat{V} : v(q)c_4c_3c_2^\dagger; c_1^\dagger$

When it acts, to the left and right part, we have

$$\langle \Psi_{43\bar{2}} | = \langle \Psi_0 | c_4 c_3 c_2^\dagger; \quad \text{and} \quad |\Psi_1\rangle = c_1^\dagger |\Psi_0\rangle$$

respectively, where $\langle \Psi_{43\bar{2}} | \Psi_1 \rangle \neq 0$, and $[H, n_p] = 0$.

The correspondence between particle and hole (acts to Fermi state)

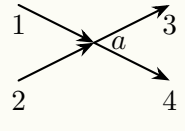
$$c_{p < p_F}^\dagger |FS\rangle = 0, \quad \text{and} \quad c_{-p > p_F} |FS\rangle = 0 \quad (7.45)$$

where we can define the operator

$$\tilde{\gamma}_p = \begin{cases} c_{p < p_F}^\dagger, \\ c_{-p > p_F} \end{cases}$$

the minus sign is respected to the Fermi surface. which will always annihilate a state, also for the operator $\gamma_p^\dagger$, always create a state.

Scattering If we change the comma to the center, i.e.,


 $\hat{V} : v(q)c_4c_3; c_2^\dagger c_1^\dagger \quad (7.46)$

which c_4 and c_3 act to the left, and the others acts to the right. Then, we have

$$a(1 \rightarrow \bar{2} + 3 + 4) = a(1 + 2; 3 + 4)$$

From which we can obtain the samp matrix element $\hat{\gamma}_p^\dagger$

$$\langle \Psi_0 | c_4 c_3; c_2^\dagger c_1^\dagger | \Psi_0 \rangle$$

It is the same wherever the comma is.

7.4.2 Correspondence between between & holes

The operator and its Hermite conjugate

$$\begin{cases} \gamma \rightarrow c_p^\dagger, & \epsilon_p, p, \\ \gamma^\dagger \rightarrow c_{-p}, & \epsilon_p, -p. \end{cases}$$

where p is measured with respect / relative to k_F . We have “ $1 \leftrightarrow 1$ ” correspondence, need the same.

Remark. Since the Fermi surface form a circle, the area in and out the circle should have the same density of states. When $T \rightarrow 0$ (or $\epsilon \rightarrow 0$) valid, they have the same number of states.

If we only take a small curve of the surface, then the difference $\Delta n_p - \Delta n_f \sim \kappa$, the curvature of the curve. Which is about *Differential geometry*

$$\frac{d^2\rho}{dp^2} \sim K^2$$

The Fermi Golden rule

$$I[\delta n_p] = \frac{2\pi}{\hbar} \sum_{2,3,4} |a(12;34)|^2 [-n_1(1-n_4)(1-n_3)n_2 + (1-n_1)(1-n_2)n_3n_4] (2\pi\hbar)^3 \times \delta(p_1 + p_2 - p_3 - p_4) \delta(\epsilon_1 + \epsilon_2 - \epsilon_3 - \epsilon_4) \quad (7.47)$$

where the counter is just counting the *available states*.

$$1 \begin{matrix} \nearrow 2 \\ \rightarrow 3 \\ \searrow 4 \end{matrix}, \quad \text{also for the Hermite conjugate, i.e., the inverse process.}$$

$$I[\delta n_p] \text{ is the dynamical process of } n_1 \begin{cases} -mn_1 \rightarrow \\ +pn_1 \rightarrow \end{cases} n_1. \text{ at thermal equilibrium } I[\delta n_p] = 0.$$

$$\delta n_p = n_p - n_p^{(0)} \quad \text{caused by } \delta^{(0)}\epsilon_p$$

Then

$$I[\delta n_p] = \underbrace{-\Gamma \delta n_1}_{\text{relaxion}} + \dots,$$

where $\Gamma = -\delta I / \delta n_1$. Also, we have

$$\Gamma = \frac{2\pi}{\hbar} \sum_{2,3,4} |a(1,2,3,4)|^2 [n_2^{(0)}(1-n_3^{(0)})(1-n_4^{(0)}) + (1-n_2^{(0)})n_3^{(0)}n_4^{(0)}] \quad (7.48)$$

The sum can be extend to integral

$$\sum_{234} \rightarrow \int_{(p_2, p_3, p_4)} \delta(\{p_i\}) \delta(\epsilon_1 + \epsilon_2 - \epsilon_3 - \epsilon_4) \int d\epsilon$$

Also, the sum term is proportion to

$$\sum_{2,3,4} |a(1,2,3,4)|^2 \propto |N^*(0)|^3$$

and the residue proportion to $\frac{1}{2}(\epsilon_1^2 + (\pi k_B T)^2)$. It can be written as

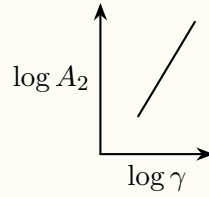
$$\Gamma = (m^*)^3 \cdot \frac{1}{8\pi^4 \hbar^6} \langle \frac{W(0, \phi)}{2\omega_0} \rangle_{\Omega} [\epsilon^2 + (k_B \pi T)^2]. \quad (7.49)$$

where W stands for the integral.

7.4.3 Kadowaki-Woods ratio

$$\rho(T) = \rho_0 + A_2 T^2 \underbrace{\pm A_1 T}_{T \propto E_F \sim 10^4 \text{ K/M}}, \quad \gamma = \left. \frac{C_v}{T} \right|_{T \rightarrow 0} \propto m^* \quad (7.50)$$

The ratio

$$\frac{A_2}{\gamma^2} \sim \text{Const}$$


7.4.4 Microscopic basis Landau Fermi Liquid Theory

Due to the GML theory

$$|\phi\rangle = U|\Psi_0\rangle, \quad |\tilde{k}\sigma\rangle = U|k\sigma\rangle \xrightarrow{\text{canonical transition}} U c_{k\sigma}^\dagger U^\dagger |\phi\rangle$$

Denote

$$Z_k = |\langle \tilde{k}_0 | c_{k\sigma}^\dagger | \phi \rangle|^2$$

at noninteracting $Z_k = 1$.

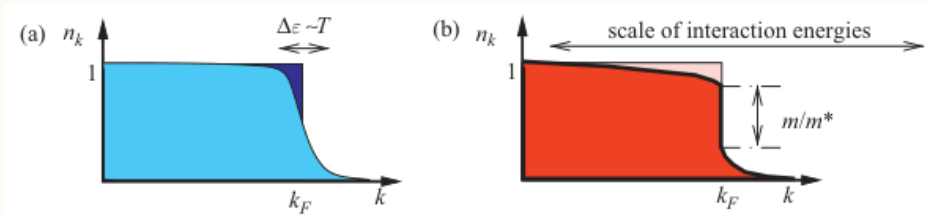
$$c_{k\sigma}^\dagger \approx \sqrt{Z_k} a_{k\sigma}^\dagger + \sum_{k_4 k_3 = k_2 + k} A_1(\quad) a_k^\dagger a_3^\dagger a_2 + \dots$$

$$a_{k\sigma}^\dagger \approx \frac{1}{\sqrt{Z_k}} c_{k\sigma}^\dagger + \sum c_4^\dagger c_3^\dagger c_2 + \dots$$

Then, the factor

$$A(k, \omega) \cong \frac{1}{\pi} \text{Im}[G(k, \omega - i\delta)] = \underbrace{\text{qp peak}}_{Z_{k\sigma} \delta(\omega - \epsilon_k)} + \sum_{\lambda \neq k\sigma} |M_\lambda|^2 \delta(\omega - \epsilon_\lambda) \quad (7.51)$$

The sum term is continuum, incoherent port. The curve of A respect to ω becomes from “square” to a peak after interaction.



with $m/m^* \sim Z_{k_F}$.

Homework #9 [2025-11-04]

Problem 9.1. Calculate the Landau parameters to leading order in $\lambda_{1,2}$ for a Fermi liquid with the contact interactions

- (a) $V(\mathbf{x} - \mathbf{x}') = \lambda_1 \delta^{(3)}(\mathbf{x} - \mathbf{x}')$.
- (b) $V(\mathbf{x} - \mathbf{x}') = -\lambda_2 \nabla^2 \delta^{(3)}(\mathbf{x} - \mathbf{x}')$ (so that $V(q) = \lambda_1 q^2$ in Fourier space).
- (c) Taking the results of (a) and (b) literally, sketch the regions of the λ_1, λ_2 phase diagram where the Fermi surface becomes unstable.

Solution. For a spin-independent two-body interaction $V(\mathbf{q})$, the Landau interaction function in the forward direction

$$f_{\mathbf{p}\sigma, \mathbf{p}'\sigma'} = [V(0) - V(|\mathbf{p} - \mathbf{p}'|)] \delta_{\sigma\sigma'}.$$

Decomposing into spin-symmetric f^s and spin-antisymmetric f^a parts

$$f^s(\theta) = V(0) - \frac{1}{2}V(q), \quad f^a(\theta) = -\frac{1}{2}V(q),$$

where $q = |\mathbf{p} - \mathbf{p}'| = 2k_F \sin(\theta/2)$, $\theta = \langle \mathbf{p}, \mathbf{p}' \rangle$. Expanding in Legendre polynomials

$$f^s(\theta) = \sum_l f_l^s P_l(\cos \theta), \quad f^a(\theta) = \sum_l f_l^a P_l(\cos \theta).$$

Hence, the Landau parameters are

$$F_l^s = N(0) f_l^s, \quad F_l^a = N(0) f_l^a.$$

where $N(0) = \frac{mk_F}{2\pi^2}$ is the density of states per spin at the Fermi level.

- (a) In the case of $V(\mathbf{x} - \mathbf{x}') = \lambda_1 \delta^{(3)}(\mathbf{x} - \mathbf{x}')$.

Do the Fourier transform

$$V(\mathbf{q}) = V(0) = \lambda_1,$$

then

$$f_l^s(\theta) = f_0^s = \lambda_1 - \frac{\lambda_1}{2} = \frac{\lambda_1}{2}, \quad f_l^a(\theta) = f_0^a = -\frac{\lambda_1}{2}.$$

Hence, the Landau parameters are

$$F_0^s = \frac{N(0)\lambda_1}{2}, \quad F_0^a = -\frac{N(0)\lambda_1}{2}, \quad F_{l>0} = 0.$$

- (b) In the case of $V(\mathbf{x} - \mathbf{x}') = -\lambda_2 \nabla^2 \delta^{(3)}(\mathbf{x} - \mathbf{x}')$.

Do the Fourier transform

$$V(\mathbf{q}) = \lambda_2 q^2, \quad \text{and} \quad V(0) = 0,$$

then

$$f^s(\theta) = f^a(\theta) = -\frac{\lambda_2}{2} q^2 \stackrel{q^2=2k_F^2(1-\cos\theta)}{=} -\lambda_2 k_F^2 (1 - \cos \theta).$$

Expanding in Legendre polynomials

$$f^s(\theta) = -\lambda_2 k_F^2 + \lambda_2 k_F^2 \cos \theta = -\lambda_2 k_F^2 + \lambda_2 k_F^2 P_1(\cos \theta),$$

Then

$$f_0^s = f_0^a = -\lambda_2 k_F^2, \quad f_1^s = f_1^a = \lambda_2 k_F^2, \quad f_{l \geq 2} = 0.$$

Hence, the Landau parameters are

$$F_0^s = F_0^a = -N(0)\lambda_2 k_F^2, \quad F_1^s = F_1^a = N(0)\lambda_2 k_F^2, \quad F_{l > 1} = 0.$$

(c) When both interactions are present, the Landau parameters add linearly

$$F_0^s = \frac{N(0)\lambda_1}{2} - N(0)\lambda_2 k_F^2, \quad F_0^a = -\frac{N(0)\lambda_1}{2} - N(0)\lambda_2 k_F^2, \quad F_1^s = F_1^a = N(0)\lambda_2 k_F^2.$$

The Pomeranchuk stability conditions are

$$1 + \frac{F_l^s}{2l+1} > 0, \quad 1 + \frac{F_l^a}{2l+1} > 0.$$

For $l = 0$: $1 + F_0^s > 0$, $1 + F_0^a > 0$; For $l = 1$: $1 + \frac{F_1^s}{3} > 0$, $1 + \frac{F_1^a}{3} > 0$. Define two coefficients $A = \frac{N(0)}{2}$, $B = N(0)k_F^2$, then

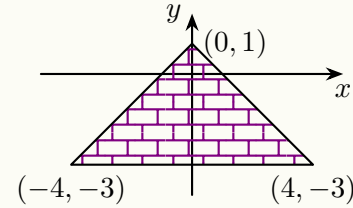
$$F_0^s = A\lambda_1 - B\lambda_2, \quad F_0^a = -A\lambda_1 - B\lambda_2, \quad F_1^s = F_1^a = B\lambda_2.$$

Let $x = A\lambda_1$, $y = B\lambda_2$, the stability conditions become

$$\text{i. } 1 + x - y > 0 \quad \text{ii. } 1 - x - y > 0 \quad \text{iii. } 1 + \frac{y}{3} > 0.$$

Do the linear programming, the stable region in the (x, y) -plane is a triangle on the right. In terms of λ_1, λ_2 , the stable region is inside the triangle with vertices

$$\left(-\frac{4}{A}, -\frac{3}{B}\right), \quad \left(\frac{4}{A}, -\frac{3}{B}\right), \quad \left(0, \frac{1}{B}\right).$$



Problem 9.2. Test your understanding of Landau's mass renormalization formula by generalizing it to include the effect of a magnetization. Suppose we introduce a second vector potential into (6.86)

$$A(\theta) \underset{|\mathbf{k}| \rightarrow 0}{\sim} \int_{k_F - k \cos \theta}^{k_F} dq \frac{2i\pi a^2}{\omega - v_F(|\mathbf{k} + \mathbf{q}| - q)} = \frac{2i\pi a^2 k \cos \theta}{\omega - v_F k \cos \theta}. \quad (6.86)$$

that couples to the spin current, writing

$$\mathcal{H}[\mathbf{A}_N, \mathbf{W}] = \sum_{\sigma} \int d^3x \frac{1}{2m} \psi_{\sigma}^{\dagger}(x) [(-i\hbar \nabla - \mathbf{A}_N - \sigma \mathbf{W})^2] \psi_{\sigma}(x) + \hat{V}.$$

Whereas $\mathbf{A}_N$ couples to the current of particles, $\mathbf{W}$ couples to the (z component of the) spin current. Assume that V conserves spin current.

- (a) By comparing the bare shift of the energies

$$\delta\epsilon_{\mathbf{p}\sigma}^{(0)} = -\frac{\mathbf{p}}{m} \cdot (\mathbf{A}_N + \sigma \mathbf{W}),$$

with the shift that results from interaction feedback,

$$\delta\epsilon_{\mathbf{p}\sigma} = -\frac{\mathbf{p}}{m^*} \cdot \mathbf{A}_N - \sigma \frac{\mathbf{p}}{m_s^*} \cdot \mathbf{W},$$

show that there are two different mass renormalizations,

$$\frac{m}{m^*} = \frac{1}{1 + F_1^s}, \quad \frac{m}{m_s^*} = \frac{1}{1 + F_1^a}.$$

- (b) Show that, when the Fermi liquid is polarized, the masses of the “up” and “down” quasiparticles are now different, and given by

$$\frac{1}{m_\sigma^*} = \frac{1}{m} \left[\frac{1}{1 + F_1^s} + \frac{M}{1 + F_1^a} \right], \quad (\sigma = \uparrow, \downarrow),$$

where the magnetization $M = n_\uparrow - n_\downarrow$ is the difference of “up” and “down” densities.

Solution.

- (a) *Proof.* In the steady state, the distribution function is a shifted Fermi sphere, then

$$\delta n_{\mathbf{p}\sigma} = -\frac{\partial n^0}{\partial \epsilon} (\mathbf{p} \cdot \mathbf{u}_\sigma),$$

where $\mathbf{u}_\sigma$ is the drift velocity for σ . Substitute it and $\delta\epsilon_{\mathbf{p}\sigma}^{(0)}$ to get the full quasiparticle energy shift

$$\delta\epsilon_{\mathbf{p}\sigma} = \delta\epsilon_{\mathbf{p}\sigma}^{(0)} + \sum_{\mathbf{p}'\sigma'} f_{\mathbf{p}\sigma, \mathbf{p}'\sigma'} \delta n_{\mathbf{p}'\sigma'} = -\frac{\mathbf{p}}{m} \cdot (\mathbf{A}_N + \sigma \mathbf{W}) - \sum_{\mathbf{p}'\sigma'} f_{\mathbf{p}\sigma, \mathbf{p}'\sigma'} \frac{\partial n^0}{\partial \epsilon'} (\mathbf{p}' \cdot \mathbf{u}_{\sigma'}).$$

Due to isotropy, the interaction term simplifies. The Landau interaction function is decomposed into spin-symmetric and spin-antisymmetric parts

$$f_{\mathbf{p}\sigma, \mathbf{p}'\sigma'} = f^s(\theta) \delta_{\sigma\sigma'} + f^a(\theta) (\sigma\sigma').$$

where θ is the angle between $\mathbf{p}$ and $\mathbf{p}'$. Expanding in Legendre polynomials

$$f^s(\theta) = \sum_l f_l^s P_l(\cos \theta), \quad f^a(\theta) = \sum_l f_l^a P_l(\cos \theta).$$

The Landau parameters are defined as:

$$F_l^s = N(0) f_l^s, \quad F_l^a = N(0) f_l^a,$$

where $N(0)$ is the density of states at the Fermi level per spin. After evaluating the angular integrals, the energy shift becomes

$$\delta\epsilon_{\mathbf{p}\sigma} = -\frac{\mathbf{p}}{m} \cdot (\mathbf{A}_N + \sigma \mathbf{W}) + \frac{1}{3} [F_1^s (\mathbf{p} \cdot \mathbf{u}_\sigma) + F_1^a \sigma (\mathbf{p} \cdot (\mathbf{u}_\uparrow - \mathbf{u}_\downarrow))].$$

In the steady state, the effective force on quasiparticles vanishes, i.e., $\delta\epsilon_{p\sigma} = \mathbf{p} \cdot \mathbf{u}_\sigma$. Comparing with the formula above, we can obtain $\mathbf{u}_\sigma$

$$\mathbf{u}_\sigma = -\frac{1}{m^*} \mathbf{A}_N - \sigma \frac{1}{m_s^*} \mathbf{W}.$$

Then, comparing coefficients yields the desired relations

$$\frac{m}{m^*} = \frac{1}{1 + F_1^s}, \quad \frac{m}{m_s^*} = \frac{1}{1 + F_1^a}.$$

- (b) *Proof.* When the system is magnetized ($M = n_\uparrow - n_\downarrow \neq 0$), the Fermi surfaces for spin-up and spin-down are different. The drift velocities become

$$\mathbf{u}_\uparrow = \mathbf{u} + \mathbf{u}_s, \quad \mathbf{u}_\downarrow = \mathbf{u} - \mathbf{u}_s,$$

where $\mathbf{u}$ is the charge drift velocity and $\mathbf{u}_s$ is the spin drift velocity. The energy shifts are

$$\delta\epsilon_{p\uparrow} = -\frac{\mathbf{p}}{m^*} \cdot \mathbf{A}_N - \frac{\mathbf{p}}{m_s^*} \cdot \mathbf{W}, \quad \delta\epsilon_{p\downarrow} = -\frac{\mathbf{p}}{m^*} \cdot \mathbf{A}_N + \frac{\mathbf{p}}{m_s^*} \cdot \mathbf{W}.$$

The effective masses for each spin species are then:

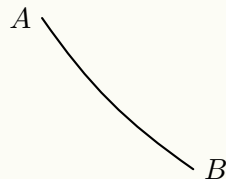
$$\frac{1}{m_\sigma^*} = \frac{1}{m} \left[\frac{1}{1 + F_1^s} + \sigma \frac{M}{1 + F_1^a} \right], \quad \text{for } \sigma = \uparrow, \downarrow,$$

where M is the magnetization.

CHAPTER 8 (WD)Particle-hole-pairs (PHPs)

CHAPTER 9 The semi-classical (true) variational approach (WD)

Starting from the two problems



$$\delta \int d\tau = 0$$



$$\delta \int dl F = 0$$

We actually let $\delta \int d\mathcal{L} = 0$, i.e., the Euler-Lagrangian EOM. If it contains the interaction term, then we have the *Differential-integro-equation*

9.1 Variational calculation for free fermions

$$F[n(E)] = \tilde{E}[n(E)] - TS[n(E)] = \int dE \phi(E, n)$$

where $S = -n \ln n - (1 - n) \ln(1 - n)$. The particle number conservation

$$N = \int dE g(E) n(E), \quad \tilde{E} = \int dE E n(E)$$

We have to minimize the density function

$$\frac{\delta F}{\delta n(E)} \Rightarrow \frac{\partial \phi}{\partial n} = \dots \Rightarrow n^0(E) = \frac{1}{e^{(E-M)/T} + 1} \quad (9.1)$$

also for $\frac{\delta F}{\delta \mu}$.

9.1.1 Generic variation around n^0

The energy

$$E_0 = \int \frac{d^a k}{(2\pi)^a} n_k^0(\epsilon(k))$$

where δ expanding some functional (function of functions)

$$\delta n_k(\epsilon(k)) = \nu^0(k) + \underbrace{\sum_{i=x_i} \frac{\partial n}{\partial k_i} \Big|_{n_k^1 \rightarrow n^0}}_{\delta(k-k)F, \text{ The quasi-particle weight}} \nu_{k_i}^{(1)} + \frac{1}{2} \sum_{i,j} \frac{\partial^2 n}{\partial k_i \partial k_j} \underbrace{\nu_{k_i k_j}^{(2)}}_{\frac{\partial^2 n}{\partial k_i \partial k_j}}$$

To the energy,

$$\delta E_0 = \delta \int \frac{dk}{2\pi} n_k \epsilon_k = \int \frac{dk}{2\pi} \epsilon_k \delta n_k$$

To put δn_k into it,

$$\frac{\delta E}{\delta n} = \frac{\delta}{\delta n} (\delta E_0 + V(\delta n_k)) = 0$$

Now, focus on $T = 0$, then $n_k^{(0)} = \Theta(\epsilon(k) - \mu)$, $\epsilon_{k_F} - \mu = 0$

$$\frac{\partial n_k^{(0)}}{\partial k} = \delta(\epsilon(k) - \mu) \partial_{k_i} \epsilon|_{k \rightarrow k_F}$$

Then, the second derivative

$$\frac{\partial^2 n^{(0)}}{\partial k_i \partial k_j} = \left[\underbrace{\delta(\epsilon - \mu)}_{\delta(k_{\perp} - k_{F,\perp})} \partial_{k_i k_j}^2 \epsilon(k)|_{k \rightarrow k_F} + \delta'(\epsilon - \mu) \partial_{k_i} \epsilon \partial_{k_j} |_{k \rightarrow k_F} \right]$$

Note that $\delta(k_{\perp} - k_{F,\perp}) \neq \delta(k - k_r)$, since the shape of the Fermi surface is arbitrary. where we can separate the motion around the Fermi surface into the parallel and perpendicular terms. In the simple case single connected Fermi surface

$$k_F(\theta) = (k_{r,F}(\theta) \cos \theta, k_{r,F}(\theta) \sin \theta)$$

where $k_{r,F}(\theta) = k_{r,F_0} + k^0 \cos \theta + \dots$

Now, we can build the connection between energy and

$$\delta E = \int d\theta dk_r k_r (\epsilon - \mu) (\nu^{(0)} + \delta(k_r - k_{r,F}(\theta))) (\partial_{k_r} \epsilon \nu_{k_r} + k_r^{-1} \partial_{\theta} \epsilon \nu_{\theta} + \frac{1}{2} \partial_{k_r}^2 \epsilon \nu_{k,k_r} + \dots) + \frac{1}{2} \delta'(\epsilon(k) - \mu) (\quad) \quad (9.2)$$

where

$$\nu_{k_r} = \frac{\nu_{k_x}}{\cos \theta} + \frac{\nu_{k_y}}{\sin \theta}, \quad \nu_{k_{\theta}} = -\frac{\nu_{k_x}}{\sin \theta} + \frac{\nu_{k_y}}{\cos \theta}$$

where the derivative to the δ -function can be expressed as

$$\delta'(\epsilon - \mu) = \partial_{\epsilon} \delta(k_r(\epsilon) - k_{r,F}(\epsilon)) = (\partial_{\epsilon} k_r(\epsilon) - \partial_{\epsilon} \theta(\theta) \partial_{\theta} k_{r,F}) \delta'(k_r - k_{r,F})$$

and

$$\epsilon(k) \Rightarrow k(\epsilon) = \epsilon^{(-1)}(k) = [(\partial_{k_r} \epsilon)^{-1} + (\partial_{\theta} \epsilon)^{-1} \partial_{\theta} k_{r,F}] \delta'(\dots)$$

The δE can have a compact form

$$\delta E = \delta E^{(0)} + \int d\theta dk_r k_r \delta^{(T)}(k_r - k_{r,F}) [(\epsilon - \mu) \partial_{k_r} F + \partial_{k_r}(\epsilon F)] \quad (9.3)$$

where

$$F = [(\partial_{k_r} \epsilon)^{-1} - (\partial_{\theta} \epsilon)^{-1} \partial_{\theta} k_{r,F}] \frac{1}{2} (\partial_{k_r} \epsilon)^2 \nu_{k_r, k_{\theta}} z + \dots + (\quad) \nu_{k_r, k_r} + (\quad) \nu_{k_{\theta} k_{\theta}}$$

9.1.2 Understanding Fermi Liquid: Particle-hole-pairs (LPHPS)

Bosonization – 1D case.

The “vacuum” state $|\mathbf{0}\rangle_0$, where $c_k|\mathbf{0}\rangle = 0$, with $k > 0$. The particle number N ,

$$:\hat{A}\hat{B}\hat{C}:=ABC - \langle \mathbf{0}|ABC|\mathbf{0}\rangle$$

For Bosonic, particle-hole-pair operators

$$b_q^\dagger = \frac{1}{\sqrt{n_q}} \sum_{k=-\infty}^{\infty} c_{k+q}^\dagger c_k, \quad b_q = -\frac{i}{\sqrt{n_q}} \sum c_{k-q}^\dagger c_k$$

where $q = \frac{2\pi}{L}n_q$, $(b_q^\dagger)^\dagger \rightarrow b_q \rightarrow b_{-q}^\dagger$. The particle number

$$N_q = b_q^\dagger b_q$$

If $q > 0$, then without the term $b_{-q}^\dagger$.

The commutator

$$[b_q, b_{q'}] = \delta_{q,q'},$$

$$[b_q^\dagger, b_{q'}^\dagger] = \sum \frac{1}{n_1} (: c_{k+q-q'}^\dagger c_k - c_{k+q}^\dagger c_{k+q'} :) = \sum_{k=-\infty}^{\infty} \frac{1}{n_q} (: c_k^\dagger c_k - c_{k+q}^\dagger c_{k+q} :) = \delta_{qq'} + \mathcal{O}(1/k_F)$$

where $|\mathbf{0}\rangle$ is bosonic ground state

$$b_q|\mathbf{N}_0\rangle = 0$$

Look at a typical Fermonic Hamiltonian

$$\mathcal{H} = \sum E_p n_p + \frac{\lambda}{2} \sum V(q) \underbrace{c_{p-q}^\dagger c_{p'+q}^\dagger c_{p'} c_p}_{\frac{\lambda}{2} \sum_{q,k_F,k'_F} V(q) k_{q,k_F}^\dagger b_{q,k_F'}} \quad (9.4)$$

The first sum term

$$[n_p, b_q^\dagger] = \pm \delta_{p,q} n_q$$

Then, we obtain the spectral generating algebra

$$\sum_{q>0} q b_{q,k_F}^\dagger b_{q,k_F} + \frac{\lambda}{2} \sum_{q,k_F,k'_F} b_{q,k_F}' b_{q,k_F'} \equiv \mathcal{H}_{\text{eff}}$$

i.e., the integral equation of θ . The energy

$$E(q) \begin{cases} \rightarrow q/m^*, \\ \rightarrow \text{Collative hole zero sound} \end{cases}$$

and the matrix form

$$\begin{pmatrix} q & V(1) & k_f k_{F'} \\ & \ddots & \ddots \\ \ddots & & q \end{pmatrix}$$

From $|\mathbf{N}_0\rangle \rightarrow |\Phi_0\rangle$, $n^0 \rightarrow \delta n_p$ jump Z from Z_1 to Z_k ,

$$\langle c^\dagger c^\dagger c c \rangle \rightarrow \langle b_{qk_F}^\dagger b_{qk_{F'}} \rangle \propto 1 - Z_k$$

Identify the *Low energy excitations* (of $|\mathbf{N}_0\rangle$). The interaction will tend to lower the excited energy. The interaction will populate the occupy excitations. If do the canonical

$$U|\Phi_0\rangle \Rightarrow (\dots b^\dagger b^\dagger \dots)|\mathbf{N}_0\rangle$$

Homework #10 [2025-11-11]

Problem 10.1. Interpreting the Fermi liquid via a Schrieffer-Wolff transformation. We start from a translationally invariant, spin-1/2 fermion system

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_{\text{int}},$$

where

$$\mathcal{H}_0 = \sum_{k,\sigma} \epsilon_k c_{k\sigma}^\dagger c_{k\sigma}, \quad \mathcal{H}_{\text{int}} = \frac{1}{2V} \sum_{k,k',q,\sigma,\sigma'} V_q c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{k\sigma}.$$

The Fermi sea $|\text{FS}\rangle$ is the ground state of $\mathcal{H}_0$, with all states $|k| < k_F$ filled. We assume V_q is weak and smooth near $q = 0$.

To do a canonical transformation for a Fermi liquid state, we want to integrate out (cancel out) **off-shell** excitations. This is different from previous cases, where the original interacting terms are canceled out entirely. Instead, for a Fermi liquid state, we write $\mathcal{H}_{\text{int}}$ schematically as

$$\mathcal{H}_{\text{int}} = \mathcal{H}_{\text{diag}} + \mathcal{H}_{\text{off}},$$

where

- $\mathcal{H}_{\text{diag}}$ conserves the number of quasiparticles near the Fermi surface (i.e., forward scattering, density-density type).
- $\mathcal{H}_{\text{off}}$ mixes sectors with different numbers of particle-hole pairs (e.g., it creates or destroys particle-hole pairs relative to the Fermi sea).

Formally,

$$\mathcal{H}_{\text{diag}} = \frac{1}{2V} \sum_{\substack{k,k',q \\ \text{both } k, k'+q \text{ near } k_F}} V_q c_{k+q}^\dagger c_{k'-q}^\dagger c_{k'} c_k,$$

and $\mathcal{H}_{\text{off}}$ is the rest.

Such separation can be formally done via the semi-classical variational method A simplified version is written

$$\mathcal{H}_{\text{diag}} = \frac{1}{2V} \sum_{\substack{k,k',q \\ \text{both } k = |k+q| = |k'-q| = |k'| = k_F, q \rightarrow 0}} V_q c_{k+q}^\dagger c_{k'-q}^\dagger c_{k'} c_k.$$

So you can see $\mathcal{H}_{\text{off}}$ is just $\mathcal{H}_{\text{int}}$ with at least one momentum that is away from the Fermi surface. We can just keep it in the form of $\mathcal{H}_{\text{int}}$.

We choose $\mathcal{S}$ such that

$$\mathcal{H}_{\text{off}} + [\mathcal{S}, \mathcal{H}_0] = 0.$$

This means $\mathcal{S}$ is chosen to cancel the leading-order off-diagonal part of $\mathcal{H}_{\text{int}}$ under the transformation.

Then the transformed Hamiltonian becomes

$$\mathcal{H}' = \mathcal{H}_0 + \mathcal{H}_{\text{diag}} + \frac{1}{2}[\mathcal{S}, \mathcal{H}_{\text{off}}] + \mathcal{O}(V^3),$$

where the new effective Hamiltonian is block-diagonal up to order V^2 .

Constructing $\mathcal{S}$ explicitly.

The commutator $[\mathcal{S}, \mathcal{H}_0]$ acts as

$$[\mathcal{S}, \mathcal{H}_0] = \sum_{\alpha\beta} S_{\alpha\beta}(\epsilon_\alpha - \epsilon_\beta)c_\alpha^\dagger c_\beta,$$

so to satisfy $[\mathcal{S}, \mathcal{H}_0] = -\mathcal{H}_{\text{off}}$, we can write

$$\mathcal{S} = \sum_{\alpha\beta} \frac{(\mathcal{H}_{\text{off}})_{\alpha\beta}}{\epsilon_\alpha - \epsilon_\beta}.$$

This is the Schrieffer-Wolff generator, which mixes states differing in unperturbed energy. For the present problem, $\mathcal{S}$ connects a bare fermion state $c_{k\alpha}^\dagger|\text{FS}\rangle$ to configurations with one additional particle-hole pair.

Now, answer the following questions.

- (a) Evaluate $[\mathcal{S}, c_{k\sigma}^\dagger]$;
- (b) Transforming the creation operator - “ressing” the bare fermion. The transformed fermion creation operator is

$$\tilde{c}_{k\sigma}^\dagger = e^{\mathcal{S}} c_{k\sigma}^\dagger e^{-\mathcal{S}} = c_{k\sigma}^\dagger + [\mathcal{S}, c_{k\sigma}^\dagger] + \frac{1}{2}[\mathcal{S}, [\mathcal{S}, c_{k\sigma}^\dagger]] + \dots$$

to the first order in V

$$\tilde{c}_{k\sigma}^\dagger \approx c_{k\sigma}^\dagger + [\mathcal{S}, c_{k\sigma}^\dagger].$$

- (c) Compute $\tilde{c}_{k\sigma}^\dagger|\text{FS}\rangle$.
- (d) Compute quasiparticle weight $Z_k \equiv |\langle\text{FS}|\tilde{c}_k^\dagger\tilde{c}_k^\dagger|\text{FS}\rangle|^2$ from the canonical transformation.

Solution.

- (a) Starting from the identity of $\mathcal{S}$,

$$[\mathcal{S}, \mathcal{H}_0] = -\mathcal{H}_{\text{off}}.$$

To get the matrix elements of $\mathcal{S}$, substitute the commutator above into the states $\langle m|$ and $|n\rangle$ ($m \neq n$)

$$-\langle m|\mathcal{H}_{\text{off}}|n\rangle = \langle m|[\mathcal{S}, \mathcal{H}_0]|n\rangle = \langle m|\mathcal{S}|n\rangle E_n - E_m \langle m|\mathcal{S}|n\rangle = (E_n - E_m) \langle m|\mathcal{S}|n\rangle,$$

where $\langle m|\mathcal{H}_0 = \langle m|E_m$ and $\mathcal{H}_0|n\rangle = E_n|n\rangle$. Then,

$$\langle m|\mathcal{S}|n\rangle = \frac{\langle m|\mathcal{H}_{\text{off}}|n\rangle}{E_m - E_n} \quad \text{for } m \neq n,$$

where we set the diagonal elements $\langle n|\mathcal{S}|n\rangle = 0$. By inserting $|m\rangle\langle n|$ to form the identity $\mathbb{1}$, the operator $\mathcal{S}$ in Dirac notation can be expressed as

$$\mathcal{S} = \sum_m \sum_n |m\rangle \frac{\langle m|\mathcal{H}_{\text{off}}|n\rangle}{E_m - E_n} \langle n|$$

So, we can generate $\mathcal{S}$ from the four operators in $\mathcal{H}_{\text{off}}$, which is equivalent to the four operators in $\mathcal{H}_{\text{int}}$ but without the diagonal elements, that is

$$\mathcal{S} = \sum_{k,k',q,\sigma,\sigma'} S_{kk'q}^{\sigma\sigma'} c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{k\sigma},$$

Then, substitute it into the condition $[\mathcal{S}, \mathcal{H}_0] = -\mathcal{H}_{\text{off}}$ to determine $S_{kk'q}^{\sigma\sigma'}$. To distinguish the indices, we need to write $\mathcal{H}_0$ as

$$\mathcal{H}_0 = \sum_{p,\tau} \epsilon_p c_{p\tau}^\dagger c_{p\tau}$$

Then, the commutator is

$$[\mathcal{S}, \mathcal{H}_0] = \sum_{k,k',q,\sigma,\sigma'} S_{kk'q}^{\sigma\sigma'} \sum_{p,\tau} \epsilon_p [c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{k\sigma}, c_{p\tau}^\dagger c_{p\tau}].$$

Compute the kernel first

$$\begin{aligned} [c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{k\sigma}, c_{p\tau}^\dagger c_{p\tau}] &= [c_{k+q,\sigma}^\dagger, c_{p\tau}^\dagger c_{p\tau}] c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{k\sigma} + c_{k+q,\sigma}^\dagger [c_{k'-q,\sigma'}^\dagger, c_{p\tau}^\dagger c_{p\tau}] c_{k'\sigma'} c_{k\sigma} \\ &\quad + c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger [c_{k'\sigma'}, c_{p\tau}^\dagger c_{p\tau}] c_{k\sigma} + c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} [c_{k\sigma}, c_{p\tau}^\dagger c_{p\tau}]. \end{aligned}$$

The four terms

- i. $[c_{k+q,\sigma}^\dagger, c_{p\tau}^\dagger c_{p\tau}] = -c_{p\tau}^\dagger \delta_{p,k+q} \delta_{\tau,\sigma}$
- iii. $[c_{k'\sigma'}, c_{p\tau}^\dagger c_{p\tau}] = c_{p\tau} \delta_{k',p} \delta_{\sigma',\tau}$
- ii. $[c_{k'-q,\sigma'}^\dagger, c_{p\tau}^\dagger c_{p\tau}] = -c_{p\tau}^\dagger \delta_{p,k'-q} \delta_{\tau,\sigma'}$
- iv. $[c_{k\sigma}, c_{p\tau}^\dagger c_{p\tau}] = c_{p\tau} \delta_{kp} \delta_{\sigma\tau}$

where we used the anti-commutative properties of the Fermions

$$\{c_{i,j}, c_{i',j'}^\dagger\} = \delta_{i,i'} \delta_{j,j'}, \quad \{c_{i,j}, c_{i',j'}\} = \{c_{i,j}, c_{i',j'}^\dagger\} = 0.$$

Substitute them into the second sum of the commutator

$$\begin{aligned} \sum_{p,\tau} \epsilon_p [c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{k\sigma}, c_{p\tau}^\dagger c_{p\tau}] &= - \sum_{p,\tau} \epsilon_p c_{p\tau}^\dagger \delta_{p,k+q} \delta_{\tau,\sigma} c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{k\sigma} \\ &\quad - \sum_{p,\tau} \epsilon_p c_{k+q,\sigma}^\dagger c_{p\tau}^\dagger \delta_{p,k'-q} \delta_{\tau,\sigma'} c_{k'\sigma'} c_{k\sigma} \\ &\quad + \sum_{p,\tau} \epsilon_p c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{p\tau} \delta_{k',p} \delta_{\sigma',\tau} c_{k\sigma} \\ &\quad + \sum_{p,\tau} \epsilon_p c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{p\tau} \delta_{kp} \delta_{\sigma\tau}. \end{aligned}$$

Due to the sifting property of the δ -function, the second sum of the commutator becomes (here take the minus sign out since $[\mathcal{S}, \mathcal{H}_0] = -\mathcal{H}_{\text{off}}$)

$$\sum_{p,\tau} \epsilon_p [c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{k\sigma}, c_{p\tau}^\dagger c_{p\tau}] = -(\epsilon_{k+q} + \epsilon_{k'-q} - \epsilon_{k'} - \epsilon_k) c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{k\sigma}.$$

Hence, the commutator $[\mathcal{S}, \mathcal{H}_0]$ becomes

$$-\mathcal{H}_{\text{off}} = [\mathcal{S}, \mathcal{H}_0] = - \sum_{k,k',q,\sigma,\sigma'} S_{kk'q}^{\sigma\sigma'} (\epsilon_{k+q} + \epsilon_{k'-q} - \epsilon_{k'} - \epsilon_k) c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{k\sigma}$$

To compare with the expression of $\mathcal{H}_{\text{off}}$

$$\mathcal{H}_{\text{off}} = \frac{1}{2V} \sum_{k,k',q,\sigma,\sigma'} V_q c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{k\sigma} \quad \text{where } k \neq k', \sigma \neq \sigma'.$$

Then, we arrive at

$$\begin{aligned} \sum_{k,k',q,\sigma,\sigma'} S_{kk'q}^{\sigma\sigma'} c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{k\sigma} &= \mathcal{S} \\ &= \frac{1}{2V} \sum_{k,k',q,\sigma,\sigma'} \frac{V_q}{\epsilon_{k+q} + \epsilon_{k'-q} - \epsilon_k - \epsilon_{k'}} c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{k\sigma}. \end{aligned}$$

This is the second-quantized form of $\mathcal{S}$. Now, calculate the commutator $[\mathcal{S}, c_{k\sigma}^\dagger]$. Calculate the kernel

$$[c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{k\sigma}, c_{k\sigma}^\dagger]$$

first. To distinguish the indices, we take $c_{k\sigma}^\dagger \rightarrow c_{p\tau}^\dagger$

$$\begin{aligned} [c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} c_{k\sigma}, c_{p\tau}^\dagger] &= c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger (c_{k'\sigma'} \{c_{k\sigma}, c_{p\tau}^\dagger\} - \{c_{k'\sigma'}, c_{p\tau}^\dagger\} c_{k\sigma}) \\ &= c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} \delta_{kp} \delta_{\sigma\tau} - c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger \delta_{k'p} \delta_{\sigma'\tau} c_{k\sigma}, \end{aligned}$$

where $c_{p\tau}^\dagger$ only “knock” on $c_{k'\sigma'}$ and $c_{k\sigma}$ effectively due to the anti-commutative properties of the Fermions. We define the function

$$\Phi(p_1, p_2, q) = \frac{1}{\epsilon_{p_1+q} + \epsilon_{p_2-q} - \epsilon_{p_1} - \epsilon_{p_2}}.$$

Then, the commutator becomes

$$\begin{aligned} [\mathcal{S}, c_{p\tau}^\dagger] &= \sum_{k,k',q,\sigma,\sigma'} \frac{V_q \Phi(k, k', q)}{2V} (c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} \delta_{kp} \delta_{\sigma\tau} - c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger \delta_{k'p} \delta_{\sigma'\tau} c_{k\sigma}) \\ &= \sum_{k',q,\sigma'} \frac{V_q \Phi(p, k', q)}{2V} c_{p+q,\tau}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} - \sum_{k,q,\sigma} \frac{V_q \Phi(k, p, -q)}{2V} c_{k+q,\sigma}^\dagger c_{p-q,\tau}^\dagger c_{k\sigma}. \end{aligned}$$

Swap dummy variables in the second sum: $k \rightarrow k', \sigma \rightarrow \sigma', q \rightarrow -q$, and use $c_{p+q}^\dagger c_{k'-q}^\dagger = -c_{k'-q}^\dagger c_{p+q}^\dagger$

$$[\mathcal{S}, c_{p\tau}^\dagger] = \sum_{k',q,\sigma'} \frac{V_q \Phi(p, k', q)}{2V} c_{p+q,\tau}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'} + \sum_{k,q,\sigma} \frac{V_{-q} \Phi(k', p, -q)}{2V} c_{p+q,\tau}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'},$$

where $\Phi(p, k', q) = \Phi(k', p, -q)$. Combine the two terms, and restore the indices $c_{p\tau}^\dagger \rightarrow c_{k\sigma}^\dagger$

$$[\mathcal{S}, c_{k\sigma}^\dagger] = \sum_{k',q,\sigma'} \frac{V_q + V_{-q}}{2V} \frac{c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'}}{\epsilon_{k+q} + \epsilon_{k'-q} - \epsilon_k - \epsilon_{k'}}.$$

(b) Substitute the result from (a) directly

$$\tilde{c}_{k\sigma}^\dagger \approx c_{k\sigma}^\dagger + \sum_{k',q,\sigma'} \frac{V_q + V_{-q}}{2V} \frac{c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'}}{\epsilon_{k+q} + \epsilon_{k'-q} - \epsilon_k - \epsilon_{k'}}.$$

(c) Substitute the result from (b) directly

$$\tilde{c}_{k\sigma}^\dagger |\text{FS}\rangle = c_{k\sigma}^\dagger |\text{FS}\rangle + \sum_{k',q,\sigma'} \frac{V_q + V_{-q}}{2V} \frac{c_{k+q,\sigma}^\dagger c_{k'-q,\sigma'}^\dagger c_{k'\sigma'}}{\epsilon_{k+q} + \epsilon_{k'-q} - \epsilon_k - \epsilon_{k'}} |\text{FS}\rangle$$

(d) Evaluate the overlap $\langle \text{FS} | c_k \tilde{c}_k^\dagger | \text{FS} \rangle$ term by term

i. The zeroth order $\mathcal{O}(V^0)$

$$\langle \text{FS} | c_k c_k^\dagger | \text{FS} \rangle = 1.$$

Here assume $|k| > k_F$, so $c_k c_k^\dagger = 1$ in the Fermi sea.

ii. Denote the operator

$$\chi_k \equiv [\mathcal{S}, c_k^\dagger].$$

The first order $\mathcal{O}(V^1)$

$$\langle \text{FS} | c_k \chi_k | \text{FS} \rangle = \langle \text{FS} | c_k [\mathcal{S}, c_k^\dagger] | \text{FS} \rangle = 0,$$

where χ_k creates a *two-particle, one-hole state* relative to the Fermi sea. Applying c_k still leaves a net excitation, and the matrix element vanishes since the states are orthogonal ($\mathcal{S}$ is off-diagonal).

iii. The second order $\mathcal{O}(V^2)$

$$\frac{1}{2} \langle \text{FS} | c_k [\mathcal{S}, \chi_k] | \text{FS} \rangle = \frac{1}{2} \langle \text{FS} | c_k [\mathcal{S}, [\mathcal{S}, c_k^\dagger]] | \text{FS} \rangle = -\frac{1}{2} \langle \text{FS} | \chi_k^\dagger \chi_k | \text{FS} \rangle = -\frac{1}{2} |\chi_k | \text{FS} \rangle|^2.$$

Since the generator $\mathcal{S}$ is anti-Hermitian ($\mathcal{S}^\dagger = -\mathcal{S}$), the second-order projection simplifies to $-\frac{1}{2} \langle \chi_k^\dagger \chi_k \rangle_{\text{FS}}$. Substituting the explicit form of χ_k into the weight, we obtain

$$Z_k = 1 - \frac{1}{4V^2} \sum_{k',q,\sigma'} \frac{|V_q + V_{-q}| \theta(k_F - |k'|) \theta(|k+q| - k_F) \theta(|k'-q| - k_F)}{(\epsilon_{k+q} + \epsilon_{k'-q} - \epsilon_k - \epsilon_{k'})^2} < 1,$$

which brings a quasiparticle weight less than 1.

CHAPTER 10 Week Coupling (Canonical) Perturbation Theory

Starting from the equation of motion of the Green's function

$$i \partial_t G_\epsilon \left[c_i, c_j^\dagger \right]_{s_i^+ s_j^-} = \delta_{ij} \quad (10.1)$$

where the canonical means that the RHS is a δ -function, and $[s_i^+, s_j^-] = 2s_i^z \delta_{ij}$. the operators satisfy Wick's Theorem

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \quad [a_i, a_j^\dagger] = \delta_{ij}$$

10.1 Wick Theorem

For Gell-Mann-Low, the limitation

$$\lim_{\alpha \rightarrow 0} \frac{u_\alpha^D(0, -\infty) |\eta_0\rangle}{\langle \eta_0 | u_\alpha^D(0, -\infty) | \eta_0 \rangle} = \lim_{\alpha \rightarrow 0} \frac{|\Psi_\alpha^D(0)\rangle}{\langle \eta_0 | \Psi_\alpha^D(0) \rangle} =$$

we can convert to the obverse $\langle A^H(t) \rangle$ by computing from the GML theory

$$\begin{aligned} \langle E_0 | A^H(t) | E_0 \rangle &= \lim_{\alpha \rightarrow 0} \frac{1}{\langle \eta_0 | S_\alpha | \eta_0 \rangle} \sum_{n=0}^{\infty} \frac{1}{2^n n!} \left(-\frac{i}{\hbar} \right)^n e^{-\alpha(|t_1| + \dots + |t_n|)} \\ &\quad \times \int_{-\infty}^{\infty} dt_1 \dots dt_n \langle \eta_0 | T_\epsilon \left\{ \hat{V}(t_1) \hat{V}(t_2) \dots \hat{V}(t_n) A^H(t) \right\} | \eta_0 \rangle \end{aligned} \quad (10.2)$$

where

$$V = \frac{1}{2} \sum_{\substack{\alpha\beta \\ \gamma\delta}} \langle \alpha\beta | \hat{V} | \gamma\delta \rangle a_\alpha^\dagger a_\beta^\dagger a_\gamma a_\delta$$

The obverse

$$\langle \alpha\beta | \hat{V} | \gamma\delta \rangle = \langle \beta\alpha | \hat{V} | \delta\gamma \rangle$$

For any observables

$$\langle E_0 | A(t) B(t^*) \dots | E_0 \rangle$$

which can be converted to

$$\dots \int_{-\infty}^{\infty} dt_1 \dots dt_n \langle \eta_0 | T_\epsilon \left\{ V(t_1) \dots V(t_n) \underset{a_i(t_i)}{A}(t) \underset{a_f^\dagger(t_f)}{B}(t') \dots \right\} | \eta_0 \rangle$$

To calculate it, consider

$$T_\epsilon(O_1(t_1) O_2(t_2) O_3(t_3) O_4(t_4))$$

which has the order number $4! = 24$.

10.2 Normal product

The “book-keeping” of counting: For a given $|\eta_0\rangle$, the fermi wave vector k_F , then

$$\gamma_k = \begin{cases} c_k^\dagger |\eta_0\rangle = 0, & |k| < k_F \\ c_k |\eta_0\rangle = 0, & |k| > k_F \end{cases}$$

where $\gamma_k |\eta_0\rangle = 0$. Also for $\gamma_k^\dagger |\eta_0\rangle$.

For N -ordering: If we have an arbitrary sequence

$$N(\gamma_1 \gamma_2^\dagger \cdots \gamma_n) \rightarrow (\gamma^+ \gamma^+ \cdots |\gamma \gamma \gamma)$$

where some of the γ s have † . The normal ordering operator N brings the † to the front.

$$\langle \eta_0 | N(\cdots) | \eta_0 \rangle = 0$$

Example 10.2.1. $N(\gamma_1 \gamma_2^\dagger \gamma_3) = (-1) \gamma_2 \gamma_1 \gamma_3 = (-1)^2 \gamma_2^\dagger \gamma_3 \gamma_1$.

Example 10.2.2.

$$c_1 c_2 c_2^\dagger c_3^\dagger = (-1)^3 c_3^\dagger c_1 c_2 c_2^\dagger = (-1)^5 c_3^\dagger c_2 c_2^\dagger c_1 = (-1)^5 c_3^\dagger (1 - c_2^\dagger c_2) c_1 = (-1)^6 (c_3^\dagger c_2^\dagger c_2 c_1 + (-1)^5 (c_3^\dagger \delta_{22} c_1))$$

So, the normal ordering should be

$$N(\cdots) \underset{\text{operator manipulation}}{=} () c^\dagger c^\dagger \cdots c + c^\dagger \cdots c + () \cdots$$

10.3 Contraction

Define the contraction of two operators

$$\overline{A(t)B(t')} = T_\epsilon(A(t)B(t')) - N(A(t)B(t')) \quad (10.3)$$

The operator identity

$$\langle \eta_0 | T_\epsilon(A(t)B(t') - N(A(t)B(t'))) | \eta_0 \rangle$$

10.4 Operator level of “contraction”

Example 10.4.1 (Two operators). We just look at the RHS of the contraction

$$\overline{\gamma(t)\gamma^\dagger(t')} = T_\epsilon(\gamma(t)\gamma^\dagger(t')) - N(\gamma(t)\gamma^\dagger(t'))$$

On the operator level. Where

$$\theta(t-t')\gamma(t)\gamma^\dagger(t') - \theta(t'-t)\gamma^\dagger(t')\gamma(t) - [\theta(t-t') + \theta(t'-t)]\gamma^\dagger(t')\gamma(t) = \theta(t-t')\gamma(t)\gamma^\dagger(t')$$

Then, we have to define the actual contraction

$$\langle \eta_0 | \overline{\gamma \gamma^\dagger} | \eta_0 \rangle = iG^{0,c}[\gamma, \gamma^\dagger]$$

Consider 4 time-ordered term

$$1\ 2\ 3\ 4 : O_1(t_1)O_2(t_2)O_3(t_3)O_4(t_4)$$

have $4! = 24$ T-ordering: complete permutation of index. Only look at the two terms

$$1234 + 2134 + \dots = (12 + 21)34 + T_\epsilon(12)(34) \xrightarrow{\text{Contraction tricks}} (\overline{12} + N(12))(34)$$

So, the observable

$$\langle \eta_0 | \quad | \eta_0 \rangle = ig_{12}^{0c}(34) + \langle \eta_0 | N(12)(34) | \eta_0 \rangle$$

So, the two terms gives

$$1234 + 2134 \rightarrow ig_{12}(34), \quad 1243 + 2143 \rightarrow ig_{12}(43),$$

The combines to ig_{12} and ig_{34} , i.e., $T_\epsilon(\overline{1234})$. Also for $1324 + 3124$ and $1342 + 3142$, which can be combined into $T_\epsilon(\overline{1324})$. The LHS have $4!$ terms in total, and we will have $4 \times 3/2 = 6$ contractions

$$g_{12}g_{34} \quad g_{13}g_{24} \quad g_{14}g_{23}$$

We shall compute

$$\begin{aligned} T_\epsilon(123 \dots 2N) &= N(12 \dots 2N) + \left(\begin{array}{c} N\text{-product} \\ \text{with ONE contraction} \end{array} \right) N(\overline{1234} \dots) + N(\overline{1324} \dots 2N) \\ &+ \left(\begin{array}{c} N\text{-product} \\ \text{with TWO contraction} \end{array} \right) + N(\overline{123456} \dots 2N) + \dots + \{\text{total contractions}\} \quad (10.4) \end{aligned}$$

Example 10.4.2 ($T_\epsilon(123456)$). Apply the result of 1234

$$T_\epsilon(1234)(56), \quad T_\epsilon(C_6^4) \times (C_6^2) \rightarrow C_6^4 \times \text{res}_{1243} \times (C_6^2)$$

i.e., pick 4 operators out of 6.

Do the induction $2n/2n + 2$ -Wick's

$$(\overline{12}(34) + N(12)(34)) \cdot (\overline{56} + N(56)) \rightarrow \prod_{1, \dots, 1}^n (\overline{i_1 i_2} + N(\quad))$$

Together, we have n -terms

$$1; m \underbrace{\quad}_{N_{m+1,n}(\quad)}$$

we can put into normal-ordering of arbitrary terms

$$\gamma \gamma^\dagger \dots \gamma^\dagger = \gamma^\dagger \gamma^\dagger (\delta_{ij} - \gamma^\dagger \gamma)$$

where we consider δ_{ij} as one-contraction, and $|\eta_0\rangle$ is the ground state.

Underlying structure $N(\quad)|\eta_0\rangle = 0$ for Wick's theorem ($|\eta_0\rangle$ Gaussian states).

The canonical algebra

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \quad \gamma \gamma^\dagger = T_\epsilon - N(\dots) \rightarrow s^2 \text{Tr}(s^+, s^-)$$

One is canonical algebra, and one is the Gaussian states. The main idea is to convert $\{s^+, s^-\}$ into canonical bosons: Schwinger Boson, Maleev, Pvmakoff.

For $A = \mathbb{I}$, it becomes

$$\boxed{\dots} \Rightarrow \langle \eta_0 | U_\alpha(t, t') | \eta_0 \rangle_{t, t' \rightarrow \infty}$$

and $S_\alpha = U_\alpha(+\infty, -\infty)$. Do the Wick's rotation $t \rightarrow i\tau$, $S_\alpha \rightarrow Z$. $\langle \Psi_0 | \Psi_0 \rangle$ and $\langle \eta_0 | \Psi_D \rangle$.

10.5 Feynman Diagram (of what)

10.6 $\langle \eta_0 | U | \eta_0 \rangle$: Vacuum diagrams

We rewrite the term into Taylor series form

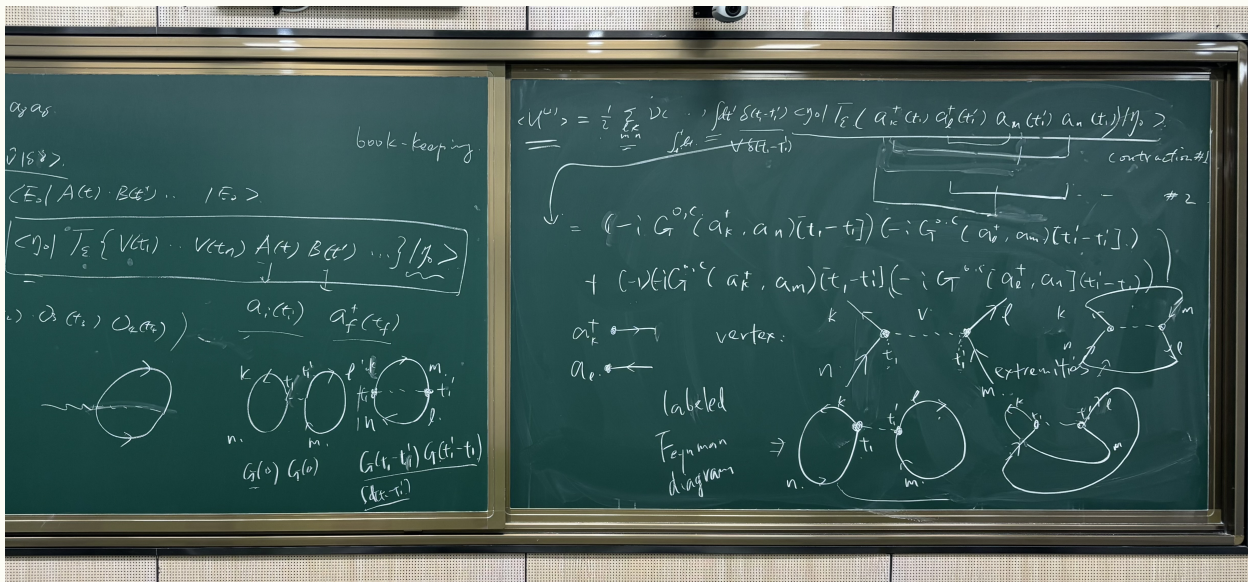
$$\langle \eta_0 | U | \eta_0 \rangle = 1 + \sum_{n=1}^{\infty} \langle \eta_0 | U_\alpha^{(n)}(t, t') | \eta_0 \rangle \quad (10.5)$$

For $n = 1$

$$V(t_1) = \frac{1}{2} \sum_{\substack{kl \\ nm}} \nu(kl, nm) \int dt'_1 \delta(t_1 - t'_1) a_k^\dagger(t_1) a_l^\dagger(t'_1) a_m(t'_1) a_n(t_1)$$

where the index is just the matrix element $\langle kl | v | nm \rangle$.

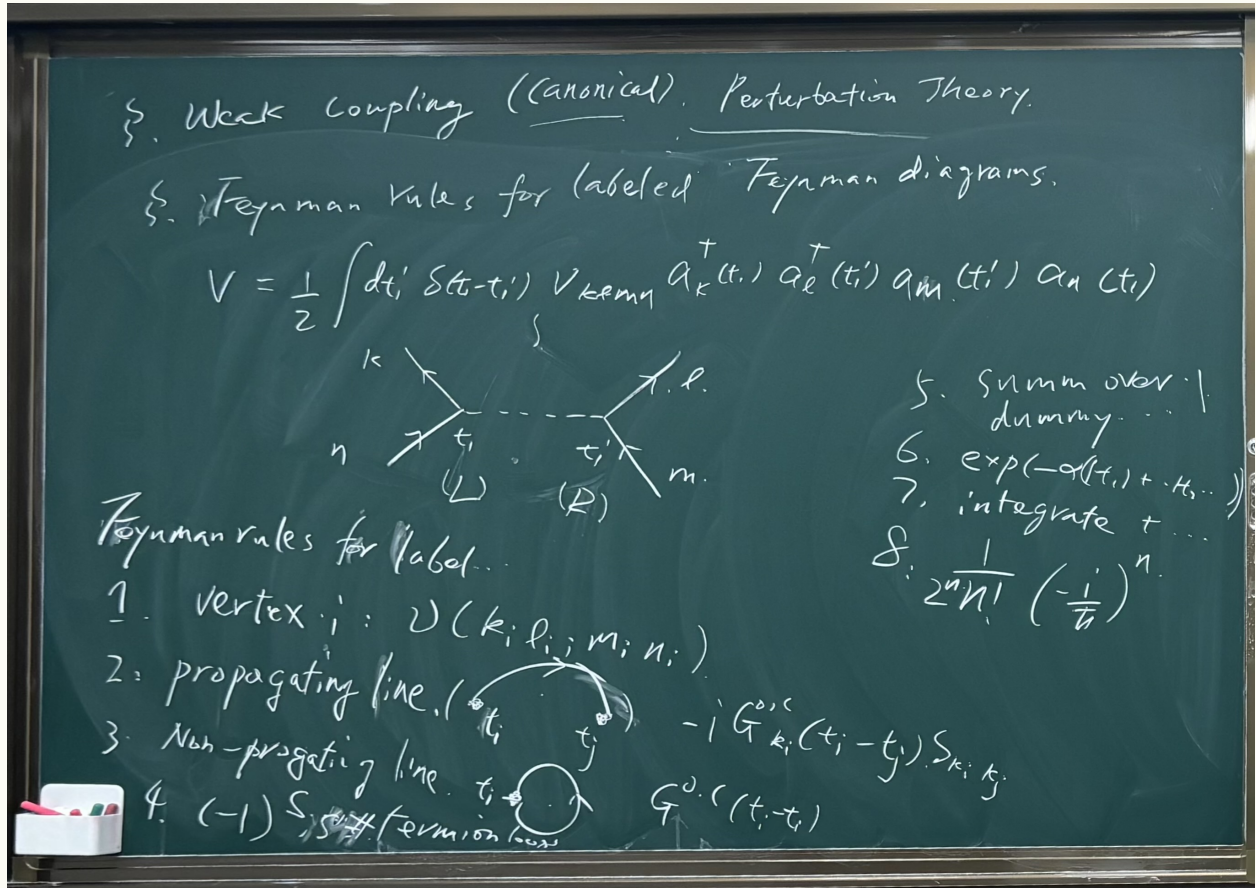
$$\begin{aligned} \langle u^{(1)} \rangle &= \frac{1}{2} \sum_{\substack{lk \\ mn}} \nu(\dots) \int dt' \delta(t_1 - t'_1) \langle \eta_0 | T_\epsilon (a_k^\dagger(t_1) a_l^\dagger(t'_1) a_m(t'_1) a_n(t_1)) | \eta_0 \rangle \\ &= (-iG^{0,C}(a_k^\dagger, a_n)[t_1 - t_1]) (-iG^{0,C}(a_l^\dagger, a_m)[t'_1 - t'_1]) \\ &\quad + (-1) (-iG^{0,C}(a_k^\dagger, a_m)[t_1 - t_i]) (-iG^{0,C}(a_l^\dagger, a_n)[t'_1 - t_1]) \end{aligned}$$



The operators and the vertices $a_k^\dagger : \bullet \longrightarrow$ $a_k : \bullet \longleftarrow$ which form through the labeled Feynman diagram.

10.7 Feynman rules for labeled Feynman diagrams

$$V = \frac{1}{2} \int dt'_1 \delta(t - t'_1) V_{klmn} a_k^\dagger(t_1) a_l^\dagger(t'_1) a_n(t'_1) a_n(t_1) \quad (10.6)$$



For $\langle lk | \hat{V} | mn \rangle$, if one switch $\langle k(L)l(R) | \hat{V} | n(L)m(R) \rangle$, then the diagram will be flipped: permuting extremities ($L \leftrightarrow R$).

(a) Vertex i : $\nu(k_i l_i; m_i n_i)$

(d) $(-1)^S$, S # fermion loop.

(b) Propagating line:

(e) Summ order dummy ...

$$\begin{array}{c} \bullet \quad \bullet \\ t_i \quad t_j \end{array} \quad -i G_{k_i}^{0,c}(t_i - t_j) \delta_{k_i k_j}$$

(f) $\exp(-\alpha(|t_1| + |t_2| + \dots))$

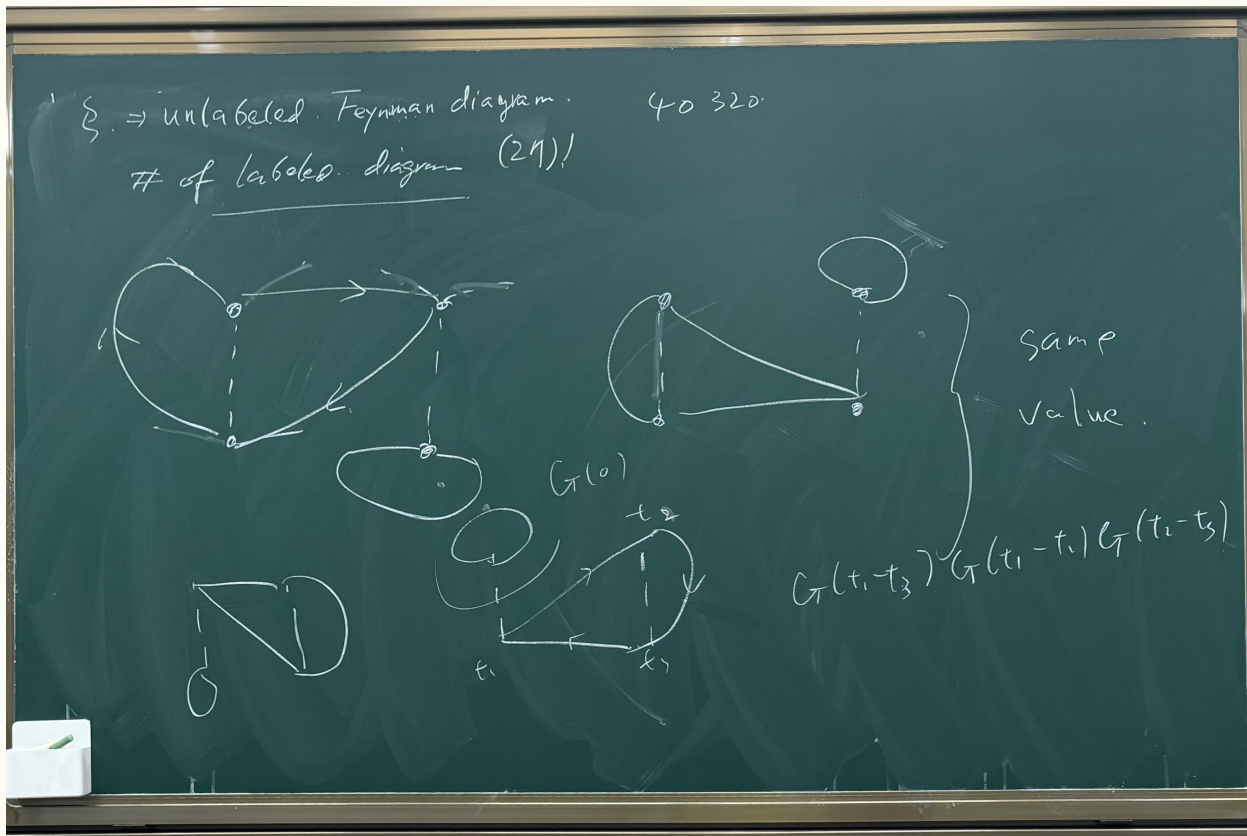
(c) Non-propagating line:

(g) Integrate t ...

$$t_i \quad \text{loop} \quad G^{0,c}(t_j - t_i)$$

(h) $\frac{1}{2^n n!} \left(-\frac{i}{\hbar}\right)^n$

10.8 Unlabeled Feynman diagram # of labeled diagram $(2\eta)!$



we can integrate over

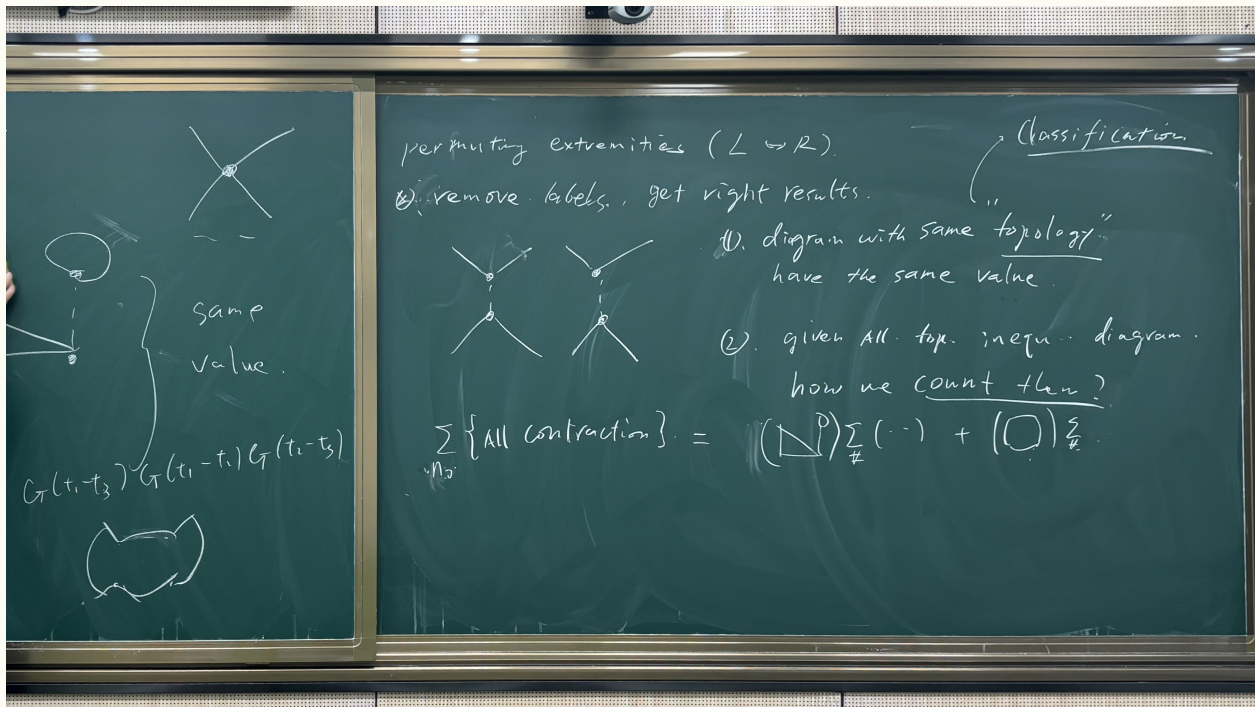
$$G(t_1 - t_3)G(t_1 - t_2)G(t_2 - t_3) \quad (10.7)$$

in these Feynman diagrams.

10.9 Remove labels, get right results

- (a) Diagram with same “topology” (means classification in math) have the same value
- (b) Given all top. inequ diagram how we count them?

$$\sum_{n_0} \{\text{All contraction} = \} \quad (10.8)$$



Transformations of $\Gamma \rightarrow \Gamma'$, but leave it's value unchanged.

Symmetry factor.

We put the labes backing: k, l, n, m .

Transformation Leaves two structures

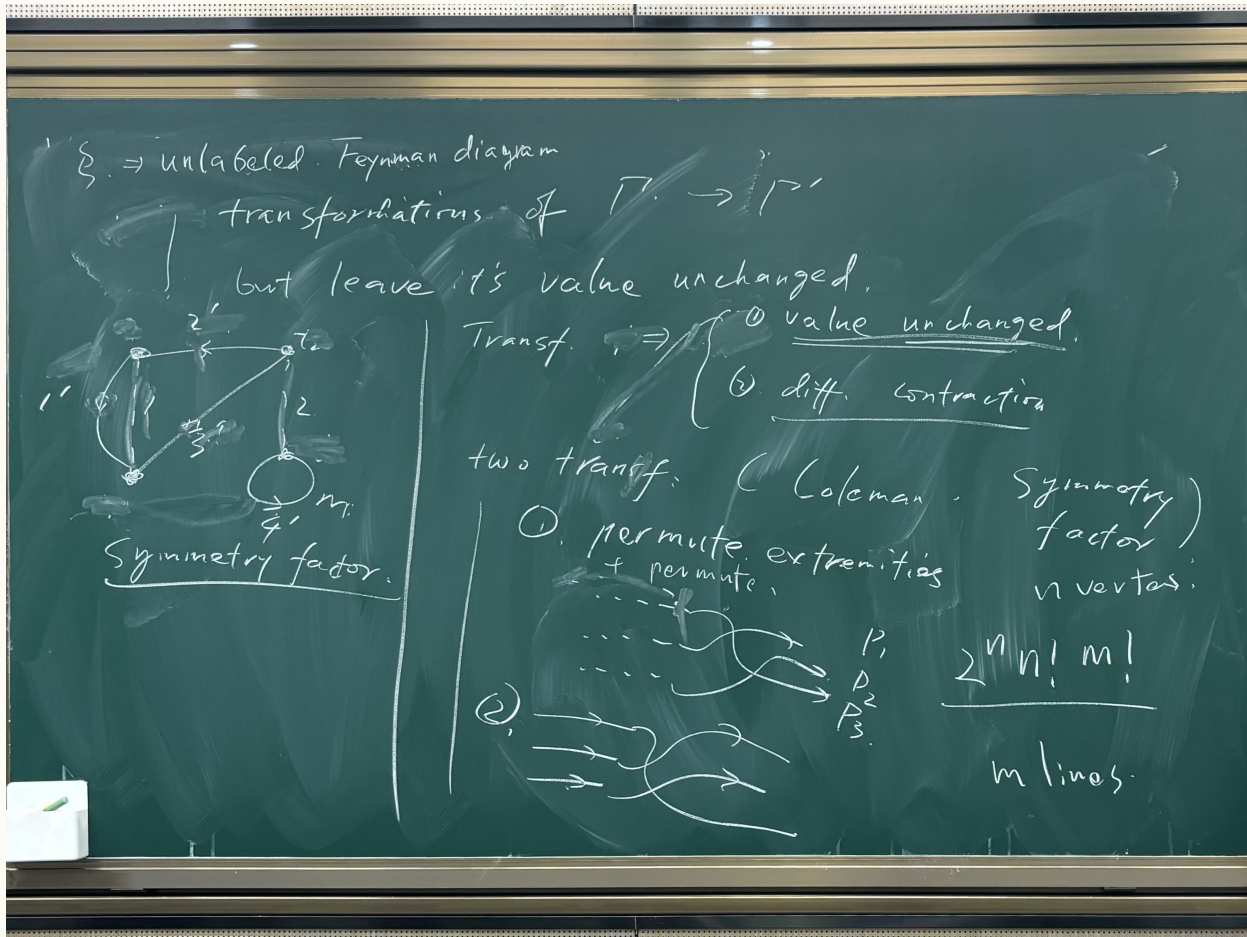
(a) Value unchanged

(b) Diff contraction.

Two type of transformations ([Ref] Coloman: symmetry factor):

(a) Permute extremities.

(b) Propagating lines.



If we have n vertices and m lines, and this will give $2^n n! m!$ possibilities.

All these transf leave value unchanged, but

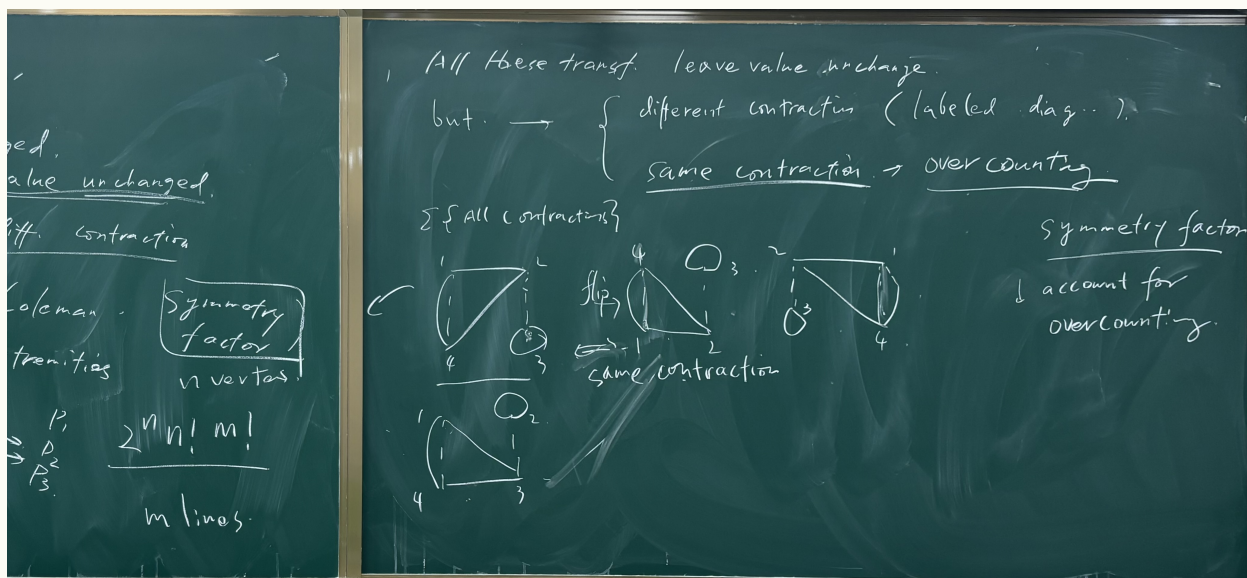
(a) Different contraction (labeled diag)

(b) Same contraction $\rightarrow$ over counting

The sum

$$\sum \{\text{All contractions}\}$$

Symmetry factor: account for overcounting. The flips come from the transformations



Final step Obtain the symmetry factor S .

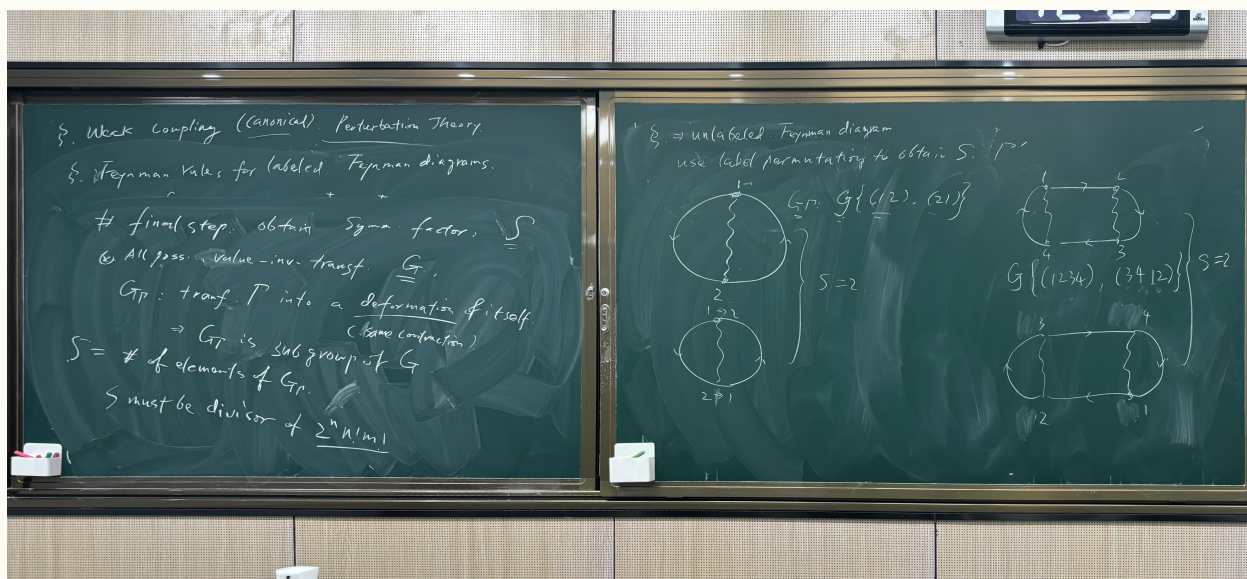
- All pass: value-inv.transf G .
- G_Γ : Transf Γ into a deformations (same contraction) of itself. Hence, G_Γ is sub group of G .
- $S = \#$ of elements of G_Γ , S must be divisor of $2^n n! m!$.

10.10 Unlabeled Feynman diagram

Using label permutation to obtain S , Γ'

$$G_\Gamma : \mathcal{G}\{(12), (21)\}$$

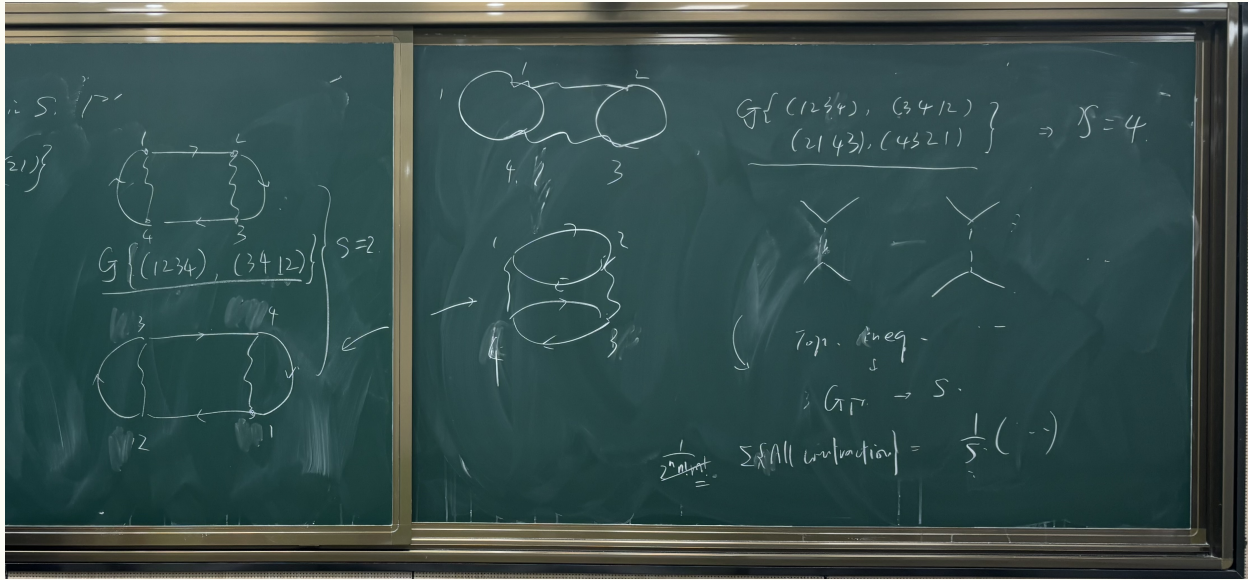
means $1 \rightarrow 2$ and $2 \rightarrow 1$.



These diagrams have $S = 2$.

Example 10.10.1. Two vertices diagram $\mathcal{G}\{(1234), (3412)\}$. Using label Perturbation: $1 \rightarrow 3, 2 \rightarrow 4, 3 \rightarrow 1, 4 \rightarrow 2$.

Example 10.10.2 $((43\ 12) \notin \mathcal{G})$. Diff contraction



Example 10.10.3. Compare with two diagrams

$$\mathcal{G}\{(1234), (3412), (2143), (4321)\} \rightarrow S = 4$$

Combines them Topo ineq.

$$G_{\Gamma} \rightarrow S, \quad \frac{1}{2^n n! m!} \sum \{n || \text{contraction}\} = \frac{1}{s} (\dots)$$

which cancels by “All poss value-inv-transf”, for $v = \frac{1}{2}$.

Homework #11 [2025-11-18]

Problem 11.1.

- (a) Write the time-ordered product

$$T_\epsilon \{a_{k\sigma}(t_1)a_{l\sigma'}^\dagger(t_2)a_{m\sigma}(t_3)a_{n\sigma'}^\dagger(t)3\}$$

in terms of normal products and suitable contractions.

- (b) Express the expectation value of the time-ordered product in part (a) in the ground state $|\eta_0\rangle$ of the unperturbed system in terms of products of the free causal Green's functions.

Solution.

Problem 11.2. Evaluate explicitly the expectation value of the time-ordered product in part 1 of **Problem 11.1** in the ground state $|\eta_0\rangle$ of the unperturbed system for the special case $k = l = m = n, \sigma = \sigma'$ for

- (a) $t_1 > t_2 > t_3$,

- (b) $t_1 > t_3 > t_2$.

Check the results by direct calculation of the expectation values, i.e., without using Wick's theorem.

Solution.

- (a)

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